K-FOG: A Symmetric Multi-Role Game Calculus

Elazar Gershuni, Technion

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Abstract

We introduce K-FOG, a symmetric K-player calculus extending first-order logic. Formulas evaluate to outcome tuples in $\{0,1\}^K$ rather than truth values. The core syntax is a minimal, prenex-style language with player-labeled quantifiers ($P_i x.\Phi$), and we extend it with IF-style binders ($P_i x \setminus V.\Phi$) to constrain information flow.

For $K = 2$, two operations mediate between classical FOL and K-FOG: *abstraction* embeds classical prenex formulas as symmetric, one-hot (zero-sum) games, and *adversarial projection* recovers classical truth values by selecting a player's viewpoint. We analyze the core symmetries—rotation (player swap) and complement (payoff negation)—and show that classical negation is an emergent property of the two-player, zero-sum fragment. We also give a framework for structural combinators that build complex games from simpler components without extending the core language.

Because IF-style binders describe who may see which move, we show that a linear K-FOG quantifier prefix can be read as a linear, non-branching multiparty session protocol. We present two translations to MPST-style global types and adopt the visibility-restricting one so that the $\setminus V$ constraints become explicit in the protocol.

Finally, we give game-theoretic semantics defined by structural recursion, an abstraction–projection round-trip, and a complexity analysis showing that the core is PSPACE-complete while the IF-extension is NEXPTIME-hard.

K-FOL offers a single game-semantic core in which (i) the standard 2-player FO evaluation game appears as the zero-sum fragment via abstraction/projection, and (ii) the fully cooperative, single-specification style of session/choreography checking appears as the all-goals-equal fragment. In that sense it unifies a classical and a protocol-verification reading. We call this an adversarial verification perspective because the same semantic machinery can talk about non-aligned players as well.

1 Introduction

First-order logic (FOL) admits a game-semantics reading: existential quantifiers are choices of a "verifier," universal quantifiers are choices of a "falsifier," and negation swaps the two roles. This is useful, but the two quantifiers encode only two adversarial roles and do not naturally extend to $K > 2$ players with distinct objectives.

We introduce K-FOG as a symmetric variation for K players. Each choice is explicitly labeled by the player making it ($P_0, \ldots, P_{K-1}$), and the operators preserve this symmetry. A K-FOG formula denotes a finite perfect-information game: a prefix of player-labeled quantifiers $P_i x.\,\Phi$ indicates which player chooses each variable, and a terminal K-tuple of classical formulas $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$ determines the outcome $(v_0, \ldots, v_{K-1})$, where $v_i = 1$ means that player i achieves its objective.

Classical truth reappears only in the $K = 2$ specialization by choosing a viewpoint: *adversarial projection* at player $i \in \{0,1\}$ asks "can player i force $v_i = 1$?", yielding an ordinary truth value. Conversely, ordinary FOL quantifiers are recovered by *abstraction*, which embeds a classical formula φ as the zero-sum game $\langle \varphi, \neg\varphi \rangle$. In this fragment, *rotation* (swapping the two players) behaves as classical negation.

A useful way to situate our work is against refined multiparty session types (R-MPST) [81]. In that line of work, one first writes a single global interaction script that says who communicates

when and what each participant is allowed to see, and then refines individual steps with logical conditions so that standard operational questions (e.g. "are the runs of this protocol admissible?") can be asked. We adopt exactly this protocol/logic separation, but we keep the protocol in a broadcast-style linear prefix with IF-style hiding and we fix ordinary first-order logic as the default goal language — choices made so that the same object can also be read as a finite, visibility-constrained game. Because the terminal of a K-FOG formula is a K-tuple of per-role goals, we can, in addition to the usual trace-validity questions, pose explicitly counterfactual, solution-concept questions over the very same script: who can force under these visibility restrictions, which coalitions can succeed together, whether a given profile is Nash-style stable, and so on. Our MPST translation shows that the interaction part can still be projected to a familiar session-typed view, while the abstraction–refinement embedding shows that per-player logical questions are recoverable. Thus the difference is not in rejecting the R-MPST pattern, but in exploiting it for both trace-level verification and game-theoretic analysis under one global description.

The contributions of this paper are:

- A symmetric K-player calculus (K-FOG) with outcomes given by K first-order goals evaluated to $\{0, 1\}$ and a minimal prefix (prenex-style) syntax.
- A detailed algebraic analysis of the $K = 2$ specialization, including rotation and complement, and the FOL retract via abstraction–projection.
- An extension with IF-style information-hiding binders ($P_i x \setminus V. \Phi$) that make the visibility relation explicit.
- A linear, non-branching translation from K-FOG prefixes to MPST-style global protocols, including a visibility-restricting variant that makes the $\setminus V$ constraints explicit.
- A complexity analysis showing that the K-FOG core is PSPACE-complete and the IF-extension is NEXPTIME-hard.

This is a *horizontal* paper: our aim is to fix the core K-player calculus and situate it among neighbouring formalisms (game-semantics for FOL, IF-style binders, MPST-style global protocols, and structural combinators). We leave deeper treatments of each formalism to subsequent work.

The remainder of this paper is organized as follows. Section 2 defines the syntax, game-theoretic semantics, and forceability relation for the K-FOG core. Section 3 defines the S_K group action. Section 11 presents multiple computational readings (procedural, functional, Gallina-style dependent types, and a minimal core calculus with parametric success notions) and explains how K-FOG combines MPST-like protocols with multi-headed game-theoretic outcomes. Section 4 analyzes the $K = 2$ specialization in detail. Section 7 introduces the IF-style extension and the structural operators (including tensor-like compositions and coalition projections). Section 12 relates K-FOG prefixes to MPST-style global protocols and presents the two translation schemes. Section 13 establishes the complexity bounds. Section 14 discusses related work.

2 Core Calculus: Syntax and Semantics

2.1 Syntax

Fix a classical first-order signature $\Sigma = (S, \mathcal{F}, \mathcal{R})$ (sorts, function symbols, and relation symbols) and an integer $K \in \mathbb{N}$. Formulas of K-FOG are generated by

$$\begin{array}{rcll} \Phi & ::= & \langle \varphi_0, \ldots, \varphi_{K-1} \rangle & \text{(terminal } K\text{-tuple)} \\ & | & P_i x. \, \Phi & \text{(player-}i\text{ binding)} \quad i \in \{0, \ldots, K-1\}, \end{array}$$

where each φ_i is a classical first-order formula over Σ.

Thus every formula has the shape

$$P_{i_1} x_1. \, P_{i_2} x_2. \, \cdots \, P_{i_n} x_n. \, \langle \varphi_0, \ldots, \varphi_{K-1} \rangle,$$

i.e. a finite sequence of labeled binders followed by exactly one terminal.

Free variables. A variable is free in Φ if it occurs free in some φ_i and is not bound in the prefix. A formula is *closed* if it has no free variables.

Alpha-equivalence. $P_i x.\, \Phi \equiv_\alpha P_i x'.\, \Phi[x \mapsto x']$ for x' fresh.

Roles and coherence. Each binding is owned by some player index i, and all occurrences of the same index are governed by a single strategy (coherence). The terminal K-tuple encodes objectives: player i succeeds if φ_i holds under the final assignment.

Visibility order. In this base fragment, the visible information at a move is exactly the set of earlier moves in syntactic (left-to-right) order. Later extensions introduce explicit syntax for constraining the dependence/visibility relation.

Underlying visibility DAG. Each K-FOG formula induces a directed acyclic graph (DAG) of information flow: the nodes are the binder occurrences in the prefix, and there is an edge from one binder to any later binder that is permitted to see its value (i.e. not excluded by a visibility/slash set). We regard two games as equivalent if they coincide up to (i) permutation of player indices, (ii) α-renaming of bound variables, and (iii) isomorphism of this visibility DAG.

Perfect recall (standing assumption). We assume *perfect recall*. For each role i: (i) *Action recall*: the local strategy at any binder owned by i may depend on all of i's earlier choices; and (ii) *Knowledge memory*: any variable value that was visible to i at an earlier point remains available at all later binders of i, unless excluded by explicit independence/visibility constraints.

2.2 Histories and Strategies

Fix a structure $\mathcal{M}$ for Σ with domain $|\mathcal{M}|$.

Definition 2.1 (Pure strategies)**.** For a formula Φ and a player i, a *pure strategy* σ_i is a function that, for every subformula Ψ of Φ of the form $\Psi = P_i x.\, \Phi'$ and every assignment s over the free variables of Ψ, returns a value $a \in |\mathcal{M}|$:

$$\sigma_i(\Psi, s) \in |\mathcal{M}|$$

This function represents player i's choice for x given the visible history (encoded in s) at the moment of the choice. A *strategy profile* is a K-tuple $\sigma = (\sigma_0, \ldots, \sigma_{K-1})$.

2.3 Evaluation Semantics

Definition 2.2 (Semantics)**.** Let Φ be a K-FOG formula, $\mathcal{M}$ a structure, s an assignment for the free variables, and σ a strategy profile. The outcome

$$[\![\Phi]\!]_{\mathcal{M},s,\sigma} \in \{0,1\}^K$$

is defined by structural recursion, where $[\,\cdot\,]$ maps true to 1 and false to 0:

$$\begin{aligned} [\![\langle \varphi_0, \ldots, \varphi_{K-1} \rangle]\!]_{\mathcal{M},s,\sigma} &:= \big([\,\mathcal{M}, s \models \varphi_0\,], \ldots, [\,\mathcal{M}, s \models \varphi_{K-1}\,] \big) \\ [\![P_i x., \Phi]\!]_{\mathcal{M},s,\sigma} &:= [\![\Phi]\!]_{\mathcal{M},s[x \mapsto a],\sigma}, \quad \text{where } a = \sigma_i(P_i x., \Phi, s) \end{aligned}$$

For closed Φ, we write $[\![\Phi]\!]_{\mathcal{M},\sigma} := [\![\Phi]\!]_{\mathcal{M},\emptyset,\sigma}$.

2.4 Forceability

Definition 2.3 (Individual forceability)**.** For player $i \in \{0, \ldots, K-1\}$ define

$$F_i(\Phi, \mathcal{M}, s) := \exists \sigma_i\ \forall \sigma_0 \cdots \forall \sigma_{i-1}\ \forall \sigma_{i+1} \cdots \forall \sigma_{K-1}\ \big([\![\Phi]\!]_{\mathcal{M},s,\sigma} \big)_i = 1.$$

The *forceability vector*

$$F(\Phi, \mathcal{M}, s) := \big(F_0(\Phi, \mathcal{M}, s), \ldots, F_{K-1}(\Phi, \mathcal{M}, s) \big) \in \{0,1\}^K$$

records which players can guarantee their objective.

Remark 2.4 (Adversarial reading). The quantification in Definition 2.3 treats all players $j \neq i$ as potential adversaries of player i: their strategies are universally quantified and are not restricted to those that would help them achieve their own objectives. In particular, $F_i(\Phi, \mathcal{M}, s)$ depends only on player i's component of the outcome tuple and on the interaction structure, not on the other components. If we were to replace all φ_j for $j \neq i$ in the terminal with 1 or with 0, the value of F_i would be unchanged.

3 Symmetric Group Action

Player indices are nominal; this is captured by an action of S_K.

3.1 Permutation of Formulas, Strategies, and Outcomes

Definition 3.1 (Permutation action)**.** For $\pi \in S_K$, define $\pi \cdot \Phi$ by:

$$\pi \cdot \langle \varphi_0, \ldots, \varphi_{K-1} \rangle := \langle \varphi_{\pi^{-1}(0)}, \ldots, \varphi_{\pi^{-1}(K-1)} \rangle,$$
$$\pi \cdot (P_i x.\, \Phi) := P_{\pi(i)} x.\, (\pi \cdot \Phi).$$

Define the action on strategy profiles and outcome vectors by

$$(\pi \cdot \sigma)_i := \sigma_{\pi^{-1}(i)}, \qquad \pi \cdot (v_0, \ldots, v_{K-1}) := (v_{\pi^{-1}(0)}, \ldots, v_{\pi^{-1}(K-1)}).$$

Because we kept the prefix order unchanged, the syntactic visibility relation ("all earlier moves") is preserved under π.

Theorem 3.2 (Semantic equivariance)**.** *For any $\pi \in S_K$, structure $\mathcal{M}$, assignment s, and strategy profile σ,*

$$[\![\pi \cdot \Phi]\!]_{\mathcal{M},s,\pi\cdot\sigma} = \pi \cdot [\![\Phi]\!]_{\mathcal{M},s,\sigma}.$$

Corollary 3.3 (Forceability equivariance)**.** *For any $\pi \in S_K$ and i,*

$$F_{\pi(i)}(\pi \cdot \Phi, \mathcal{M}, s) \iff F_i(\Phi, \mathcal{M}, s).$$

Definition 3.4 (Adversarial projection)**.** For $i \in \{0, \ldots, K-1\}$ the adversarial projection $\exists\forall_i(\Phi)$ of a K-FOG formula Φ is the first-order formula defined by

$$\begin{aligned} \underset{i}{\exists\forall}(\langle \varphi_0, \ldots, \varphi_{K-1} \rangle) &:= \varphi_i, \\ \underset{i}{\exists\forall}(P_i x.\, \Phi) &:= \exists x.\, \underset{i}{\exists\forall}(\Phi)\,, \\ \underset{i}{\exists\forall}(P_j x.\, \Phi) &:= \forall x.\, \underset{i}{\exists\forall}(\Phi) \quad (j \neq i). \end{aligned}$$

Lemma 3.5 (Soundness w.r.t. individual forceability)**.** *For any structure $\mathcal{M}$, assignment s, player i, and* K-FOG *formula Φ,*

$$\mathcal{M}, s \models \underset{i}{\exists\forall}(\Phi) \quad \iff \quad F_i(\Phi, \mathcal{M}, s).$$

3.2 Outcome Classes

Fix a background theory T. For a terminal $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$ we distinguish:

- **Zero-sum (one-hot).** $\sum_i [\,\mathcal{M}, s \models \varphi_i\,] = 1$ for all $\mathcal{M}, s$. This covers the classical 2-player fragment via abstraction–projection (see Section 4).
- **Jointly realizable.** $\varphi_0 \wedge \cdots \wedge \varphi_{K-1}$ is T-consistent. Players have distinct objectives, but they are not mutually exclusive. Forceability then tells us which of them can be guaranteed unilaterally.
- **Independent.** All 2^K Boolean combinations of $\{\varphi_0, \ldots, \varphi_{K-1}\}$ are T-consistent. This is the fully non-interfering case and admits tensor-style factorization when the arenas also decompose.

- **Conflicting.** $\exists i \neq j$ with $T \models \varphi_i \to \neg\varphi_j$. The zero-sum (one-hot) case is a special instance.
- **Constant.** Each φ_i is T-equivalent to $\top$ or to $\bot$.

Since K-FOG formulas have a single terminal, a formula inherits its terminal's class.

Remark 3.6. Zero-sum formulas are the natural target for embedding classical FOL (Theorem 4.2).

Remark 3.7 (No general determinacy). For $K > 2$ and arbitrary terminals, the forceability vector $\mathcal{F}(\Phi, \mathcal{M}, s)$ can be any element of $\{0,1\}^K$. In particular, multiple players may simultaneously force their objectives, or none may be able to do so.

3.3 Outcome-insensitive binders as cheap talk

A bound variable x in Φ is *outcome-insensitive* if for all assignments s, s' differing only on x we have the same outcome vector $v(\Phi, s) = v(\Phi, s') \in \{0,1\}^K$. Intuitively, x never affects payoffs/truth. Nevertheless, if some later binder can *observe* x (i.e. is not declared independent of x), x functions as *cheap talk*: it can coordinate which equilibrium is played among several that are already available.

3.4 Algebraic Structure

The general K-FOL calculus carries a small but useful family of syntactic symmetries. They all come from two ingredients: (i) the S_K-action that permutes player indices, and (ii) a uniform componentwise complement on objectives. For $K = 2$ this specializes to the familiar *rotation* together with componentwise complement, which we will later use in the 2-FOL development.

Componentwise complement. We define a uniform, coordinatewise negation of objectives:

$$\widetilde{\langle \varphi_0, \ldots, \varphi_{K-1} \rangle} := \langle \neg\varphi_0, \ldots, \neg\varphi_{K-1} \rangle, \qquad \widetilde{P_i x.\, \Phi} := P_i x.\, \widetilde{\Phi}.$$

This operator $\widetilde{\cdot}$ is an involution, $\widetilde{\cdot}^2 = \mathrm{id}_{\text{K-FOG}}$, and it is S_K-equivariant:

$$\pi \cdot \widetilde{\Phi} = \widetilde{(\pi \cdot \Phi)} \qquad (\pi \in S_K).$$

Equivalently, $\widetilde{\cdot}$ is the unique uniform player-symmetric nontrivial involution on outcome tuples.

Remark 3.8 (Negation under cooperation). If the outcome is *cooperative*, then every player's adversarial projection sees exactly classical negation.

Remark 3.9 (No zero-sum preservation). Componentwise complement does not preserve zero-sum games for $K > 2$; as a simple counterexample, take a game with the constant outcome $(true, false, false)$, complementing it you get $(false, true, true)$

Remark 3.10 (No global De Morgan duality). The complement $\widetilde{\cdot}$ preserves player roles and only flips objectives, so under adversarial projection it yields

$$\underset{i}{\exists\forall}\left(\widetilde{P_i x.\, \Phi}\right) \equiv \exists x. \neg \underset{i}{\exists\forall}(\Phi)\,,$$

which is in general not equivalent to

$$\neg \underset{i}{\exists\forall}(P_i x.\, \Phi) \equiv \forall x. \neg \underset{i}{\exists\forall}(\Phi)\,.$$

Thus $\widetilde{\cdot}$ is a purely algebraic, player-symmetric symmetry and not a logical negation.

Remark 3.11 (Witnesses, not a classification). Theorems 3.9 and 3.10 each establish their negative claim by a single counterexample: they show $\widetilde{\cdot}$ can fail to preserve a given outcome class or duality, not for which outcome classes, if any, it always succeeds. We do not know a classification of the terminals (or outcome classes) for which componentwise complement acts as a logical negation under adversarial projection, nor for which it preserves zero-sumness. This is restated as an open question in Section 15, alongside the closely related question of what group of transformations on $\{0,1\}^K$ generalizes rotation/complement for $K > 2$ (Theorem 3.12).

Remark 3.12 (Outcome classes as regions of the payoff space). The outcome classes of Section 3.2 can be read geometrically by viewing a terminal's truth pattern, for fixed $\mathcal{M}, s$, as a vertex $v \in \{0,1\}^K$ of the Boolean hypercube (equivalently, an indicator vector in $\mathbb{R}^K$). Zero-sum (one-hot) terminals are exactly those confined to the K standard basis vectors $e_0, \ldots, e_{K-1}$, i.e. the K mutually orthogonal "pure conflict" axes; the fully cooperative terminal $\langle \varphi, \ldots, \varphi \rangle$ is confined to the diagonal $\{(0,\ldots,0),(1,\ldots,1)\}$, a single axis shared by all players. Under this reading, $\widetilde{\cdot}$ is the antipodal map $v \mapsto \mathbf{1} - v$ on the hypercube, which fixes the diagonal setwise (matching Theorem 3.9: it fixes the diagonal but does not fix the one-hot vertices for $K > 2$), and for $K = 2$ the rotation swapping players $0, 1$ coincides with a $90°$ rotation of the square $\{e_0, e_1\}$-plane. This suggests a general question, left open here: what is the natural group of transformations of $\{0,1\}^K$ (or its convex hull) interpolating between the one-hot and diagonal extremes, and what game-theoretic meaning — partial coalitions, correlated objectives, degree of cooperativeness — attaches to its orbits for $K > 2$? A related, more quantitative decomposition already exists for utility-based games: every finite game decomposes uniquely into a *potential* component (aligned incentives, akin to the cooperative diagonal) and a *harmonic* component (zero-sum-like) via an orthogonal projection on the space of payoff functions [15]; relating that decomposition to the Boolean/outcome setting here is left for future work.

4 Specialization to Two Players (2-FOG)

The case $K = 2$ yields the two-player fragment 2-FOG. This fragment is the natural target for defining a precise relationship with classical (prenex) FOL. Working in fully prenex form is standard in logic: classical proof procedures such as resolution routinely assume a prenex presentation. We do the same, except that we keep the quantifiers/player-binders as a visible global script rather than pushing them into an implicit semantics. Extending the language to handle first-order intuitionistic logic would require additional constructs, which we defer to future work.

The syntax and semantics are those of K-FOG specialized to $K = 2$:

$$\Phi \ ::= \ \langle \varphi_0, \varphi_1 \rangle \ \mid \ P_0 x.\, \Phi \ \mid \ P_1 x.\, \Phi.$$

The semantics, forceability, and S_2 group action (rotation) are direct specializations of the definitions in Definition 2.2, Definition 2.3, and Definition 3.1. Throughout this section, $i \in \{0,1\}$ and $1 - i$ denotes the opposite role.

4.1 Abstraction (2-player, one-hot)

Let ψ be a prenex classical formula $Q_1 x_1 \cdots Q_n x_n.\ \varphi$ with φ quantifier-free. We define *abstraction* A_i compositionally on the prenex structure. Fix $i \in \{0,1\}$:

$$\begin{aligned} \mathsf{A}_i(\varphi) &:= \langle \varphi,\ \neg\varphi \rangle \quad \text{(for quantifier-free } \varphi\text{)} \\ \mathsf{A}_i(\exists x.\, \psi) &:= P_i x.\, \mathsf{A}_i(\psi) \\ \mathsf{A}_i(\forall x.\, \psi) &:= P_{1-i} x.\, \mathsf{A}_i(\psi) \end{aligned}$$

This embeds prenex FOL into the $K = 2$ fragment as a one-hot game where seat i pursues φ and the other seat pursues its negation.

Proposition 4.1 (Canonical 2-player embedding)**.** *In the two-player fragment, once we fix player for the adversarial projection, there is—up to swapping the two players—exactly one way to embed an ordinary FOL formula φ as a 2-player game: it must be $\mathsf{A}(\varphi) = \langle \varphi, \neg\varphi \rangle$ on atoms, extended compositionally to the Boolean connectives and with $\exists x.\varphi$ mapped to the binder of the refining player and $\forall x.\varphi$ to the binder of the other player.*

Any alternative like $\langle \varphi, 0 \rangle$ breaks the requirement that the other *projection see $\neg\varphi$, and any alternative shape breaks the rotation-negation compatibility once adversarial projection is assumed to respect rotation; hence the embedding is unique under these conditions. (Proof in Appendix A).*

Rotation. Instead of general permutations, for $K = 2$ the group $S_2 = \{\mathrm{id}, (0\,1)\}$ has a single nontrivial element. We call the action of $(0\,1)$ *rotation* and write τ:

$$\tau(\langle\varphi_0, \varphi_1\rangle) := \langle\varphi_1, \varphi_0\rangle, \qquad \tau(P_i x.\,\Phi) := P_{1-i} x.\,\tau(\Phi)\,.$$

This is an involution: $\tau^2 = \mathrm{id}$, and it is the $K = 2$ instance of the general S_K-action. We will use τ explicitly in the 2-FOL algebra.

Generated symmetry with complement. Because $\widetilde{\cdot}$ commutes with every player permutation, the symmetries of the form "permute then (optionally) complement" form a direct product $S_K \times \mathbb{Z}_2$ acting on K-FOG:

$$(\pi, 0) \cdot \Phi := \pi \cdot \Phi, \qquad (\pi, 1) \cdot \Phi := \pi \cdot \widetilde{\Phi}.$$

4.2 Embedding Theorems and Algebraic Identities

Theorem 4.2 (FOL as retract)**.** $\exists\!\!\forall_i \circ \mathsf{A}_i = \mathrm{id}_{\mathrm{FOL}}$ *and* $\exists\!\!\forall_{1-i} \circ \mathsf{A}_i = \neg$.

Proof sketch. Structural induction on prenex FOL formulas. The base case $\mathsf{A}_i(\varphi) = \langle\varphi, \neg\varphi\rangle$ gives $\exists\!\!\forall_i(\cdot) = \varphi$ and $\exists\!\!\forall_{1-i}(\cdot) = \neg\varphi$. The inductive step maps $\exists/\forall$ to P_i/P_{1-i}, which adversarial projection maps back to $\exists/\forall$ and $\forall/\exists$ respectively. □

Theorem 4.3 (Group action and intertwining)**.** *The operators satisfy the following intertwining laws:*

$$\tau \circ \mathsf{A}_i = \mathsf{A}_i \circ \neg, \qquad \underset{i}{\exists\!\!\forall} \circ \tau = \underset{1-i}{\exists\!\!\forall}\,.$$

Proof sketch. For abstraction: $\tau(\mathsf{A}_i(\varphi)) = \tau(\langle\varphi, \neg\varphi\rangle) = \langle\neg\varphi, \varphi\rangle$. $\mathsf{A}_i(\neg\varphi) = \langle\neg\varphi, \neg\neg\varphi\rangle = \langle\neg\varphi, \varphi\rangle$. The inductive step for quantifiers follows similarly. For adversarial projection: $\exists\!\!\forall_i(\tau(\langle\varphi_0, \varphi_1\rangle)) = \exists\!\!\forall_i(\langle\varphi_1, \varphi_0\rangle) = \varphi_1 = \exists\!\!\forall_{1-i}(\langle\varphi_0, \varphi_1\rangle)$. □

Lemma 4.4 (Rotation as classical negation on one-hot)**.** *If every terminal of* Φ *is one-hot (i.e.,* $\varphi_1 \equiv \neg\varphi_0$*), then*

$$\underset{i}{\exists\!\!\forall}(\tau(\Phi)) \equiv \neg\, \underset{i}{\exists\!\!\forall}(\Phi)\,.$$

Proof. By induction. For the base case $\Phi = \langle\varphi, \neg\varphi\rangle$, $\exists\!\!\forall_i(\tau(\Phi)) = \exists\!\!\forall_i(\langle\neg\varphi, \varphi\rangle) = \neg\varphi$, which is $\neg\,\exists\!\!\forall_i(\Phi)$. The inductive step for $P_i x.\Phi'$ follows from the IH: $\exists\!\!\forall_i(\tau(P_i x.\Phi')) = \exists\!\!\forall_i(P_{1-i} x.\tau(\Phi')) = \forall x.\,\exists\!\!\forall_i(\tau(\Phi')) \equiv \forall x.\neg\,\exists\!\!\forall_i(\Phi') \equiv \neg\exists x.\,\exists\!\!\forall_i(\Phi') = \neg\,\exists\!\!\forall_i(P_i x.\Phi')$. The P_{1-i} case is dual. □

5 IF-style binders for roles

We refine the player binder to carry an *independence set*, in the style of Independence-Friendly (IF) logic [47]. The primitive syntax is

$$\Phi ::= \langle\varphi_0, \ldots, \varphi_{K-1}\rangle \;\mid\; P_i x \setminus V.\,\Phi$$

where $i \in \{0, \ldots, K-1\}$ is the role making the move, x is the chosen object, and V is a finite set of *previously bound variables from which this choice must be independent*. Informally, *"player i chooses a value for x, and i's choice may not depend on the values already bound to the variables in V."*

Why variable-level, not role-level. An earlier version of this section attached the independence set to *roles* instead ($P_i x/J$, $J \subseteq \{0, \ldots, K-1\}$: "the future moves of roles in J must not depend on this choice"), matching the natural reading of a broadcast being withheld from certain participants. That grammar is strictly less expressive than what the rest of the paper needs and, in places, already assumed: the DQBF-hardness reduction of Proposition 13.2 requires each existential binder x_j — all owned by the *same* role — to exclude an arbitrary, binder-specific subset H_j of the previously bound universal variables. Role-level exclusion cannot express this

once two binders share an owner, because the number of distinct exclusion sets available is bounded by the number of role subsets, independent of how many variables that role has bound; a $K = 2$ construction has only two role-level exclusion sets available ($\emptyset$ or the other role) no matter how many existential variables P_0 binds. The tensor combinator $\otimes$ of Section 6.1.2 has the same requirement and already used variable-level exclusion sets without this section licensing it. We correct that mismatch here by making variable-level exclusion the primitive and deriving role-level exclusion from it.

Role-level exclusion, as a derived notation. For a formula $\Psi = P_{k_1} z_1 \setminus V_1. \ldots . P_{k_n} z_n \setminus V_n. \langle \vec{\varphi} \rangle$, a variable x, and a set of roles J, define

$$\mathrm{hide}(\Psi, x, J) := P_{k_1} z_1 \setminus V_1'. \ldots . P_{k_n} z_n \setminus V_n'. \langle \vec{\varphi} \rangle, \qquad V_\ell' := \begin{cases} V_\ell \cup \{x\} & k_\ell \in J \\ V_\ell & k_\ell \notin J. \end{cases}$$

Then, for $J \subseteq \{0, \ldots, K-1\} \setminus \{i\}$,

$$P_i x / J. \, \Phi \;:=\; P_i x \setminus \emptyset. \, \mathrm{hide}(\Phi, x, J)$$

recovers the role-level reading exactly: x is added to the slash-set of *every* later binder owned by a role in J, so no future move of any role in J can depend on x. (The same style of rewriting, run in the opposite direction — removing a variable from later slash-sets rather than adding one — is vis-add in Section 6.1.1.) This is the notation used by the MPST translation of Section 12, whose well-formedness condition ($V \subseteq \mathcal{P} \setminus \{i\}$, i.e. every binder's exclusion set is already role-uniform) should be read as restricting that translation to formulas built from this derived form, not as a property of K-FOG formulas in general.

Semantics. The semantics of this binder is identical to $P_i x. \, \Phi$, but it adds a *constraint on valid strategies*. A strategy profile σ is *valid* for Φ if, for every binder $P_i x \setminus V. \, \Phi'$, the strategy function σ_i is independent of V.

Formally, for any two assignments s_1, s_2 over the free variables of $P_i x \setminus V. \, \Phi'$,

$$\big(\forall v \notin V, \; s_1(v) = s_2(v)\big) \implies \sigma_i(P_i x \setminus V. \, \Phi', s_1) = \sigma_i(P_i x \setminus V. \, \Phi', s_2).$$

(Note: The implication states that if two assignments agree on all variables *except* possibly for those in V, the strategy must produce the same choice. This formally captures independence from V.)

The definitions of evaluation $[\![\cdot]\!]$ and forceability F_i (Definition 2.2 and Definition 2.3) are unchanged, but the quantification $\exists \sigma_i$ is now restricted to *valid* strategies.

This is a syntactic, not a semantic, guarantee. Validity constrains the *strategy function's argument list*: x's chosen value cannot be computed as a function of V. It does not follow that the *outcome* — whether ϕ_i ends up true — is independent of V; ϕ_i can still relate x and V through the terminal formula itself, in exactly the way a classical IF-logic formula's truth conditions can involve two syntactically independent variables through a shared atomic subformula [77]. This is not a defect: it is what the Proposition 13.2 DQBF reduction exploits (the matrix f entangles x_j and the universal variables it cannot see), and it is the expected, IF-logic-standard reading. It does mean $\setminus V$ alone gives no epistemic guarantee that role i's later behavior, or the outcome, is uncorrelated with V in any stronger sense — only that i's choice function cannot name V as an input. A genuinely stronger, more epistemic notion of independence, closer to a guarantee about what is knowable rather than only about a function's argument list, would need to be built from something like Section 7.1's identifiability machinery (Id_u), which tracks what is *derivable* from public commitments rather than only what a binder is syntactically permitted to name. We have not built this stronger notion or compared its forceability theory to the one in this paper.

5.1 Syntax Sugar: Where-Clauses

For convenience we allow binders to carry an explicit local condition, in the style of refined MPST. A binder

$$P_i x : \psi. \Phi$$

is required to use only variables that were bound earlier in the prefix and are visible to player i (i.e. not hidden from i by a slash-set). Its meaning is defined by desugaring to the core calculus as follows:

1. ψ is conjoined to player i's terminal component, expressing that i is responsible for establishing it;
2. for every other player j, every subformula of j's terminal component that depends on x is wrapped under $\psi \to (\cdot)$, expressing that j may rely on i having established ψ.

Intuitively, a where-clause records at the point of the move which property the owner promises to make true, and makes this promise available as an assumption to later players.

6 Structural Operations: A General Framework

Before introducing specific binary operations like $\otimes$ and $|\rangle$, we develop a general theory of *structural operations* on K-FOG formulas. The intuition is simple: a structural operation combines games by interleaving their syntactic structure (the prefix) and composing their outcomes (the terminal), but never reorders moves *within* a single component game or inspects semantic properties like forceability or equilibrium structure.

Informal overview. Think of each K-FOG formula as an abstract syntax tree: a sequence of player-labeled binders followed by a terminal tuple. A *structural combinator* merges multiple such trees by:

1. **Shuffling prefixes**: interleaving the binders while preserving the relative order within each component,
2. **Managing information flow**: optionally adding IF-style independence constraints ($\backslash V$) to control visibility,
3. **Composing terminals**: building the outcome tuple from the component outcomes via Boolean operations.

Crucially, these operations are *syntactic* and *local*: they operate on the structure of formulas without examining game-theoretic properties or restructuring individual components internally.

Player renaming convention. When combining formulas $\Phi^{(1)}, \ldots, \Phi^{(m)}$ over disjoint signatures, we automatically rename players to fresh, disjoint blocks of indices. If $\Phi^{(i)}$ is a K_i-FOL formula, its players are renumbered to $\{\sum_{j<i} K_j, \ldots, \sum_{j\le i} K_j - 1\}$, so the combined formula is a K'-FOL formula where $K' = \sum_{i=1}^{m} K_i$. This ensures player identities remain distinct; merging players across games is handled separately via coalition projection (Definition 6.10).

Definition 6.1 (Structural combinator)**.** A **structural combinator** is an operator C that takes $m \ge 2$ formulas $\Phi^{(1)}, \ldots, \Phi^{(m)}$ over pairwise disjoint signatures $\sigma_1, \ldots, \sigma_m$ and produces a formula Ψ over $\sigma = \bigcup_{i=1}^{m} \sigma_i$ satisfying the following conditions.

After player renaming, write each component as

$$\Phi^{(i)} \;=\; P_{k_1^{(i)}} x_1^{(i)}. \cdots P_{k_{n_i}^{(i)}} x_{n_i}^{(i)}. \langle \varphi_0^{(i)}, \ldots, \varphi_{K_i-1}^{(i)} \rangle.$$

(SC1) Shuffle structure. The output prefix is determined by a *shuffle*

$$\sigma : \{1, \ldots, N\} \to \bigcup_{i=1}^{m} \{i\} \times \{1, \ldots, n_i\},$$

where $N = \sum_{i=1}^{m} n_i$, satisfying:

- σ is a bijection (every input binder appears exactly once),

- For each game i, if $\sigma(j) = (i, k)$ and $\sigma(j') = (i, k')$ with $j < j'$, then $k < k'$ (order preservation within components).

The output is

$$\Psi \;=\; P_{\ell_1} y_1 \setminus V_1. \cdots P_{\ell_N} y_N \setminus V_N. \langle \psi_0, \ldots, \psi_{K'-1} \rangle,$$

where, if $\sigma(j) = (i, k)$, then $y_j = x_k^{(i)}$, $\ell_j = k_k^{(i)}$, and $V_j \subseteq \{y_1, \ldots, y_{j-1}\}$ (IF-constraints may reference only earlier variables in the linear prefix).

(SC2) Terminal compositionality. For each output index $\ell \in \{0, \ldots, K' - 1\}$ fix a Boolean formula

$$B_\ell\big(\{p_j^{(i)} \mid 1 \le i \le m,\ 0 \le j \le K_i - 1\}\big)$$

in propositional variables $p_j^{(i)}$ indexed by the components and their players. Given input formulas

$$\Phi^{(i)} \;=\; P_{k_1^{(i)}} x_1^{(i)}. \cdots . P_{k_{n_i}^{(i)}} x_{n_i}^{(i)}. \langle \varphi_0^{(i)}, \ldots, \varphi_{K_i-1}^{(i)} \rangle \quad (1 \le i \le m),$$

the output terminal is

$$\langle \psi_0, \ldots, \psi_{K'-1} \rangle \quad \text{where} \quad \psi_\ell \;:=\; B_\ell\big(\{\varphi_j^{(i)}\}_{i,j}\big).$$

We require *block uniformity*: the Boolean formulas $(B_0, \ldots, B_{K'-1})$ must treat the component games symmetrically. Permuting entire player blocks in the output corresponds to permuting which component game's formulas appear in the Boolean construction, ensuring no artificial preference for specific component positions.

(SC3) Player preservation.

- The output uses exactly the player indices $\{0, \ldots, K' - 1\}$ from the renamed inputs (no new indices introduced),
- Each binder position preserves its player index: if $\sigma(j) = (i, k)$, then $\ell_j = k_k^{(i)}$ (no player duplication or fusion),
- The output terminal has exactly K' components (arity matches the player count).

Remark 6.2 (Information flow vs. shuffle choice). Condition (SC1) separates two orthogonal design choices: the *shuffle* σ (which positions get which binders) and the *IF-pattern* $(V_1, \ldots, V_N)$ (which information is actually visible). The same shuffle with different IF-patterns yields combinators with identical syntactic structure but radically different information flow. This distinction is central to the relationship between $|\rangle$ and $\otimes$ (see §6.1.2).

Remark 6.3 (Cross-game outcome dependencies). The Boolean formulas B_ℓ in (SC2) explicitly permit a player's objective to depend on objectives from multiple input games. For instance, when combining two 2-FOL formulas Φ and Ψ with terminals $\langle \varphi_0, \varphi_1 \rangle$ and $\langle \psi_0, \psi_1 \rangle$, we can define

$$B_0 = p_0^{(1)} \wedge p_1^{(2)}, \quad B_1 = p_1^{(1)} \wedge p_0^{(2)}, \quad B_2 = \top, \quad B_3 = \top$$

to obtain the output terminal $\langle \varphi_0 \wedge \psi_1, \varphi_1 \wedge \psi_0, \top, \top \rangle$. This enables constructions like the cross-diagonal product $\circledast$ and models scenarios where a physical player occupies different roles in different subgames.

Lemma 6.4 (Normal form)**.** *Every structural combinator is uniquely determined (up to α-equivalence of bound variables and up to block-preserving permutations of player indices) by the triple $(\sigma, (V_1, \ldots, V_N), (B_0, \ldots, B_{K'-1}))$ consisting of:*
1. A bijective shuffle σ as in (SC1),
2. An IF-pattern $(V_1, \ldots, V_N)$ with $V_j \subseteq \{y_1, \ldots, y_{j-1}\}$,
3. A tuple of Boolean formulas $(B_0, \ldots, B_{K'-1})$ as in (SC2).
Conversely, any such triple satisfying (SC1)–(SC3) defines a structural combinator.

This normal form shows that structural combinators have no hidden degrees of freedom beyond the three explicit design choices.

Remark 6.5 (A categorical semantics is plausible but unchecked). The three-part normal form above (shuffle, IF-pattern, terminal-combining Boolean functions) is exactly the kind of data a *lens-* or *optic*-based categorical semantics is built to organize, and the compositional-game-theory literature already builds a symmetric monoidal category of open games with sequential and parallel composition strikingly similar in shape to $|\rangle$ and $\otimes$ below. We have not attempted this connection; see Section 14.5 for what we do and do not know about it, including a mismatch (Nash-style equilibrium selection vs. our forceability semantics) that would need to be resolved first.

Proposition 6.6 (Combinator closure)**.** *The class of structural combinators is closed under composition: if C_1 and C_2 are structural combinators, then $C_1(\Phi, C_2(\Psi, \Theta))$ is equivalent to a single structural combinator $C_3(\Phi, \Psi, \Theta)$ acting directly on the original components.*

Proof sketch. Composing shuffles yields a shuffle (by transitivity of order preservation and composition of bijections). Composing Boolean formulas on terminals yields a Boolean function. The union of two prefix-respecting IF-patterns is prefix-respecting: if V_j references only earlier variables and V'_k does likewise in the nested structure, their combined pattern in the flattened prefix references only earlier variables. Player preservation (SC3) is maintained because composition does not introduce, duplicate, or merge player indices. □

Decomposability. Sometimes we want to know whether the combined game still exposes each component verbatim. Call a structural combinator *decomposable* if, for every input tuple $(\Phi^{(1)}, \ldots, \Phi^{(m)})$, each component $\Phi^{(i)}$ can be read back from the output by: (i) selecting exactly those binders whose source in the shuffle was component i, (ii) deleting from their IF-sets only variables coming from other components, and (iii) projecting the terminal to the entries corresponding to component i's players. A sufficient and necessary condition for this simple reconstruction is: *prefix transparency* (the IF-sets V_j never reference variables from the same component i when $\sigma(j) = (i, k)$) and *terminal transparency* (every input terminal formula $\varphi_j^{(i)}$ appears verbatim among the output terminals, i.e., $B_\ell = p_j^{(i)}$ for some ℓ). The basic products ($|\rangle$ and $\otimes$ in Example 6.7 below) both satisfy these conditions.

Structural morphisms (unary operations). While structural combinators merge multiple games (arity $m \geq 2$), we also consider *structural morphisms* that transform a single game (arity $m = 1$). Some morphisms preserve the player set structure (like componentwise complement), while others intentionally violate (SC3) by restructuring player identities through fusion or relabeling. Examples include:

- The permutation action $\pi \cdot \Phi$ for $\pi \in S_K$ (Definition 3.1), which relabels all binders and permutes the terminal (violates (SC3) by changing player indices),
- Coalition projections $\mathsf{Co}_\pi^\wedge$ and $\mathsf{Co}_\pi^\vee$, which reduce a K-FOL formula to a 2-FOL formula by merging roles according to a surjection $\pi : \{0, \ldots, K-1\} \twoheadrightarrow \{A, B\}$ (violates (SC3) by fusing players),
- The componentwise complement $\widetilde{\Phi}$, which negates all terminal formulas while preserving the player set and prefix structure (preserves (SC3)).

We reserve the term *combinator* for operators satisfying (SC3) and *morphism* for unary operators (some of which may violate (SC3)). The full algebra of K-FOG is built by composing combinators and morphisms.

Example 6.7 (Prototypical combinators)**.** Two fundamental structural combinators illustrate the extremes of information flow:

1. The **sequential (leaky)** product $\Phi \,|\rangle\, \Psi$ uses a concatenation shuffle with empty IF-pattern ($V_j = \emptyset$ for all j), allowing Ψ-moves to observe all Φ-moves.

2. The **tensor (independent)** product $\Phi \otimes \Psi$ uses the same concatenation shuffle but adds IF-constraints to enforce strategic independence: Ψ-strategies must be independent of all Φ-moves.

Both use the same shuffle and concatenate terminals; only the IF-pattern differs. Precise definitions and properties appear in §6.1.2.

Example 6.8 (Non-decomposable: cross-diagonal product)**.** The cross-diagonal construction $\Phi \circledast \Psi$, which we define later in Definition 6.10 via coalition projection, is *not* decomposable because it violates terminal transparency. Its terminal

$$\langle \varphi_0 \wedge \psi_1, \varphi_1 \wedge \psi_0 \rangle$$

mixes objectives from both components: neither φ_0 nor ψ_1 appears as a standalone output terminal. This mixing is intentional—$\circledast$ models scenarios like Battleship where each physical player occupies different roles in different subgames, and recovery of the original component games is neither possible nor desired.

What structural operations exclude. To clarify the boundary of the class, structural operations do *not*:

- Reorder moves within a component game (violates (SC1) order preservation),
- Inspect game-theoretic properties like forceability or determinacy (violates syntactic locality),
- Create outcomes that depend on meta-properties like "number of quantifiers controlled by player 0" (such dependencies cannot be expressed as Boolean functions of component terminals),
- Duplicate a single player into multiple roles with independent strategies (violates (SC3) for combinators; some morphisms may do this).

These restrictions ensure that structural operations are *predictable*: understanding the combined game requires only understanding each component plus the explicit combination rules given by $(\sigma, (V_j), (B_0, \ldots, B_{K'-1}))$.

6.1 Derived operators

The core language stays prenex, single-terminal K-FOL. We keep it small and add useful operations *externally*, by rewriting prefixes and terminals.

6.1.1 Clock and Megaphone Points

We sometimes want to split a single linear interaction into two phases, so that all binders of the second phase are guaranteed to "come after" the first phase in the visibility order. Since our syntax is purely prenex, we realize such phase boundaries as ordinary binders owned by a distinguished, purely structural player.

Preliminaries. Fix once and for all a ground sort unit with a single element $()$. Let Clk denote a fresh player index not occurring in the original game (i.e. if Φ uses players $\{0, \ldots, K-1\}$, then $\mathsf{Clk} = K$). We assume that terminals never mention Clk.

For a formula Ψ and a variable name z, write

$$\text{vis-add}(\Psi, z)$$

for the formula obtained from Ψ by making z *visible* to every binder of Ψ, i.e. if a binder of Ψ is written $P_i x \setminus V.\, \Psi'$, we rewrite it to $P_i x \setminus (V \setminus \{z\}).\, \text{vis-add}(\Psi', z)$. Intuitively, no binder is allowed to hide from z.

More generally, for a finite set S of variables we define $\text{vis-add}(\Psi, S)$ by removing S from every slash-set.

We write $\text{vars}(\Phi)$ for the set of variables bound in Φ.

Clock point. Given a K-player formula

$$\Phi \;=\; P_{i_1} x_1.\, \ldots .\, P_{i_n} x_n.\, \langle \varphi_0, \ldots, \varphi_{K-1} \rangle,$$

its *clocked* version is the $(K+1)$-player formula

$$\mathsf{Clk}(\Phi) \;:=\; P_{\mathsf{Clk}} c : \mathsf{unit}.\, \Phi$$

where c is fresh and Clk is the new structural player. Since the clock binder is last, it sees all variables bound in Φ.

Clock sequencing. Given two formulas Φ and Ψ over the same player set $\{0, \ldots, K-1\}$, their *clock-sequencing*

$$\Phi \triangleright \Psi$$

is defined as

$$\Phi \triangleright \Psi \;:=\; P_{\mathsf{Clk}} c : \mathsf{unit}.\, \text{vis-add}(\Psi, c)$$

placed after Φ, i.e. concretely

$$\Phi \triangleright \Psi \;:=\; \big(\Phi \cdot P_{\mathsf{Clk}} c : \mathsf{unit}.\, \text{vis-add}(\Psi, c)\big),$$

where "$\cdot$" denotes concatenation of the two prefixes. Intuitively, the clock is a value-less choke-point: it comes after Φ, and we force all binders in Ψ to be able to see it, so Ψ is unambiguously in the "second phase."

Megaphone point. Given a K-player formula

$$\Phi \;=\; P_{i_1} x_1.\, \ldots .\, P_{i_n} x_n.\, \langle \varphi_0, \ldots, \varphi_{K-1} \rangle,$$

its *megaphoned* version is the $(K{+}1)$-player formula

$$\mathsf{Meg}(\Phi) \;:=\; \Phi \cdot P_{\mathsf{Meg}}\, x_1.\; \ldots .\, P_{\mathsf{Meg}}\, x_n.\, \langle \varphi_0, \ldots, \varphi_{K-1} \rangle$$

where Meg is a fresh role and each u_{x_k} is fresh. Since the megaphone binders are placed *after* Φ, they see all x_k; we assume a fixed echo strategy $\sigma_{\mathsf{Meg}}(u_{x_k}) = x_k$ (the role is non-strategic). This adds no new objectives, only copies of earlier values.

Megaphone sequencing. Given two formulas Φ and Ψ over the same player set $\{0, \ldots, K{-}1\}$, define their *megaphone sequencing*, written

$$\Phi \mathbin{!} \Psi,$$

by inserting per-variable echoes of Φ at the cut and making Ψ see them:

$$\Phi \mathbin{!} \Psi \;:=\; \Phi \cdot P_{\mathsf{Meg}}\, x_1.\; \ldots .\, P_{\mathsf{Meg}}\, x_m.\Psi,$$

where $\{x_1, \ldots, x_m\}$ are the variables bound in the left part Φ. Intuitively, the megaphone *re-utters* x and is made visible to all binders of Ψ, so later moves can condition on the previously hidden x's.

Informal view. In the visibility graph, a clock introduces a single node that every later node must see, but that carries no data. A megaphone introduces a single node after which the visibility relation is reset to "everybody sees everything so far."

6.1.2 Sequencing vs. tensor products of disjoint games

We now provide the complete formal definitions of the two prototypical combinators introduced in Example 6.7, specialized to the case of disjoint arenas.

In this subsection we fix the following setting.

Disjoint arenas. $\mathcal{A}_1$ and $\mathcal{A}_2$ share no relation or function symbols. Terminals of the first component mention only variables of the first component, and likewise for the second.

Broadcast, left-to-right play. The combined game is played in the order induced by the written prefix. Every move sees all earlier moves.

Pure strategies. A strategy for player i is a total function from the visible history (subject to IF-constraints, when present) to a value.

Symmetric-capable players. Every player can realize any pure strategy that is admissible under the visibility regime (no computational or oracle asymmetries are modeled in the main results).

We introduce two external binary operators on K-FOL formulas over disjoint arenas: an *asymmetric, leaky* one (a pipe) and a *symmetric, independence-enforcing* one (a tensor).

Sequencing (leaky) product. Let Φ be a K_1-FOL formula over arena $\mathcal{A}_1$ and let Ψ be a K_2-FOL formula over arena $\mathcal{A}_2$, with $\mathcal{A}_1$ and $\mathcal{A}_2$ disjoint. Write

$$\Phi \equiv P_{i_1}x_1 \ldots P_{i_m}x_m.\, \langle \varphi_0, \ldots, \varphi_{K_1-1} \rangle, \qquad \Psi \equiv P_{j_1}y_1 \ldots P_{j_n}y_n.\, \langle \psi_0, \ldots, \psi_{K_2-1} \rangle,$$

after renaming the roles of Ψ to $\{K_1, \ldots, K_1+K_2-1\}$. We define the *sequencing product*

$$\Phi \mid\rangle \Psi$$

to be the (K_1+K_2)-FOL formula

$$P_{i_1}x_1 \ldots P_{i_m}x_m.\, P_{j_1}y_1 \ldots P_{j_n}y_n.\, \langle \varphi_0, \ldots, \varphi_{K_1-1}, \psi_0, \ldots, \psi_{K_2-1} \rangle.$$

That is, we just concatenate the prefixes and the terminal tuples. Because the prefix order is "all x first, then all y," every move in the Ψ-component can *see* all x-values chosen in the Φ-component. Information therefore flows one way, from Φ to Ψ, but not back.

Tensor (independent) product. To remove this one-sided leak we use the IF-style binder. Let

$$X := \{x_1, \ldots, x_m\}, \qquad Y := \{y_1, \ldots, y_n\}.$$

We define the *tensor* of Φ and Ψ by

$$\Phi \otimes \Psi \;:=\; \big(P_{i_1}x_1 \setminus Y. \ldots P_{i_m}x_m \setminus Y.\big)\, \big(P_{j_1}y_1 \setminus X. \ldots P_{j_n}y_n \setminus X.\big)\, \langle \varphi_0, \ldots, \varphi_{K_1-1}, \psi_0, \ldots, \psi_{K_2-1} \rangle.$$

Here every x-move is explicitly independent of *all* Y, and every y-move is explicitly independent of *all* X. Valid strategies for $\Phi \otimes \Psi$ must therefore choose x's without seeing any y's and choose y's without seeing any x's. This is the genuinely non-signalling combination.

Forceability invariance. Forceability asks whether a winning strategy *exists*, not whether it can be feasibly computed or executed. At this idealized level, the extra visibility in the leaky product does not change the answer.

Proposition 6.9 (forceability ignores information flow for disjoint games)**.** *Let $\mathcal{A}_1, \mathcal{A}_2$ be disjoint arenas. Then for every structure $\mathcal{M}$, assignment s, and player i,*

$$F_i(\Phi \otimes \Psi, \mathcal{M}, s) \;=\; F_i(\Phi \mid\rangle \Psi, \mathcal{M}, s).$$

Proof. The two products have the same terminal $\langle \varphi_0, \ldots, \varphi_{K_1-1}, \psi_0, \ldots, \psi_{K_2-1} \rangle$. By disjointness, each φ_k mentions only X and each ψ_ℓ mentions only Y.

($\otimes \Rightarrow |\rangle$): any strategy profile valid for the tensor is also a strategy profile for the leaky product, which only reveals *more* history.

($|\rangle \Rightarrow \otimes$): fix i. By symmetry of the construction it suffices to treat $i \in \{K_1, \ldots, K_1+K_2-1\}$ (the Ψ-side). Suppose i has a pure winning strategy σ_i in $\Phi \mid\rangle \Psi$. Since i's objective ψ_j depends only on Y, any dependence of σ_i on X-values is observationally irrelevant for winning. Define σ_i' by: for any history h visible to player i in $\Phi \otimes \Psi$, pick some extension h' in $\Phi \mid\rangle \Psi$ with the same Y-projection and set $\sigma_i'(h) := \sigma_i(h')$. Since σ_i wins regardless of X, every such σ_i' witnesses forceability in the tensor game. □

Realizability (the operational view). Proposition 6.9 concerns the existence of winning strategies, not their feasibility. In real protocols, where players have bounded computation, limited randomness, or restricted oracle access, the two products separate: the leaky product allows observing another player's expensive computations, random samples, or privileged information, while the tensor forbids such observation. Forceability is insensitive to these resource constraints; realizability is not.

Operational reading. The tensor $\Phi \otimes \Psi$ is written as one prenex block, but the IF-constraints make the X-moves and the Y-moves *strategically* independent. This supports two equivalent readings:

1. **Static**: any linearization that respects the constraints yields the same outcome.
2. **Nondeterministic/concurrent**: any enabled binder may move next; because of independence the interleaving does not affect the terminal.

Under this second reading, $\otimes$ behaves like a genuine "parallel" combination, in contrast to the leaky pipe $|\rangle$, which bakes the one-sided information flow into the prefix order.

Definition 6.10 (Coalition forming)**.** Let Φ be a K-role game and let

$$\pi : \{0, \ldots, K-1\} \twoheadrightarrow \{0, \ldots, M-1\}$$

be a surjection with $1 \le M < K$. For each j, the set $\pi^{-1}(j)$ is the j-th *block* of original roles.

We form coalitions along π in two steps:

1. **Relabel binders.** In every subterm $P_i x.\, \Psi$ of Φ, replace P_i by $P_{\pi(i)}$. After this step, each coalition j acts as a single role that makes all moves of the roles in its block $\pi^{-1}(j)$.
2. **Combine terminal goals per block.** If Φ ends in $\langle \psi_0, \ldots, \psi_{K-1} \rangle$, then for each block $\pi^{-1}(j)$ pick a fixed Boolean way of combining the goals in that block (for example: "all of them must hold" or "at least one must hold") and replace the terminal by the M-tuple whose j-th component is that combination of the goals of $\pi^{-1}(j)$.

We write the resulting M-role game as $\mathsf{Co}_\pi^{\mathsf{agg}}(\Phi)$. We require $M < K$ so this actually merges at least two roles; when $M = K$ this construction reduces to a mere permutation.

Remark 6.11 (Merging is lossless only under broadcast). When Φ has no $\backslash V$ binders, relabeling a block of roles to a single role loses no information about who can see what, because every binder already sees all earlier moves regardless of which label is attached to it. This is exactly what makes coalitional forceability collapse to ordinary two-role forceability of $\mathsf{Co}_\pi^\wedge(\Phi)$ (Theorem 13.5): under broadcast visibility, a coalition is nothing more than a relabeled single role. The same relabeling stops being information-preserving once slash-sets can name specific roles, which is where we conjecture — but do not prove — that genuine $K \ge 3$ coalitional content appears (Section 13.3).

Two instances will be used throughout. For the 2-coalition case, let $\pi : \{0, \ldots, K-1\} \twoheadrightarrow \{0, 1\}$. Then

$$\mathsf{Co}_\pi^\wedge(\Phi) := \mathsf{Co}_\pi^\wedge(\Phi), \qquad \mathsf{Co}_\pi^\vee(\Phi) := \mathsf{Co}_\pi^\vee(\Phi).$$

Implication projection. Once a K-role game has been formed into two coalitions via a map $\pi : \{0, \ldots, K-1\} \twoheadrightarrow \{0, 1\}$ (using, e.g., $\mathsf{Co}_\pi^\wedge$ or $\mathsf{Co}_\pi^\vee$), the result is a 2-role game. On such 2-role games we can read a conditional relationship between the two coalition objectives.

Define the *implication projection*

$$\underset{0\to 1}{\to} : \text{2-FOG} \to \text{FOL}$$

by structural recursion:

$$\begin{aligned}
(\langle \alpha, \beta \rangle) \restriction_{0\to 1}^{\exists-\forall} &:= \alpha \to \beta, \\
(P_0 x.\, \Phi) \restriction_{0\to 1}^{\exists-\forall} &:= \exists x.\, (\Phi) \restriction_{0\to 1}^{\exists-\forall}, \\
(P_1 x.\, \Phi) \restriction_{0\to 1}^{\exists-\forall} &:= \forall x.\, (\Phi) \restriction_{0\to 1}^{\exists-\forall}\ .
\end{aligned}$$

Thus coalition 0's moves are treated as existential choices and coalition 1's moves as universal challenges; at the terminal we test exactly "if 0's bundled objective holds, then so does 1's."

Combining coalition forming and implication projection, for any K-role game Φ and any surjection $\pi : \{0, \ldots, K-1\} \twoheadrightarrow \{0, 1\}$ we obtain an ordinary first-order formula

$$\big(\mathsf{Co}_\pi^\wedge \Phi\big) \restriction_{0\to 1}^{\exists-\forall} \qquad \text{or} \qquad \big(\mathsf{Co}_\pi^\vee \Phi\big) \restriction_{0\to 1}^{\exists-\forall},$$

to be read as:

> under the interaction discipline of Φ, whenever coalition 0 achieves its (conjunctive / disjunctive) bundled objective, coalition 1 also succeeds.

This lets us reason about the original multi-party protocol in plain FOL: first coalesce roles according to π, then project the resulting 2-role game to the implication shape.

This is the 2-role analogue of the adversarial projection: we follow the same $\exists/\forall$ discipline, but test an implication at the end instead of a single coalition's goal.

Coalition-mediated multiplication (2-FOL). When $\Phi^{(1)}$ and $\Phi^{(2)}$ are *2-FOL* formulas, we can compose them without leaving 2-FOL by:

$$\Phi^{(1)} \circledast \Phi^{(2)} \;:=\; \mathsf{Co}_{\pi_\times}^\wedge\big(\, \Phi^{(1)} \otimes \Phi^{(2)} \,\big),$$

where

$$\pi_\times^{-1}(0) = \{0^{(1)}, 1^{(2)}\}, \quad \pi_\times^{-1}(1) = \{1^{(1)}, 0^{(2)}\}.$$

We first form the 4-role product, then collapse the 4 roles to 2 physical players via the chosen coalition. Different coalition maps give different binary operations; we keep them *external* so the core grammar stays minimal.

Remark. All five operators above preserve the game-theoretic reading but they do *not* all preserve the arity K, so forcing them into the core grammar would complicate typing and substitution. Keeping them as derived/meta operators lets us talk about permutation, complement, tensor, and coalition-building uniformly.

Diagonal multiplication (pointwise). Take the *diagonal* coalition

$$\pi_\parallel^{-1}(A) = \{P_0^{(1)}, P_0^{(2)}\}, \qquad \pi_\parallel^{-1}(B) = \{P_1^{(1)}, P_1^{(2)}\}.$$

Then

$$\mathsf{Co}_{\pi_\parallel}^\wedge(\Phi^{(1)} \otimes \Phi^{(2)}) \quad \text{has terminal} \quad \langle \varphi_0 \wedge \psi_0,\ \varphi_1 \wedge \psi_1 \rangle,$$

i.e. we just conjoin the "same" role in the two games. When the two arenas are literally the same, this recovers pointwise conjunction.

Cross-diagonal multiplication. Take instead the *cross* coalition

$$\pi_\times^{-1}(A) = \{P_0^{(1)}, P_1^{(2)}\}, \qquad \pi_\times^{-1}(B) = \{P_1^{(1)}, P_0^{(2)}\}.$$

Define

$$\Phi^{(1)} \circledast \Phi^{(2)} \;:=\; \mathsf{Co}^{\wedge}_{\pi_\times}(\Phi^{(1)} \otimes \Phi^{(2)}).$$

Unfolding the terminal we get

$$\Phi^{(1)} \circledast \Phi^{(2)} \quad \text{has terminal} \quad \langle \varphi_0 \wedge \psi_1,\ \varphi_1 \wedge \psi_0 \rangle,$$

and the binders are relabeled according to $\pi_\times$, so the result is again a 2-FOL formula.

Rotation compatibility. The cross coalition is stable under swapping the two 2-FOL components, hence

$$\Phi \circledast \Psi \;\equiv_\tau\; \Psi \circledast \Phi, \qquad \tau(\Phi \circledast \Psi) \;=\; \tau(\Phi) \circledast \tau(\Psi)\,.$$

Example 6.12 (Battleship)**.** Battleship is naturally the cross-diagonal product of two independent 2-player games:

$$\text{Alice attacks Bob} \quad \text{and} \quad \text{Bob attacks Alice}.$$

Physical Alice must win in her *attacker* role in the first game and in her *defender* role in the second. This is exactly[1] the $\pi_\times$ aggregation:

$$\text{Battleship} \;=\; (\text{Alice} \to \text{Bob}) \;\circledast\; (\text{Bob} \to \text{Alice}).$$

Proposition 6.13 (Cross-diagonal forceability factors, for separable objectives)**.** *Let $\Phi^{(1)}, \Phi^{(2)}$ be 2-FOL formulas over disjoint arenas. Then*

$$F_A(\Phi^{(1)} \circledast \Phi^{(2)}, \mathcal{M}) \iff F_0(\Phi^{(1)}, \mathcal{M}) \wedge F_1(\Phi^{(2)}, \mathcal{M}),$$

and symmetrically for F_B with the roles swapped.

Proof sketch. Role $A = \{0^{(1)}, 1^{(2)}\}$ has binders drawn from both components, and $\otimes$'s independence constraints mark every $0^{(1)}$-binder (in the $\Phi^{(1)}$-arena) independent of every $\Phi^{(2)}$-variable, and every $1^{(2)}$-binder independent of every $\Phi^{(1)}$-variable. So A's own binders are mutually restricted — a $\setminus V$-crossing *within* the coalition, in the sense of the remark following Theorem 13.5 below. But by disjointness of arenas, A's terminal component $\varphi_0 \wedge \psi_1$ splits into a part mentioning only $\Phi^{(1)}$-variables and a part mentioning only $\Phi^{(2)}$-variables, so neither half of A's objective could have benefited from seeing across the restriction even if it were allowed to. A merged strategy for A is therefore exactly the unconstrained pair of an optimal $\Phi^{(1)}$-role-0 strategy and an optimal $\Phi^{(2)}$-role-1 strategy, chosen independently, and A wins iff both halves do. □

Remark 6.14 (A syntactic crossing is not, by itself, a loss). Theorem 6.13 shows a coalition can have a $\setminus V$-crossing among its own members and still merge losslessly, as long as the members' objectives do not *couple* across the restriction: here φ_0 and ψ_1 each depend on only one side, so withholding cross-visibility costs nothing. This sharpens (and corrects an overstatement in an earlier draft of) the discussion in Section 13.3: the syntactic pattern "a binder of $i \in S$ excludes some $j \in S$" is necessary-looking but not sufficient for genuine loss; what should matter is whether the excluded information would actually have helped satisfy the coalition's (possibly cross-component) joint objective. We do not have a general sufficient condition for when it does.

[1]The real game does not require consistency between the two Alice roles or the two Bob roles, so it is a 4-player game, where only the final winning condition (who wins first) connects the two groups of roles. The real game would also require more complex hide/reveal scheme to force the defending player to reveal the correct board.

6.2 Splicing (Overlap)

We add a combinator that glues a suffix of Φ to a prefix of Ψ by identifying a boundary of moves and their visibility.

Data. Let $G_\Phi = (V_\Phi, E_\Phi)$ and $G_\Psi = (V_\Psi, E_\Psi)$ be the visibility DAGs (nodes labeled by $(\text{name}, \text{player}, \text{dom})$). A *splice instance* is a finite labeled DAG $I = (V_I, E_I)$ with injective label-preserving homomorphisms

$$\iota_L : I \hookrightarrow G_\Phi, \qquad \iota_R : I \hookrightarrow G_\Psi.$$

Side conditions.

(S1) **Type/role agreement:** for all $v \in V_I$, $\text{dom}_\Phi(\iota_L v) = \text{dom}_\Psi(\iota_R v)$ and $\text{player}_\Phi(\iota_L v) = \text{player}_\Psi(\iota_R v)$.

(S2) **Boundary shape:**

- (*Suffix in* Φ) no edge enters $\iota_L(V_I)$ from $V_\Phi \setminus \iota_L(V_I)$.
- (*Prefix in* Ψ) no edge leaves $\iota_R(V_I)$ to $V_\Psi \setminus \iota_R(V_I)$.

(S3) **Acyclic gluing:** identifying $\iota_L(v) \equiv \iota_R(v)$ for all $v \in V_I$ and taking edges $E_\Phi \cup E_\Psi$ yields an acyclic transitive closure.

Definition 6.15 (Splice)**.** Given Φ, Ψ and (I, ι_L, ι_R) satisfying (S1–S3), define

$$\Phi \triangle_{(\iota_L, \iota_R)} \Psi$$

to be the game with visibility DAG $G \coloneqq G_\Phi \cup_{\iota_L \equiv \iota_R} G_\Psi$ (disjoint union modulo identification of the images of I), labels inherited from Φ and Ψ. Its prenex script is any topological linearization of G that preserves the internal orders of Φ and Ψ and places $V_\Phi \setminus \iota_L(V_I)$ before $V_\Psi \setminus \iota_R(V_I)$. Two such scripts are equivalent modulo player permutation, α-renaming, and DAG isomorphism.

Concrete prenex construction. (Useful for inductions.) α-rename so that the images of I have identical names and other names are disjoint; list the binders of $\Phi \setminus \iota_L(V_I)$, then the identified boundary, then $\Psi \setminus \iota_R(V_I)$; visibility edges are the union after identification. (S2) prevents backward edges; (S3) ensures acyclicity.

Special cases.

- **Tensor** ($\otimes$): $I = \emptyset$ (disjoint union; no cross-visibility).
- **Clocked sequencing** ($;$): $I = \{t\}$ with t a fresh unit-typed choke point appended to Φ and required in the initial keys of Ψ.
- **Leaky/broadcast sequencing** ($\curlyvee$): I is the maximal frontier at the cut (all nodes of Φ seen by any initial node of Ψ).

Algebraic laws (up to the section's equivalence). Let ρ be any player permutation or α-renaming.

(Naturality) $\rho(\Phi)\triangle_{(\rho\circ\iota_L, \rho\circ\iota_R)}\rho(\Psi) \;\equiv\; \rho\big(\Phi\triangle_{(\iota_L,\iota_R)}\Psi\big)$.

(Unit) $\Phi\triangle_\emptyset \mathbf{1} \;\equiv\; \Phi \;\equiv\; \mathbf{1}\triangle_\emptyset\Phi$.

(Associativity) $(\Phi\triangle_{I_{12}}\Psi)\triangle_{I_{23}}\Xi \;\equiv\; \Phi\triangle_{I_{13}}(\Psi\triangle_{I_{23}}\Xi)$ whenever the boundary embeddings agree on overlaps.

Space of binary operations. Every coalition map

$$\pi : \{P_0^{(1)}, P_1^{(1)}, P_0^{(2)}, P_1^{(2)}\} \to \{A, B\}$$

gives a different binary operator on 2-FOL by the schema

$$(\Phi^{(1)}, \Phi^{(2)}) \;\mapsto\; R_\pi^\wedge(\Phi^{(1)} \otimes \Phi^{(2)}).$$

Diagonal and cross-diagonal are the two most symmetric cases; others mix the roles in less structured ways.

6.3 Player-Directed Choice

In a game-theoretic setting, branching must be attributed to a specific agent. Rather than introducing a primitive branching node, we model *player-directed choice* as an ordinary move in the prefix that selects between two linearized subgames. This keeps K-FOG in a prenex, sequence-of-moves form (in the spirit of QBF and related game logics), while still validating the usual strategic intuition of extensive-form branching.

Definition. Let Φ and Ψ be games over the same set of roles $\{0, \dots, K-1\}$, with disjoint sets of bound variables and terminals

$$\Phi \ : \ \langle \varphi_0, \dots, \varphi_{K-1} \rangle \qquad \Psi \ : \ \langle \psi_0, \dots, \psi_{K-1} \rangle.$$

Fix a role i that chooses between them. Introduce a fresh boolean variable c (not occurring in Φ or Ψ), and choose any well-typed prefix shuffle σ that interleaves the binders of Φ and Ψ while preserving the internal order and visibility constraints of each branch. We define the *player-directed choice*

$$\Phi \oplus_i \Psi \ := \ P_i c.\, (\Phi \, \sigma \, \Psi)$$

with terminal

$$\langle \chi_0, \dots, \chi_{K-1} \rangle \qquad \text{where} \qquad \chi_k \ := \ (c = 0 \wedge \varphi_k) \ \vee \ (c = 1 \wedge \psi_k).$$

Here $P_i c$ is an unslashed binder: the choice c is visible to all subsequent moves. The disjointness assumption ensures that the two linearized branches do not accidentally interfere via name clashes.

Intuition: predicated execution. Syntactically, K-FOG offers only a single linear history. The operator $\oplus_i$ encodes branching by *predicated execution* on that history:

- The full sequences of moves of both Φ and Ψ appear in the prefix and are legally played.
- The decision variable c *masks* the terminal: for a fixed profile, exactly one disjunct in χ_k is live. If $c = 0$, the terminal simplifies to φ_k; if $c = 1$, to ψ_k.

Consequently, all moves in the unchosen branch become *outcome-insensitive*: they may still be observed and responded to, but no player's payoff can depend on them once c is fixed. In game-theoretic jargon, they are at most *cheap talk*: strategically inert messages that pass through the visibility DAG without changing utilities. This linearization preserves the static information structure while inducing the same strategic payoffs as a branching game tree where player i selects between Φ and Ψ.

Optimization: leaky vs independent choice. The definition allows any admissible shuffle σ; different choices of σ correspond to different visibility of irrelevant history and can significantly affect solver complexity.

- **Leaky choice.** Taking σ to be simple sequencing (e.g. $\Phi \,|\rangle\, \Psi$) yields a *leaky* encoding: moves in the second branch can observe the entire history of the first. Strategically this is harmless—the extra information concerns outcome-insensitive cheap talk—but a generic extensive-form solver must represent strategies for the second branch as functions of this enlarged history, inflating the search space.
- **Independent choice.** Alternatively, we can use a tensor-style shuffle (e.g. $\Phi \otimes \Psi$) that *hides* the internal variables of Φ from Ψ and vice versa, while keeping c globally visible. We call this the *independent choice* encoding. It enforces syntactic independence between the branches: no move in Φ can be conditioned on internal moves of Ψ and conversely.

For verification work we adopt the independent encoding as canonical. It yields a game that is strategically equivalent to the leaky version, but admits smaller and cleaner information sets: strategy functions need only range over histories that are relevant to the chosen branch. In practice, this recovers the tight strategy spaces one would obtain from a structural branching node.

Algebraic properties. The operator $\oplus_i$ interacts smoothly with the structural morphisms of the calculus.

- **Rotation compatibility.** Role rotation τ acts only on roles, not on variables. Thus, for any rotation τ we have a strategic equivalence

$$\tau(\Phi \oplus_i \Psi) \;\simeq\; \tau(\Phi) \oplus_{\tau(i)} \tau(\Psi),$$

  where the decision variable c and its visibility are preserved on both sides.
- **Distributivity over sequencing.** Let Ξ be a continuation game whose moves occur strictly after those of Φ and Ψ, and assume that c is visible to all roles that act in Ξ. Then

$$(\Phi \oplus_i \Psi) \,|\rangle\, \Xi \;\simeq\; (\Phi \,|\rangle\, \Xi) \oplus_i (\Psi \,|\rangle\, \Xi)$$

  in the sense of *strategic isomorphism*: there is a bijection between strategy profiles that preserves all players' utilities. On the left, a single strategy for Ξ may depend on c and hence implement branch-specific behavior. On the right, we have a pair $(\sigma_{\Xi_L}, \sigma_{\Xi_R})$ of strategies, one for each copy of Ξ. The correspondence is given by

$$\sigma_\Xi \;\mapsto\; (\sigma_\Xi \restriction_{c=0},\, \sigma_\Xi \restriction_{c=1}) \qquad \text{and} \qquad (\sigma_{\Xi_L}, \sigma_{\Xi_R}) \;\mapsto\; \big(c = 0 \Rightarrow \sigma_{\Xi_L} \;\mid\; c = 1 \Rightarrow \sigma_{\Xi_R}\big).$$

  This justifies factoring out common suffixes as a compression technique: the left-hand encoding shares a single copy of Ξ without changing the strategic content of the game.

Remark: hidden choice. A natural variation is *hidden choice*, written $P_i c/J$, where J is a set of roles from whom the choice is syntactically hidden. For roles in J to remain genuinely unaware of c, the two branches must be *protocol-compatible*: along any history visible to these roles, the subsequent binders, sorts, and observable actions of Φ and Ψ must be indistinguishable (typically, structurally identical up to renaming). Otherwise, the value of c can be inferred from the visible shape of the remaining game. Hidden choice matters when analyzing games with asymmetric information and signaling.

7 Extending the Core

The core calculus assumes broadcast visibility: every move sees all past moves. We now extend this by allowing binders to specify *information constraints*, in the style of IF logic.

7.1 Property-Annotated Binders: ⟨private, public⟩

We extend the core binder so that each move simultaneously fixes a *private value* and publishes a (possibly empty) set of *public properties* about that value. Intuitively, a standard "commit" is the special case where no informative property is published at bind time.

Syntax. A binder occurrence is written

$$P_i\, x.\, [\Pi] \;\; \Phi$$

where x is the private value chosen by role i, and Π is a finite set of first-order formulas (the *public certificate*) whose free variables are drawn from the variables visible at this point *together with* x:

$$\Pi \subseteq \varphi\big(\mathrm{Vars}_{\mathrm{vis}}(x) \cup \{x\}\big).$$

We write $\top$ for the trivially true formula. The special case $[\Pi] = [\{\top\}]$ corresponds to a *pure commitment* of x (type-theory readers may think of "unit"; here we use logical $\top$).

Public theory and well-formedness. Let the underlying visibility order be the usual prefix order on binders. The *public theory* available just before a binder u is

$$\Sigma_{\mathrm{pub}}(u) \;:=\; \bigcup_{b \prec u} \Pi_b.$$

A binder is *well-formed* if every $\varphi \in \Pi$ mentions only variables in $\mathrm{Vars}_{\mathrm{vis}}(x) \cup \{x\}$ (no forward references). By construction, Σ_{pub} is monotone in the order: if $v \preceq u$ then $\Sigma_{\mathrm{pub}}(v) \subseteq \Sigma_{\mathrm{pub}}(u)$.

Identifiability and value-visibility. A predecessor value x_a is *identifiable* at binder u if there exists a term t over the variables u may name such that

$$\mathsf{Id}_u(x_a) \;\coloneqq\; \exists t \text{ over } \mathrm{Vars}_{\mathrm{vis}}(u) \text{ with } \Sigma_{\mathrm{pub}}(u) \;\vdash\; x_a = t.$$

We define the (direct) *value-visibility* edge by

$$(a \to u) \in \mathrm{vis} \quad\Longleftrightarrow\quad \mathsf{Id}_u(x_a).$$

Thus a node may depend *causally* on the fact that a chose (it sees a in the order) without yet depending on the *value*; the latter becomes legal precisely when x_a is identifiable from the accumulated public theory.

Consequences.

- **Monotonicity.** If $v \preceq u$ and $\mathsf{Id}_v(x_a)$ then $\mathsf{Id}_u(x_a)$. Hence for each a, the knowing set $K(x_a) = \{u \mid \mathsf{Id}_u(x_a)\}$ is an up-set in the prefix DAG.
- **Commit and reveal (as strengthening).** A "reveal of x_a at r" is simply that Π_r makes $\mathsf{Id}_r(x_a)$ true, e.g. by asserting $x_a = y_r$ for r's public name y_r or any other formula that functionally pins x_a.
- **Staged disclosure.** If Π contains only local facts, identifiability can start at multiple incomparable *minimal* points $r_1, \ldots, r_m$; then $K(x_a) = \bigcup_j \uparrow r_j$. This captures branch-specific early openings followed by wider disclosure later.

Conservative extension. If every binder uses $[\{\top\}]$, no x becomes identifiable before an explicit equality appears; value-visibility collapses to the base setting (no additional edges beyond those introduced by later equalities). Conversely, any pattern of edges that is a finite union of principal up-sets per source can be realized by placing equalities in the corresponding minimal binders.

Examples.

- *Pure commit.* $P_i\, x.[\top]$ publishes nothing informative: $\mathsf{Id}_u(x)$ fails for all u until a later binder asserts a property that pins x (e.g. $x = y$ or a uniqueness constraint).
- *Reveal by equality.* Later binder r introduces y and publishes $[\{x = y\}]$. Then $\mathsf{Id}_u(x)$ holds for any u that can name y.
- *Reveal by uniqueness.* If r publishes $[\{\exists! z.\, \psi(z, \bar{v}),\ \psi(x, \bar{v})\}]$, then $\Sigma_{\mathrm{pub}}(r) \vdash x = \iota z.\psi(z, \bar{v})$, so $\mathsf{Id}_r(x)$.

Remark (terminology). We write $\top$ for the trivially true property; in type-theoretic terms one may think of the *unit* type. The intended reading here is logical: a pure commitment is $\langle x, \top \rangle$.

Relation to $\backslash V$-independence. Id_u tracks what is *derivable* from accumulated public commitments, which is a different, and in one respect stronger, notion than the $\backslash V$-binder's syntactic restriction on a strategy function's argument list (Section 5): a value can be $\backslash V$-excluded from a binder's arguments while still being entailed by Σ_{pub} at that point (if some other publicly derivable fact pins it down), and conversely a value can be nameable (not $\backslash V$-excluded) while not yet identifiable, if nothing published so far determines it. The two mechanisms are not substitutes for one another, and we have not compared their forceability theories; in particular, whether a version of the Section 13.3 separation conjecture stated in terms of non-identifiability rather than $\backslash V$-exclusion behaves differently is open.

8 Strategies as Relations; Profiles and Composition as Joins

This section recasts (deterministic, pure) strategies for a K-FOG prefix as *functional relations* over the variables they are allowed to see. The payoff is conceptual clarity and a compact algebra of composition: profiles are natural joins; hiding is projection; α-renaming and player permutation

are schema renamings; clocks and broadcasts are key-shaping patterns; IF-style independence is a key-exclusion constraint; and several solution concepts become anti-join/division queries. We keep the mathematics lightweight but precise enough that a fully formal account is routine.

We treat each binder as a *typed column*, the visibility DAG as the *set of key attributes* allowed to influence that column, a pure strategy as a *deterministic fill-rule* (a function from key to value), and a strategy profile as the *natural join* of those per-binder tables. In short: *shape first, behavior second.* The game specification fixes the shape (columns, types, who can depend on whom); strategies populate it.

Why is this worth doing? Two reasons. First, it gives an intuitive picture that matches how we already think about the prenex script: we fill the row from left to right, and each move can only look at certain earlier columns. Second, it buys an algebra of composition: sequencing is a join on an interface (optionally followed by projection), clocks and broadcasts are shorthand for key-shaping, independence is a key-exclusion constraint, and several semantic queries (feasibility, forceability, "no profitable deviation" in the $\{0,1\}$ case) become small relational patterns.

What this *is* and *is not*. It *is* a faithful, lightweight reading of our deterministic, prenex fragment that can be made fully formal with routine RA machinery. It is *not* a new semantics for arbitrary utilities, stochastic choices, or concurrent branching—those require extra layers (aggregation, probability, or event-structure concurrency) that we intentionally avoid here.

How to read the section. If you only need the dictionary, use Figure 1 as a reference. If you want to see why the dictionary is sound, skim the short lemmas that follow: they show that (i) joining total, functional strategy tables yields a single completed play, and (ii) the composition and independence patterns behave as advertised.

Notation. We write D_x for the domain of variable x, $X \to Y$ for a (relational) functional dependency, and use standard RA operators: natural join ($\bowtie$), projection (π), selection (σ), renaming (ρ), and set difference ($\setminus$). Schemas are sets of attributes named by variables. Fix a prenex prefix with binders $b_1, \dots, b_n$ producing variables $x_1, \dots, x_n$ and a visibility DAG $E \subseteq \{(j{\to}k) \mid j < k\}$.

Definition 8.1 (Keys, local strategy tables)**.** For each position k let the *key* (visible history) be

$$K_k \;\coloneqq\; \{\, x_j \mid (j{\to}k) \in E \,\} \subseteq \{x_1, \dots, x_{k-1}\}.$$

A *local strategy table* for x_k is a relation

$$S_k \;\subseteq\; \prod_{x \in K_k} D_x \;\times\; D_{x_k} \qquad \text{with} \qquad K_k \to x_k.$$

Intuitively, S_k is the graph of a total function from visible histories to the move for x_k (totality is stated below).

Definition 8.2 (Legal histories and totality)**.** Let $H_1 \coloneqq \{\bullet\}$ and, inductively for $k > 1$,

$$H_k \;\coloneqq\; \pi_{K_k}\Big(\; \bowtie_{j<k} S_j \;\Big).$$

We say S_k is *total on legal histories* if for every $h \in H_k$ there exists a unique $v \in D_{x_k}$ with $(h, v) \in S_k$.

Definition 8.3 (Strategy profile as join)**.** Given local tables $S_1, \dots, S_n$ (one per binder), the *profile* is the natural join

$$S \;\coloneqq \bowtie_{k=1}^{n} S_k \;\subseteq\; \prod_{k=1}^{n} D_{x_k}.$$

Lemma 8.4 (Determinacy of play)**.** *If each S_k satisfies $K_k \to x_k$ and is total on its legal histories, then S is a singleton relation over $\{x_1, \dots, x_n\}$.*

Proof sketch. Topologically sort E; functionality and totality determine a unique value for x_k once the predecessors are fixed. □

Equivalence and normalization. Equivalence up to (i) player permutation, (ii) α-renaming, and (iii) visibility-DAG isomorphism coincides with schema isomorphism for the family $\{S_k\}$ (rename attributes; re-order does not matter). Thus, adopting the relational representation turns "equality up to those symmetries" into literal equality up to renaming.

8.1 Composition as Join + Projection

Let Φ end with variables $X^\Phi = \{x_1, \dots, x_m\}$ and Ψ begin with $Y^\Psi = \{y_1, \dots\}$. To *sequentially compose* $\Phi;\Psi$ choose an *interface* $I \subseteq X^\Phi$ and rename attributes in the first wave of Ψ so that every initial key K_{y_ℓ} is a subset of I and the domains match. Then set

$$S^{\Phi;\Psi} \;\coloneqq\; \pi_{\text{externals}}\big(S^\Phi \bowtie_I S^\Psi\big),$$

where S^Φ and S^Ψ are the joins of their local tables, and externals projects away hidden interface attributes if desired.

Two useful patterns are immediate:

- **Clock (unit choke-point).** Introduce a fresh $t : \mathbf{1}$ as the last variable of Φ and include t in every initial key of Ψ. Joining on t enforces sequencing without adding information.
- **Broadcast.** Include *all* of X^Φ in the initial keys of Ψ; the first wave of Ψ can see all prior moves. This is just a wide natural join.

Proposition 8.5 (Soundness of sequential composition)**.** *With domain-consistent interfaces, $S^{\Phi;\Psi}$ coincides with the play-by-play semantics of first resolving Φ then Ψ under the visibility wiring.* □

8.2 Independence as Key Exclusion

IF-style annotations are read as *schema constraints*.

Definition 8.6 (Slash/visibility constraints)**.** Variable-level exclusion $P_i x_k \setminus V$ is primitive (Section 5); role-level exclusion $P_i x_k / J$ is the derived notation defined there. For either, enforce:

$$\forall x_j \in V \;\; (x_j \notin K_k) \qquad \text{or} \qquad \forall j \in J \;\; (\mathrm{pl}(x_j) \in J \Rightarrow x_j \notin K_k).$$

Totality remains relative to $H_k = \pi_{K_k}(\bowtie_{j<k} S_j)$, i.e. to the *reduced* keys.

Remark 8.7 (No-signalling-by-key). If a variable is slashed from x_k, it does not appear in K_k; hence x_k cannot (directly) depend on it. Allowing a slashed variable into K_k admits signalling; forbidding it prevents such channels. □

8.3 Objectives and RA Queries

Let each player's objective be a predicate $W_i \subseteq \prod_{k=1}^n D_{x_k}$ (evaluate the terminal tuple there). Given a profile S, player i succeeds iff $\sigma_{W_i}(S) = S$. Beyond single-profile checks, several semantic questions over *sets of* strategies are naturally relational (assume finite domains for enumeration).

Feasibility. A game is feasible if there exists a family $\{S_k\}$ with each S_k total on H_k. Algorithmically: compute H_k bottom-up and check emptiness of the anti-join $H_k \setminus \pi_{K_k}(S_k)$ for all k.

Forceability (worst-case guarantee). Let $\mathcal{P}_i$ be the relation enumerating admissible tables for player i (correct schemas; total on legal histories) and $\mathcal{P}_{-i}$ the product for the others. The set of forcing strategies for i is

$$\mathcal{F}_i \;=\; \mathcal{P}_i \;\setminus\; \pi_{\mathrm{id}(S_i)}\big(\sigma_{\neg W_i}(S_i \bowtie \mathcal{P}_{-i})\big),$$

i.e. those S_i for which there is no opponent completion producing a losing play. This is an anti-join/selection pattern.

Pure Nash in the $\{0,1\}$ case. A profile S (a singleton play under Theorem 8.4) is Nash iff for each i there is no alternative S_i' such that $\sigma_{W_i}(S_i' \ltimes S_{-i})$ is nonempty while $\sigma_{W_i}(S_i \ltimes S_{-i})$ is empty. Again, a self-join plus anti-join.

8.4 Small Running Example

Two binders: x (player A) then y (player B) with $K_y = \{x\}$. Let $D_x = D_y = \{0,1\}$. Tables:

$$S_x(x) \text{ with } \emptyset \to x, \qquad S_y(x,y) \text{ with } x \to y.$$

Profile $S = S_x \ltimes S_y \subseteq \{(x,y)\}$. If $W_B = \{(x,y) \mid x = y\}$, then B's success on the profile is the check $\sigma_{W_B}(S) = S$. Sequentially compose with a Ψ whose first move z sees y: set $K_z = \{y\}$ and compute $S^{\Phi;\Psi} = \pi_{x,y,z}(S \ltimes S_z)$; hiding y is $\pi_{x,z}$.

Scope and complexity (brief). When the visibility hypergraph is acyclic, worst-case optimal joins (e.g. Yannakakis-style) evaluate profiles and composition in time linear in the input plus output size. For cyclic schemas, joins can blow up in the worst case; feasibility/forceability remain *queries* but inherit that cost. These are standard RA complexity boundaries; nothing exotic is required here.

Summary. Treating local strategies as functional relations keyed by the visibility DAG makes profiles a natural join, hiding a projection, wiring a renaming, sequencing a join on an interface, clocks/broadcasts key-shaping, and IF-style independence a key-exclusion constraint. This alignment is faithful at the granularity of our deterministic, prenex fragment, and is fully formalizable with routine RA machinery.

Generalising to mixed strategies. All of the relational scaffolding survives if we replace per-binder functions by *behavioural kernels*. Assuming perfect recall (each binder's key contains everything the player could have observed so far), a mixed strategy for the binder producing x_k is a conditional distribution $p_k(x_k \mid K_k)$ with $\sum_{v \in D_{x_k}} p_k(v \mid h) = 1$ for every *reachable* key $h \in H_k$. The profile then induces a distribution over complete plays that factorises along the visibility DAG:

$$\Pr(x_1, \ldots, x_n) \;=\; \prod_{k=1}^{n} p_k\big(x_k \,\big|\, x_{K_k}\big).$$

In database terms, keep the same schema and keys; attach a weight column to each local table, interpret *join* as weight multiplication, *selection* as zeroing weights, and *projection*/hiding as marginalization (summing weights). Independence, clocks, broadcasts, and sequential composition remain purely structural—they only shape keys—so they carry over unchanged.

Solution concepts with randomness. Objectives lift from Boolean predicates on rows to expected values under Pr. Pure-strategy checks like "no $0 \to 1$ deviation" become expected payoff inequalities: for player i, a mixed profile is Nash iff for every (unilateral) behavioral deviation $\hat{p}_i$, $\mathbb{E}_{p_i,p_{-i}}[u_i] \geq \mathbb{E}_{\hat{p}_i,p_{-i}}[u_i]$. Relationally, this is the same self-join over opponents' kernels, but the decision is an aggregate comparison rather than an emptiness test. If shared private randomness is needed across a player's binders, introduce a hidden seed variable sourced once and included in that player's later keys (the same key-shaping trick as the clock/broadcast). Normalization constraints $\sum_v p_k(v \mid h) = 1$ are required only on *legal* histories h, mirroring totality in the deterministic case.

9 Solution Concepts Beyond Forceability

The adversarial forceability predicate $F_i(\Phi, \mathcal{M}, s)$ from Section 2 is intentionally strong: it asks whether player i can guarantee her objective against *all* behaviors of the other players, including behaviors that do not optimize their own objectives. This is useful as a safety or worst-case notion, but it is important to note that $F_i(\Phi, \mathcal{M}, s)$ is a *game-level* statement: it shows that a

decoupled implementation for i *exists*, but it does not, by itself, select a concrete joint strategy profile.

Moreover, such a decoupled implementation for i need not be admissible as a component of the particular joint implementation we care about (its local spec may be decoupled but too weak for the overall system we want to assemble). For example, player i may force her objective by always choosing a default value that blocks all harmful continuations. This is a valid forcing strategy in the adversarial reading, but in a cooperative profile it prevents the other players from reaching $(1, \dots, 1)$, so it is not a good component of the joint implementation.

In this section we collect lighter—but still semantic—solution concepts. Some of them (feasibility, coalition guarantees) are also game-level and talk about what the specification *admits*; others (rationality, stability) are *profile-level* and talk about which joint choices we are willing to accept. We organize them in three layers:

1. a *feasibility layer* that asks what collections of objectives are simultaneously realizable by some profile;
2. a *rationality layer* that filters *profiles* by stability or minimal-safety requirements;
3. a *guarantee layer* that generalizes individual forceability to coalitions and explains why we mostly keep forceability at the game level.

All notions are defined for a fixed closed formula Φ, structure $\mathcal{M}$, and (implicit) initial assignment $s = \emptyset$; adding free variables back is routine.

9.1 Feasibility Layer

The feasibility layer ignores adversarial behavior and asks only what the protocol *permits*.

Definition 9.1 (Joint realizability)**.** A formula Φ is *jointly realizable* in $\mathcal{M}$ if

$$\exists \sigma \,.\, [\![\Phi]\!]_{\mathcal{M},\sigma} = (1, \dots, 1).$$

That is, there exists a single strategy profile under which every player's objective holds.

Joint realizability can be too strong when the terminal is conflicting. We therefore also consider support for arbitrary winning sets.

Definition 9.2 (Support for a player set)**.** For $W \subseteq \{0, \dots, K-1\}$ we say that Φ *supports* W in $\mathcal{M}$ if

$$\exists \sigma \,.\, \forall i \in W.\, \big([\![\Phi]\!]_{\mathcal{M},\sigma}\big)_i = 1.$$

When $W = \{i\}$ is a singleton, we simply say that Φ *supports* i.

Thus we decouple the single-player query ("is there some run where i succeeds?") from the joint query ("is there some run where all of W succeed?"). Note that

$$\text{joint realizability} \iff \text{support for } W = \{0, \dots, K-1\}.$$

9.2 Rationality Layer

We now filter profiles not only by what they achieve, but also by whether they are stable under unilateral deviations. We work with the same pure strategies as in Section 2.

Definition 9.3 (Unilateral deviation)**.** Let σ be a strategy profile and i a player. For any alternative strategy σ_i' of player i, write

$$\sigma[i \mapsto \sigma_i'] := (\sigma_0, \dots, \sigma_{i-1}, \sigma_i', \sigma_{i+1}, \dots, \sigma_{K-1})$$

for the profile obtained by replacing only i's component.

Definition 9.4 (Pure Nash outcome for Φ)**.** A strategy profile σ is a *pure Nash equilibrium* for Φ in $\mathcal{M}$ if for every player i and every alternative strategy σ_i' we have

$$\big([\![\Phi]\!]_{\mathcal{M},\sigma}\big)_i \;\geq\; \big([\![\Phi]\!]_{\mathcal{M},\sigma[i\mapsto\sigma_i']}\big)_i.$$

Since outcomes are Boolean, the only strictly improving deviation is a change from 0 to 1. Thus the usual "no profitable deviation" clause specializes here to "no player can unilaterally turn a 0 into a 1."

Forceability from Definition 2.3 can now be reused as a *baseline* rather than as the final notion.

Definition 9.5 (Individually rational profile)**.** Let $F_i(\Phi,\mathcal{M})$ be the individual forceability predicates from Definition 2.3. A profile σ is *individually rational* if

$$\forall i.\ \big([\![\Phi]\!]_{\mathcal{M},\sigma}\big)_i \;\geq\; [\,F_i(\Phi,\mathcal{M})\,].$$

That is, no player obtains less than what she can guarantee in the adversarial reading.

Individually rational profiles can be intersected with Nash profiles.

Definition 9.6 (IR-NE profile)**.** A profile is an *IR-NE* for Φ in $\mathcal{M}$ if it is both a pure Nash equilibrium (Definition 9.4) and individually rational (Definition 9.5).

This is a filtering notion: an IR-NE need not exist even when pure Nash equilibria exist and all $F_i(\Phi,\mathcal{M})$ hold.

Admissible strategies. A further rationality filter is obtained by removing dominated strategies *before* looking for equilibria.

Definition 9.7 (Dominance and admissibility)**.** Fix Φ and $\mathcal{M}$. For player i, a strategy τ_i *dominates* a strategy σ_i if

$$\forall\sigma_{-i}.\ \big([\![\Phi]\!]_{\mathcal{M},(\tau_i,\sigma_{-i})}\big)_i \;\geq\; \big([\![\Phi]\!]_{\mathcal{M},(\sigma_i,\sigma_{-i})}\big)_i$$

and for some σ_{-i} the inequality is strict. A strategy is *admissible* if it is not dominated by any other strategy of the same player.

One may then define equilibria *in admissible strategies* by restricting Definition 9.4 to profiles all of whose components are admissible.

9.3 Guarantee Layer

The original F_i treats every other player as an adversary. In a multi-player setting it is natural to extend this to coalitions.

Definition 9.8 (Coalition forceability)**.** Let $C \subseteq \{0,\dots,K-1\}$ be a nonempty set of players. The coalition C can *force* their objectives in Φ over $\mathcal{M}$, written $F_C(\Phi,\mathcal{M})$, if

$$\exists(\sigma_i)_{i\in C}\ \forall(\sigma_j)_{j\notin C}\ \text{let}\ \sigma := ((\sigma_i)_{i\in C},(\sigma_j)_{j\notin C})\ \text{in}\ \bigwedge_{i\in C}([\![\Phi]\!]_{\mathcal{M},\sigma})_i = 1.$$

Thus the coalition commits to one joint strategy profile for its members that succeeds uniformly against any behavior of the outsiders.

This is again a game-level statement: it tells us that there *exists* a joint strategy for C that succeeds against all outsiders. One could try to assemble a *profile* in which every player uses a forcing strategy (an "independently forcing profile"), but for $K > 2$ and non-cooperative terminals such a profile typically does not exist. For this reason we mostly use forceability-style notions to describe the *space of possible implementations*, and reserve the profile-level notions from Section 9.2 and Section 9.7 for judging concrete joint choices.

9.4 Symmetry-Respecting Selections

Because K-FOG is S_K-equivariant (Theorem 3.2), we can speak of solution concepts that do not break player symmetry.

Definition 9.9 (Orbit-realizable outcome)**.** Let $v \in \{0,1\}^K$. We say that v is *orbit-realizable* for Φ in $\mathcal{M}$ if for every $\pi \in S_K$ there exists a strategy profile σ_π such that

$$[\![\Phi]\!]_{\mathcal{M},\sigma_\pi} = \pi \cdot v.$$

Orbit-realizability says that realizable outcomes come in S_K-orbits; it is a way to select outcomes compatible with the nominal nature of player indices.

Note that we do not require a single profile to realize the entire orbit. Different permutations may be realized by different profiles σ_π.

9.5 Cheap-Talk-Aware Variants

Recall from Section 3.3 that a bound variable is *outcome-insensitive* if changing its value never changes the outcome vector. This suggests quotienting strategies by their behavior on such variables.

Definition 9.10 (Cheap-talk equivalence)**.** Two strategy profiles σ and σ' are *cheap-talk equivalent* if, for every outcome-insensitive binder and for every history at which that binder is reached, the two profiles may choose different values, and for every other binder and every history they choose the same value.

Any of the above notions (realizability, NE, IR-NE) can be turned into a cheap-talk-aware variant by replacing "there exists a profile" with "there exists a cheap-talk equivalence class of profiles." This removes distinctions that the specification itself declared semantically irrelevant.

9.6 Solution Concepts as Lenses on Coupling

The definitions above asked "what does this specification admit?"; we now explain why the *same* vocabulary is also useful for "how tightly is this particular implementation tied together?"

Traditionally, solution concepts such as forceability, best responses, and Nash-style stability answer questions of the form: *"does this player (or this coalition) get what they want under this interaction?"* Our logic can already phrase such questions. In our setting, however, the same concepts also serve a second purpose: they expose *how tightly the described system is tied together at a chosen profile.*

- A forceability statement (game-level) says: "this role can achieve its component even if everyone else is adversarial." This witnesses *potential non-coupling*: the specification allows a decoupled implementation.
- A Nash-style statement (profile-level) says: "there exists at least one joint profile in which every role's current move is justified by the others' current moves." This witnesses *actual coupling at that profile*: the system currently leans on itself.
- Stronger notions (pervasive dependence, brittleness) make this manifest: not only does a mutually dependent point exist, but *any* local change will be detected in the outcome.

Thus the same object—a strategy profile over one linear interaction—can be read in two directions:

1. the *goal* reading: "can this player win at all?" (existence);
2. the *connection* reading: "given this profile, how much do these components lean on each other?" (actual dependence).

All of these are counterfactual in spirit: they talk about what *would* have happened had some other player chosen a different strategy; the logic need not execute such runs, it only needs to name them.

9.7 Coupling and Stability of Profiles

It is useful to view unilateral deviations as a rewrite relation on strategy profiles: $\sigma \to_i \sigma[i \mapsto \sigma_i']$. Local (Nash) stability then says a profile is a normal form for all such rewrites, while the dependence and brittleness notions measure how sensitive the outcome is to taking one rewrite step.

Fix a closed K-player specification Φ and a structure $\mathcal{M}$. For a strategy profile $\sigma = (\sigma_0, \ldots, \sigma_{K-1})$ write

$$\mathsf{out}_\Phi(\sigma) \;:=\; [\![\Phi]\!]_{\mathcal{M},\sigma} \in \{0,1\}^K,$$

i.e. evaluation in $\mathcal{M}$ with the empty assignment. For $i \in \{0, \ldots, K-1\}$ and an alternative strategy σ_i', write

$$\sigma[i \mapsto \sigma_i'] := (\sigma_0, \ldots, \sigma_{i-1}, \sigma_i', \sigma_{i+1}, \ldots, \sigma_{K-1}).$$

We record three kinds of properties: (i) forceability-style robustness, (ii) actual dependence of outcomes on other players, and (iii) brittleness. These are not all mutually comparable, but together they delimit the range from weak sensitivity to extreme coupling (and we do not attempt to linearly order them).

(1) Forceability / adversarial robustness. Player i is *robust* if

$$\exists \sigma_i \; \forall \sigma_{-i} : \; \big(\mathsf{out}_\Phi(\sigma_i, \sigma_{-i})\big)_i = 1.$$

This is a property of the game w.r.t. player i, not of a particular profile, and it is the maximally *decoupled* situation: player i does not rely on the behaviour of others.

We now turn to properties of a *fixed* profile σ.

(2) Pairwise outcome dependence. The profile σ has *pairwise dependence* if

$$\exists i \neq j \; \exists \sigma_j' : \; \big(\mathsf{out}_\Phi(\sigma)\big)_i \neq \big(\mathsf{out}_\Phi(\sigma[j \mapsto \sigma_j'])\big)_i.$$

So at least one player's outcome is sensitive to a unilateral change by some other player.

(3) Pervasive dependence. The profile σ is *pervasively dependent* if

$$\forall i \; \exists j \neq i \; \exists \sigma_j' : \; \big(\mathsf{out}_\Phi(\sigma)\big)_i \neq \big(\mathsf{out}_\Phi(\sigma[j \mapsto \sigma_j'])\big)_i.$$

Every player is sensitive to at least one other player. This strictly strengthens pairwise dependence.

Because outcomes are binary, we order $0 < 1$ componentwise.

(4) Local (Nash-style) stability. The profile σ is *locally stable* if

$$\forall i \; \forall \sigma_i' : \; \big(\mathsf{out}_\Phi(\sigma)\big)_i \geq \big(\mathsf{out}_\Phi(\sigma[i \mapsto \sigma_i'])\big)_i.$$

In the $\{0,1\}$ setting this says: no player can unilaterally turn a 0 into a 1. This witnesses a *locally self-supporting* point of the system, but it does not by itself guarantee robustness.

We now isolate two upper-end notions that express actual brittleness.

(5) Global brittleness. The profile σ is *globally brittle* if

$$\forall i \; \forall \sigma_i' \neq \sigma_i : \; \mathsf{out}_\Phi(\sigma[i \mapsto \sigma_i']) \neq \mathsf{out}_\Phi(\sigma).$$

Any unilateral deviation by any player changes the outcome vector.

(6) Maximal brittleness (extreme case).

$$\forall i \; \forall \sigma_i' \neq \sigma_i : \; \mathsf{out}_\Phi(\sigma[i \mapsto \sigma_i']) = (0, \ldots, 0).$$

Every unilateral deviation collapses the behavior to the worst outcome, namely $(0, \ldots, 0)$. This is a conceptual upper bound.

Counterfactual character. All the above notions are inherently counterfactual: they talk about what would have happened *if* some player had chosen a different strategy, even though in the given run they did not. This mirrors the style of the logic itself, where quantifiers describe possible moves that need not be taken in one particular execution.

9.8 Properties

Throughout this subsection we fix a structure $\mathcal{M}$ and consider only closed K-player formulas Φ. All evaluations $[\![\Phi]\!]_\sigma$ and solution concepts (support, realizability, Nash, etc.) are taken w.r.t. $\mathcal{M}$.

Proposition 9.11 (Downward closure of support)**.** *Let Φ be a K-player formula. If Φ supports $W \subseteq \{0, \dots, K-1\}$, then for every $W' \subseteq W$ it also supports W'.*

Proof. By Definition 9.2, there exists a strategy profile σ such that for all $i \in W$ we have $([\![\Phi]\!]_{\mathcal{M},\sigma})_i = 1$. In particular this holds for every $i \in W' \subseteq W$, so the same σ witnesses support for W'. □

Lemma 9.12 (Joint realizability as maximal support)**.** *A formula Φ is jointly realizable iff it supports $W = \{0, \dots, K-1\}$.*

Proof. Immediate from Definition 9.1 and Definition 9.2. □

Lemma 9.13 (Equivariance of support)**.** *Let $\pi \in S_K$. If Φ supports W, then $\pi \cdot \Phi$ supports $\pi(W)$.*

Proof. By assumption, there exists σ such that for all $i \in W$, $([\![\Phi]\!]_{\mathcal{M},\sigma})_i = 1$. By S_K-equivariance of the semantics (Theorem 3.2), we have

$$[\![\pi \cdot \Phi]\!]_{\mathcal{M},\pi\cdot\sigma} = \pi \cdot [\![\Phi]\!]_{\mathcal{M},\sigma}.$$

Hence for every $j \in \pi(W)$,

$$([\![\pi \cdot \Phi]\!]_{\mathcal{M},\pi\cdot\sigma})_j = (\pi \cdot [\![\Phi]\!]_{\mathcal{M},\sigma})_j = ([\![\Phi]\!]_{\mathcal{M},\sigma})_{\pi^{-1}(j)} = 1,$$

because $\pi^{-1}(j) \in W$. So $\pi \cdot \Phi$ supports $\pi(W)$. □

Lemma 9.14 (Orbit-realizability is S_K-invariant)**.** *If $v \in \{0,1\}^K$ is orbit-realizable for Φ, then for every $\pi \in S_K$, v is also orbit-realizable for $\pi \cdot \Phi$.*

Proof. By Definition 9.9, for every $\rho \in S_K$ there is σ_ρ such that $[\![\Phi]\!]_{\mathcal{M},\sigma_\rho} = \rho \cdot v$. For $\pi \cdot \Phi$, take the family $(\pi \cdot \sigma_\rho)_\rho$. By equivariance of semantics,

$$[\![\pi \cdot \Phi]\!]_{\mathcal{M},\pi\cdot\sigma_\rho} = \pi \cdot [\![\Phi]\!]_{\mathcal{M},\sigma_\rho} = \pi \cdot (\rho \cdot v) = (\pi\rho) \cdot v,$$

so every permutation of v is realized. □

Lemma 9.15 (IR-NE is a refinement of NE)**.** *If σ is an IR-NE for Φ, then σ is a pure Nash equilibrium for Φ.*

Proof. By definition an IR-NE is the conjunction of individual rationality (Definition 9.5) and pure NE (Definition 9.4). So the claim is immediate. □

9.9 Syntax-Directed Dependence Analysis

Our semantic notions of dependence and coupling in Section 9.7 are phrased in terms of counterfactual strategy changes: a player j "matters" for player i at a profile σ if replacing only j's strategy changes i's outcome. This is precise but not directly checkable from the syntax of a K-FOG specification. In this subsection we describe a lightweight, syntax-directed analysis that over-approximates such semantic dependence.

The construction follows a well-established pattern: (i) IF-logic–style independence annotations tell us which earlier choices a move is *not* allowed to see; (ii) program-dependence graphs and backward slicing collect all syntactic sources that can, in principle, affect a given use site [84]; and (iii) noninterference-style results then relate such a static approximation to semantic, run-level security properties [36]. Similar ideas have been developed for process calculi, in particular for the π-calculus and its variants, where type systems or dependency analyses are used to rule out illicit information flow between channels or security levels (see, e.g., [49]). Our setting is simpler than the general π-calculus case because our interaction structure is a single, finite, linear script; however, the underlying idea is the same: a *static* dependence relation that is sound for the *dynamic* semantics.

Setup. Fix a closed K-player formula

$$\Phi \;=\; P_{k_1}x_1 \setminus V_1.\; P_{k_2}x_2 \setminus V_2.\; \ldots.\; P_{k_n}x_n \setminus V_n.\; \langle \varphi_0, \ldots, \varphi_{K-1} \rangle$$

and a structure $\mathcal{M}$. (Plain binders $P_{k_r}x_r.$ are understood as $P_{k_r}x_r \setminus \emptyset.$.) We view the prefix as a linear sequence of *positions* $1, \ldots, n$, each owned by a player k_r, and each possibly hiding from a set of players V_r.

Direct visibility edges. For two positions $r < s$ let $p = k_r$ and $q = k_s$. We add a *direct visibility edge* $p \to q$ from r to s iff $q \notin V_r$, i.e. player q is not hidden from the choice made at position r. This is the K-player generalization of the way IF logic records independence at the quantifier level.

Direct formula-level edges. For each player i let $\mathsf{vars}(i)$ be the set of binder positions whose variables occur free in φ_i. For every $r \in \mathsf{vars}(i)$, where position r is owned by k_r, we add a *direct formula-level edge* $k_r \to i$. This is the standard data-dependence edge as in program dependence graphs (PDGs) [30].

Combined dependence graph. Let

$$G_\Phi := (\{0, \ldots, K-1\},\; E^{\mathsf{vis}} \cup E^{\mathsf{term}})$$

be the directed graph over player indices induced by all visibility and formula-level edges. Write $p \Rightarrow_\Phi q$ for the reachability relation in G_Φ.

Syntactic relevance. For a fixed player i we define the set of *syntactically relevant positions* $\mathsf{Rel}_\Phi(i) \subseteq \{1, \ldots, n\}$ by backward closure:

$$\begin{aligned} R_0(i) &:= \{\, r \mid x_r \text{ occurs free in } \varphi_i \,\}, \\ R_{t+1}(i) &:= R_t(i) \cup \{\, r < s \mid s \in R_t(i) \text{ and player } k_s \notin V_r \,\}. \end{aligned}$$

Because the prefix is finite, this construction stabilizes; let $\mathsf{Rel}_\Phi(i)$ be the limit. This is a direct analogue of backward slicing, adapted to our visibility discipline.

Lemma 9.16 (Soundness of syntax-directed dependence)**.** *Fix Φ and $\mathcal{M}$, and let i be a player. Let r be a binder position owned by player $j \neq i$, and suppose $r \notin \mathsf{Rel}_\Phi(i)$. Then for every profile of valid strategies σ and every alternative valid strategy σ'_j,*

$$(\llbracket \Phi \rrbracket_{\mathcal{M},\sigma})_i = (\llbracket \Phi \rrbracket_{\mathcal{M},\sigma[j \mapsto \sigma'_j]})_i.$$

Proof sketch. Validity of strategies enforces that a player's move may not branch on hidden variables. The backward construction of $\mathsf{Rel}_\Phi(i)$ exactly collects all positions whose values can be made available, through a sequence of such admissible branches, to a variable that φ_i actually reads. If r is not collected, its value cannot affect any later admissible choice on this path, and hence cannot affect the valuation of φ_i. This is the same noninterference reasoning used for typed information-flow systems for process calculi, specialized to a single, linear interaction. □

On completeness. As in classical slicing and information-flow analysis, the approximation is one-sided: absence of a path implies absence of semantic influence, but not conversely. Semantic influence can fail to materialize because the structure $\mathcal{M}$ identifies values or because actual strategies ignore visible information. Relative completeness can be recovered under standard richness assumptions (distinct domain elements, strategies that distinguish visible histories, and terminals that test for those distinctions), but we do not develop this variant here.

9.10 Querying Redundancy of a Specification

The solution concepts of Section 9 let us ask whether a player, or a coalition, can *force* its objective against arbitrary behavior of the others. We now repurpose the same vocabulary to phrase a software-engineering style question:

> does this K-player specification keep providing its service even if (up to) some participants fail or are removed?

Here "providing the service" is interpreted as keeping a designated outcome component at 1. This matches the informal notion of structural redundancy: the system was provisioned with more components than the minimum required to satisfy the service goal.

Service component. Fix a closed K-player specification Φ over a structure $\mathcal{M}$, and fix a *service index* $s \in \{0, \dots, K-1\}$ whose outcome $(\llbracket \Phi \rrbracket_{\mathcal{M},\sigma})_s$ we regard as the system-level requirement ("the service is up"). If the service is itself a conjunction of several players' objectives, we may introduce an auxiliary player whose terminal formula is that conjunction and take that player to be s.

One-failure redundancy. Intuitively, the system is 1-redundant for s if *every* single player can go arbitrary (fail, misbehave, be removed) and the remaining players can still force the service bit to 1.

Definition 9.17 (1-failure service redundancy)**.** Φ is *1-redundant for s in $\mathcal{M}$* if

$$\forall j \in \{0, \dots, K-1\} \setminus \{s\}.\ \exists (\sigma_i)_{i \neq j}\ \forall \sigma_j.\ \left(\llbracket \Phi \rrbracket_{\mathcal{M},\,(\sigma_i)_{i\neq j},\sigma_j}\right)_s = 1.$$

That is, for every potentially failing participant j, the coalition of all the other participants can enforce the service component against j's arbitrary behavior.

This is exactly the coalition-forceability pattern of Section 9.3, specialized to the coalition $C = \{0, \dots, K-1\} \setminus \{j\}$ and to the single objective "s wins". The failure is modeled as quantifying *universally* over j's strategy.

General k-failure redundancy. The same shape scales to multiple simultaneous failures.

Definition 9.18 (k-failure service redundancy)**.** For $k \geq 0$, Φ is *k-redundant for s in $\mathcal{M}$* if for every $F \subseteq \{0, \dots, K-1\}$ with $|F| \leq k$,

$$\exists (\sigma_i)_{i \notin F}\ \forall (\sigma_j)_{j \in F}.\ \left(\llbracket \Phi \rrbracket_{\mathcal{M},\,(\sigma_i)_{i\notin F},(\sigma_j)_{j\in F}}\right)_s = 1.$$

In words: any subset of at most k players may deviate arbitrarily, and the remaining players can still force the service.

For $k = 0$ this reduces to ordinary forceability of the service component by the grand coalition. For $k \geq 1$ it expresses "more provision than the minimum": some participants may be missing and the service is still deliverable.

Redundancy queries. Given Φ, $\mathcal{M}$, a service index s, and a budget k, the *redundancy query*

$$\mathsf{Redundant}(\Phi, \mathcal{M}, s, k)$$

asks whether Theorem 9.18 holds. This is a semantic question of the same flavor as forceability: it quantifies over strategy profiles, but the quantifier alternation matches the intended reading "the remaining coalition can cover for the failed participants". In particular, deciding redundancy is no easier than deciding the corresponding coalition-forceability instances; over the IF-extended calculus this inherits the hardness results of Section 13.

Remarks.

- The definition is intentionally *service-directed*: we do not require the deviating participants themselves to succeed, only that the designated service bit remains 1.
- One can strengthen the notion by requiring the non-failing participants to keep *their own* objectives, not only the service's; this amounts to replacing the single component $(\cdot)_s$ in Theorem 9.18 with a conjunction over a chosen survivor set.
- Because the definition is phrased only in terms of strategy quantification, it fits cleanly into the guarantee layer of Section 9 and does not require new object-language constructs.

10 Adversarial Test Components

Tests in this setting are naturally *adversarial*: they are players whose terminal formulas are written with the intention of catching bad behavior of other players. This is slightly at odds with everyday testing terminology: here a "good" test is one that is *bad for* at least one incorrect implementation. Formally, if $\mathcal{I}$ is the set of implementations and $G \subseteq \mathcal{I}$ the intended ones, we call a test non-vacuous if it fails on some $I \in \mathcal{I} \setminus G$.

Definition 10.1 (Test suite)**.** Let $\mathcal{I}$ be the set of implementations of a given interface, and let $G \subseteq \mathcal{I}$ be the intended ones. A *test suite* is a function

$$T : \mathcal{I} \to \{\mathsf{pass}, \mathsf{fail}\}.$$

It is

- *non-vacuous* if $\exists I \in \mathcal{I} \setminus G :\ T(I) = \mathsf{fail}$;
- *satisfiable* if $\exists I \in G :\ T(I) = \mathsf{pass}$;
- *complete* if $\forall I \in \mathcal{I} :\ T(I) = \mathsf{pass} \iff I \in G$.

As with solution concepts, this is a *counterfactual* view: we talk about which incorrect implementations would be rejected, not about the single observed run.

(1) Non-vacuous test. T is *non-vacuous* if

$$\exists I \in \mathcal{I} \setminus G :\ T(I) = \mathsf{fail}.$$

At least one incorrect implementation is actually detected.

(2) Satisfiable test. T is *satisfiable* if

$$\exists I \in G :\ T(I) = \mathsf{pass}.$$

At least one correct implementation survives.

(3) Complete test. T is *complete* for $(\mathcal{I}, G)$ if

$$\forall I \in \mathcal{I} :\ T(I) = \mathsf{pass} \iff I \in G.$$

This is the ideal but rarely attainable situation.

Connection to the game view. Just as forceability and the dependence notions talk about *counterfactual* strategy profiles ("if player j had chosen a different strategy, player i's outcome would have changed"), test components talk about *counterfactual* implementations ("if we had run the misbehaving implementation, this test would have failed"). In both cases the logic does not execute these alternatives; it only provides the language to state that such counterfactual runs exist and are separated by the specification.

Consequently, what we actually specify here is not "a system that is already correct", but rather

> a system whose interaction structure contains meaningful, adversarial tests; and correctness itself is obtained only under the extra assumption that those tests do in fact succeed on the run.

11 Functional and Type-Theoretic Readings

A useful way to view the system is as the combination of two orthogonal ingredients. The first is an MPST-like, linear protocol part: a single global prefix that fixes the order of moves, the owner of each move, and the visibility/independence constraints between moves. This is the "how the interaction must happen" piece, and it is very close to the usual story for global types and their projections.

The second ingredient is a many-truths outcome: instead of a single postcondition, the terminal of a formula is a K-indexed family of FO goals, one per participant. This is the "what each party ultimately wants" piece. MPST on its own does not tell you this — it gives you structure and safety, not per-role winning conditions. By putting the two together in the same object, we can say: "here is a protocol that enforces this information pattern, and here are the (possibly incompatible) objectives of all the parties who follow it."

The game-theoretic readings (forceability, coalition success, thresholds) then sit exactly at the junction of these two parts: they look at the protocol-constrained runs, and they test which components of the multi-headed outcome can actually be realized under those constraints. So the protocol half limits what strategies can do; the outcome half tells us which truth we are trying to extract from those limited strategies.

To make these ideas concrete, we now present four equivalent computational views of K-FOG expressions, each highlighting different aspects of the protocol and outcome structure.

11.1 Procedural view: linear coroutine

Evaluation of a K-FOG formula can be understood procedurally as a single-threaded, linear walk over the prefix:

1. start with an assignment s;
2. when you reach $P_i x.\,\Phi$, you must obtain a value for x *from player* i, subject to the current visible history;
3. extend s with $x := a$;
4. proceed to the next binder;
5. at the terminal $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$, evaluate each φ_j under the final assignment.

There are two closely related ways to make the "obtain a value" step concrete:

External, coroutine-style. The formula is a generator that *yields* at $P_i x.\,\Phi$ to an external owner for player i, together with the visible history (all previously bound variables, minus those in the slash-set). Control resumes when that owner returns a value. This matches the "several stateful parties are talking" intuition.

Internal, pure-strategy-style. The formula is evaluated against a fixed strategy profile $\sigma = (\sigma_0, \ldots, \sigma_{K-1})$. At $P_i x.\,\Phi$ we *call* the pure function $\sigma_i(\text{current-prefix}, s)$ to obtain a. This keeps everything inside the semantic function and is the version we use to define $[\![\Phi]\!]_{\mathcal{M},s,\sigma}$.

Both readings are linear (no backtracking, no branching); they differ only in whether the "player move" is modeled as an external interaction or as a call to a known pure strategy. The object language is the same in both cases; only the operational account changes.

Remark 11.1 (Inductive vs. coinductive readings). The two operational accounts above are dual in the usual inductive/coinductive sense. In the internal, pure-strategy reading we assume all ingredients (structure, assignment, strategy profile) are given and we evaluate by structural recursion on the prefix; this is an inductive "fold over the game." In the external, coroutine-style reading we instead expose the next required move and wait for the environment to supply a value; here the behaviour is driven from the outside, which is closer to a coinductive process view. Both target the same linear prefix; a systematic comparison is left open.

11.2 Let-expression style: functional programs

We can render the same evaluation as a nested functional program:

```
(* P0 x. P1 y. <phi0, phi1> *)
let
  val x = player_0 ()          (* first move: empty history *)
  val y = player_1 (x)         (* second move: sees x *)
in
  (phi0 x y, phi1 x y)
end
```

Independence-friendly visibility is just "drop the hidden variables from the call":

```
(* P0 x. P1 y \ {x}. P2 z. <...> *)
let
  val x = player_0 ()
  val y = player_1 ()          (* / {x} => y does NOT see x *)
  val z = player_2 (x, y)      (* later move can see both *)
in
  terminal (x, y, z)
end
```

In general, for a prefix

$$P_0 x_0.\, P_1 x_1 \setminus V_1.\, \ldots.\, P_{n-1} x_{n-1} \setminus V_{n-1}.\, \langle \vec{\varphi} \rangle$$

we get the schematic program

```
let
  val x0 = player_0 ()
  val x1 = player_1 (visible_from [x0] minus V1)
  ...
  val xn = player_n (visible_from [...] minus Vn)
in
  (phi0 (x0,...,xn), ..., phi_{K-1} (x0,...,xn))
end
```

A K-FOG game is thus an open program with one hole per player move; a strategy profile is the tuple of functions that fill those holes.

11.3 Gallina-style: player-indexed dependent presentation

We can write the same shape in a Gallina-like pseudocode that makes player ownership and capture explicit:

```
Definition game :=
  (* P0 move: sees nothing -> () *)
  let a := P0 (_ ()) in

  (* P1 move: sees a -> (a) *)
  let b := P1 (_ (a)) in

  (* P0 again, but / b: only sees (a, ()) *)
  let c := P0 (_ (a, ())) in  (* b hidden *)

  (* P2 move: sees everything so far *)
  let d := P2 (_ (a, b, c)) in

  {|
    when (a = b)             : P0;
    when (b <> c)            : P1;
    when (a = b \/ a <> b)   : P2
  |}.
```

Interpretation:

- each line `let x := Pi (_ (history)) in` is a player-owned move with a strategy hole _;
- the `history` tuple is exactly the visible prefix for that move (prefix minus the slash-set);
- the final record is a K-indexed outcome: one logical objective per player.

Curry–Howard reading. This looks like a partially constructed proof term with several heads. The outer `let` chain is the protocol skeleton. Each `Pi (_ (history))` is a typed subgoal owned by player i, to be solved using only *that* history. A strategy profile is the collection of inhabitants for all such subgoals. The difference from the usual CH story is that we do *not* collapse to one `Prop`; we keep one per player.

11.4 Core calculus: first-order, indexed, success-parametric

For metatheory it is convenient to strip this down to a first-order core and make "success" a parameter.

Syntax. Let terms and FO formulas be standard:

$$t ::= x \mid f(t_1, \ldots, t_n)$$

$$\alpha ::= R(\vec{t}) \mid t_1 = t_2 \mid \top \mid \bot \mid \alpha \wedge \alpha \mid \alpha \vee \alpha \mid \alpha \to \alpha \mid \neg\alpha.$$

Game formulas:

$$\Phi ::= \langle \alpha_0, \ldots, \alpha_{K-1} \rangle \mid P_i x : \tau.\ \Phi \quad (i \in \{0, \ldots, K-1\}).$$

Formation. With FO context Γ:

$$\frac{\Gamma \vdash \alpha_0 \quad \ldots \quad \Gamma \vdash \alpha_{K-1}}{\Gamma \vdash \langle \alpha_0, \ldots, \alpha_{K-1} \rangle} \qquad \frac{\Gamma, x : \tau \vdash \Phi}{\Gamma \vdash P_i x : \tau.\ \Phi}.$$

This gives a global, player-symmetric spec: a player-indexed prefix ending in a K-tuple of FO goals.

Evaluation. Given a structure $\mathcal{M}$, an assignment s, and a strategy profile σ,

$$[\![\langle \alpha_0, \ldots, \alpha_{K-1} \rangle]\!]_{\mathcal{M},s,\sigma} := \big([\,\mathcal{M}, s \models \alpha_0\,], \ldots, [\,\mathcal{M}, s \models \alpha_{K-1}\,]\big),$$

$$[\![P_i x : \tau.\ \Phi]\!]_{\mathcal{M},s,\sigma} := [\![\Phi]\!]_{\mathcal{M},s[x\mapsto a],\sigma} \quad \text{where } a := \sigma_i(P_i x : \tau.\ \Phi, s).$$

So evaluation always produces a K-tuple of booleans (or $\mathsf{True}/\mathsf{False}$).

Parametric success. On top of this core we *do not* fix a single success notion. Instead we admit judgments of the form

$$\mathcal{M}, s \Vdash_{\mathsf{Succ}} \Phi$$

where Succ specifies how we quantify over strategies. One instance is *forceability*. Other instances (possibility, coalition winning, thresholds) are definable over the same evaluation.

Separation. This separates:

1. structure: what is a well-formed, player-indexed game;
2. evaluation: given strategies, what outcome vector do we get;
3. success: which outcome patterns we count as "good".

Only (1) and (2) belong to the core; (3) is meant to be chosen per application.

12 Connection to Multi-Party Session Types

This section gives a deliberately lightweight correspondence between K-FOG prefixes and MPST-style global protocols. In contrast to event-structure-based accounts that relate session types and game semantics in a fully concurrent and branching setting (e.g. [20]), we stay in a *linear, non-branching, choreography-style* fragment and make visibility explicit in the protocol itself.

This translation is purely structural: it preserves the linear order of moves and the visibility/multiparty delivery pattern, but it forgets the K-ary payoff tuple. Certain game-theoretic notions (e.g. forceability of a given role) might be representable on top of a more expressive or MPST discipline, but our translation here is deliberately shape-only and we do not attempt to recover such notions from the MPST side.

12.1 Setting

We fix th We fix the participant set

$$\mathcal{P} = \{0, 1, \ldots, K-1\}$$

to coincide with the K-FOG players. Every bound variable x is assigned an MPST ground sort U_x. We require the well-formedness condition

$$P_i x \setminus V.\,\Phi \text{ is allowed only if } i \in \mathcal{P} \text{ and } V \subseteq \mathcal{P} \setminus \{i\}.$$

Intuitively, player i chooses x, and exactly the roles in $\mathcal{P} \setminus (\{i\} \cup V)$ may observe that choice.

Following the correction in Section 5, exclusion sets are variable-level in general; this section's well-formedness condition restricts attention to formulas built from the *role-level derived notation* $P_i x/J$ defined there, i.e. formulas where every binder's exclusion set is already a union of "all of role j's variables, for j in some set." Formulas with genuinely per-variable exclusion patterns, such as the DQBF construction of Proposition 13.2, fall outside this translatable fragment; the MPST correspondence below is a statement about the role-uniform sub-language, not about K-FOG in general.

Target global types. Let Glob denote the set of *linear MPST-style global types* generated by

$$\mathsf{Glob} ::= \mathbf{end} \mid (p \to q : U)\,;\,\mathsf{Glob}$$

where $p, q \in \mathcal{P}$ and U is a ground sort. Thus Glob contains only point-to-point sends in sequence and terminates in **end**; there is no branching or concurrency. Our translation $\mathcal{T} : \text{K-FOG} \to \mathsf{Glob}$ will land in this fragment.

12.2 Core Analogy

K-FOG	**MPST**
Players $\{0, \dots, K-1\}$	Participants $\mathcal{P}$
Linear binder prefix	Global type (linear sequence)
Binder $P_i x \setminus V. \cdot$	i sends x to a subset of participants
Terminal $\langle \varphi_0, \dots, \varphi_{K-1} \rangle$	**end**

We encode only the communication skeleton; the payoff tuple is erased.

12.3 Broadcasting as Multiple Messages

Standard MPST have no built-in support for broadcasting (but see [17] and [56]). We encode broadcasting into series of messages, and add the assumption of common knowledge between recipients, which is crucial for reasoning about strategy profiles and equilibria.

For $i \in \mathcal{P}$, a sort U, and $S \subseteq \mathcal{P} \setminus \{i\}$, write

$$\mathrm{BCast}(i, U, S)$$

for the sequence of point-to-point sends from i to each $j \in S$, in some fixed but arbitrary order:

$$\mathrm{BCast}(i, U, \{j_1, \dots, j_m\}) \;=\; (i \to j_1 : U)\,;\, (i \to j_2 : U)\,;\, \dots\,;\, (i \to j_m : U).$$

Define the empty case by

$$\mathrm{BCast}(i, U, \emptyset) := \epsilon.$$

Note that BCast here is only a shorthand for a sequence of point-to-point sends. In K-FOG the corresponding binder is a single atomic game move that is immediately visible to all non-slashed roles; plain MPST does not enforce or assume this common-knowledge aspect, and we rely on the game semantics (meta level) for that.

12.4 Two Translation Schemes

It is helpful to name both the "everyone sees" encoding and the visibility-restricting one, and then fix the latter.

Definition 12.1 (Broadcast-based translation $\mathcal{T}^{\mathsf{bcast}}$)**.** Define $\mathcal{T}^{\mathsf{bcast}} : \text{K-FOG} \to \mathsf{Glob}$ by

$$\begin{aligned} \mathcal{T}^{\mathsf{bcast}}\big(\langle \varphi_0, \dots, \varphi_{K-1} \rangle\big) &:= \mathbf{end}, \\ \mathcal{T}^{\mathsf{bcast}}\big(P_i x \setminus V.\, \Phi\big) &:= \mathrm{BCast}(i, U_x, \mathcal{P} \setminus \{i\})\ ;\ \mathcal{T}^{\mathsf{bcast}}(\Phi). \end{aligned}$$

This ignores the slash set V at the protocol level; enforcing "you must not depend on V" would then be pushed to a separate strategy-validity layer.

Definition 12.2 (Visibility-restricting translation $\mathcal{T}$ (chosen))**.** Define $\mathcal{T} : \text{K-FOG} \to \mathsf{Glob}$ by

$$\begin{aligned} \mathcal{T}\big(\langle \varphi_0, \dots, \varphi_{K-1} \rangle\big) &:= \mathbf{end}, \\ \mathcal{T}\big(P_i x \setminus V.\, \Phi\big) &:= \mathrm{BCast}(i, U_x, (\mathcal{P} \setminus \{i\}) \setminus V)\ ;\ \mathcal{T}(\Phi). \end{aligned}$$

So a slashed role simply does not receive the message.

This is the analogue, in our simplified setting, of "putting the game-relevant structure in the type" rather than in an external semantic notion of strategy, in the spirit of [20], but for a strictly less expressive fragment.

12.5 Projection to Local Types

Let $\mathsf{Glob} = \mathcal{T}(\Phi)$. For each $p \in \mathcal{P}$ we take the standard MPST projection and write $\mathsf{Glob} \upharpoonright_p$. We use the conventional local notation

$$!j\langle U\rangle = \text{send to } j \text{ a value of sort } U,$$
$$?j\langle U\rangle = \text{receive from } j \text{ a value of sort } U.$$

For a single binder $P_i x \setminus V$ we obtain:

- $\mathsf{Glob} \upharpoonright_i$ contains sends $!j\langle U_x\rangle$ to every $j \in (\mathcal{P} \setminus \{i\}) \setminus V$;
- for every $j \in (\mathcal{P} \setminus \{i\}) \setminus V$, $\mathsf{Glob} \upharpoonright_j$ contains a receive $?i\langle U_x\rangle$;
- for every $j \in V$, $\mathsf{Glob} \upharpoonright_j$ contains no communication action for this binder.

Example 12.3 (Two players)**.** Let

$$\Phi = P_0 x \setminus \emptyset.\, P_1 y \setminus \{0\}.\, \langle \varphi_0, \varphi_1\rangle,$$

with $x : U_x$ and $y : U_y$. Then

$$\mathcal{T}(\Phi) = (0 \to 1 : U_x)\ ;\ \mathrm{BCast}(1, U_y, \emptyset)\ ;\ \mathbf{end} = (0 \to 1 : U_x)\ ;\ \mathbf{end}.$$

Projection yields

$$\mathcal{T}(\Phi) \upharpoonright_0 = !1\langle U_x\rangle;\ \mathbf{end}, \qquad \mathcal{T}(\Phi) \upharpoonright_1 = ?0\langle U_x\rangle;\ \mathbf{end}.$$

Player 1's move y does occur in the K-FOG game but is not communicated to player 0 in the protocol, exactly because of the slash set $\{0\}$.

A conjectured logical–session correspondence. Fix Φ over participants $\{0, \ldots, K-1\}$ and fix p. Let

$$\pi_p : \{0, \ldots, K-1\} \twoheadrightarrow \{A, B\} \quad \text{be the surjection with} \quad \pi_p(p) = A \text{ and } \pi_p(j) = B \text{ for } j \neq p.$$

Apply the adversarial projection:

$$\psi_p \ := \ \underset{p}{\exists\forall}(\Phi)\,,$$

obtaining a prenex FOL formula that expresses Φ from p's viewpoint.

On the session side, translate Φ to a global type and project it:

$$L_p \ := \ (\mathcal{T}(\Phi))! \upharpoonright_p .$$

Conjecture. For every Φ and p, the local session type L_p describes the same p-view of the interaction as the logical adversarial projection $\exists\forall$, up to role renaming and erasure of the terminal payoff. In symbols,

$$(\mathcal{T}(\Phi)) \upharpoonright_p \equiv \ \mathcal{T}^{(2)}\big(\mathsf{Co}^{\vee}_{\pi_p}(\Phi)\big),$$

and $\exists\forall_A\Big(\mathsf{Co}^{\vee}_{\pi_p}(\Phi)\Big)$ is the corresponding 2-player, FOL-level viewpoint. We do not prove this; the inductive cases match the translation rules binder-by-binder but require bookkeeping on the visibility sets. The cooperative analogue of this kind of correspondence — global MPST types as proofs in a session-typed logic — is established by the coherence-proofs account of MPST [18] (Section 14); our conjecture, if provable, would be its adversarial, per-role counterpart rather than a special case of it, since ψ_p here is one role's forceability question, not a joint coherence certificate.

12.6 Scope

This correspondence is intentionally weaker than the session-type/game-semantics correspondence based on (coincident) event structures in [20]: we do not treat branching/selection, concurrency, or the synchrony–asynchrony gap; we only show that a linear K-FOG prefix with explicit visibility can be read as a linear MPST global whose projection is standard. Since the target is an ordinary MPST global, the usual projection/duality/progress properties of MPST apply to the translated protocol; what we erase is the K-FOG terminal payoff tuple, which we map to **end**.

The point of this section is only to show that every K-FOG protocol has an MPST-shaped carrier. It is not a full-abstraction result: MPST typing of the translated protocol does not recover or validate the original per-player objectives.

12.7 A Faithful Projection into Refined MPST (Best-Effort)

We now spell out a concrete projection from K-FOG to a refined multiparty-session–style global description. The goal is not to show that K-FOG is more expressive than refined MPST, but to identify precisely which parts of our specifications survive such a projection and which parts remain specific to the game-theoretic reading.

Source. Recall that a closed K-FOG formula has the shape

$$\Phi \;=\; P_{i_1} x_1 \setminus V_1.\ \ldots.\ P_{i_n} x_n \setminus V_n.\ \langle \varphi_0, \ldots, \varphi_{K-1} \rangle$$

where each binder says "player i_k chooses a value for x_k, and roles in V_k do not see this move," and the terminal is a K-tuple of first-order goals. The semantics evaluates this prefix once, centrally, and then produces a K-vector of truth values.

Target. We project Φ to a refined MPST global $\mathcal{G}(\Phi)$ over the same participant set $\mathcal{P} = \{0, \ldots, K-1\}$. A refined MPST global here is a sequence of labelled communications, each optionally decorated with a first-order refinement that must hold at that point. Ordinary MPST uses point-to-point channels; because our source has broadcast-style visibility, we use a "multi-send" sugar in the target, which can be compiled to a sequence of ordinary sends.

Step 1: interaction projection. We project the prefix position-by-position. For a binder

$$P_i x \setminus V.\, \Psi$$

we introduce, in order, a family of sends

$$i \to j : U_x\{\rho_{i,x}\} \qquad \text{for every } j \in \mathcal{P} \setminus (\{i\} \cup V),$$

and skip the slashed roles $j \in V$ altogether. Here U_x is the sort of x. This is the refined version of the visibility-restricting translation from Section 12: everyone who is not slashed receives the value.

What remains is to say what refinement $\rho_{i,x}$ is. In full generality our terminal may mention variables not yet in scope at this binder, so we extract only the part that is *checkable now*. Concretely, let $\mathrm{FV}_{\le k}$ be the variables bound *up to* this binder (i.e. $\{x_1, \ldots, x_k\}$). For each player i, define the *binder-local projection* of φ_i at position k by

$$\pi^{\mathrm{local}}_{i,k}(\varphi_i) \;:=\; \bigwedge \{\chi \mid \chi \text{ is a conjunct of } \varphi_i \text{ and } \mathrm{FV}(\chi) \subseteq \mathrm{FV}_{\le k}\}.$$

(If φ_i is not a conjunction, first put it in an equivalent CNF restricted to earlier variables; otherwise this set is empty and the conjunction is $\top$.) We then set

$$\rho_{i,x} \;:=\; \bigwedge_{p \in \mathcal{P}} \pi^{\mathrm{local}}_{p,k}(\varphi_p).$$

So the refinement attached to this send is "all binder-local constraints that are already meaningful at this point," aggregated over all players. This is exactly the kind of per-action refinement that R-MPST carries.

Applying this clause to every binder in the prefix yields a refined global $\mathcal{G}_{\mathrm{prefix}}(\Phi)$ with the same linear order of moves and the same visibility pattern.

Step 2: per-role dummy commitments. The terminal $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$ can contain constraints that are *not* checkable at any earlier binder (for example, because they refer to all players' choices at once). To make these visible in the session world without breaking locality, we append, for every participant i, a dummy action of the form

$$i \to \text{unit} : \texttt{unit}\{\varphi_i^{\text{residual}}\}$$

where $\varphi_i^{\text{residual}}$ is the part of φ_i that was *not* captured by any of the $\rho_{-,-}$ above. Operationally this is "I perform a no-op but I assert my remaining condition." This brings each player-specific formula into the refined MPST surface form: every participant now has a final, local refinement.

At this point we have a refined MPST global

$$\mathcal{G}(\Phi) \;=\; \mathcal{G}_{\text{prefix}}(\Phi)\,;\, \prod_{i \in \mathcal{P}} \big(i \to \text{unit} : \texttt{unit}\{\varphi_i^{\text{residual}}\}\big)$$

(where the final product can be sequenced in any fixed order).

What this preserves. This projection preserves:
- the *linear order* of moves;
- the *assignment of moves to participants*;
- the *visibility pattern* expressed by slash-sets (as recipients are omitted);
- all *binder-local* logical conditions that are expressible over the variables in scope at that point;
- and the fact that there are *per-role* logical obligations at the end (via the dummy steps).

A refined-MPST toolchain that projects globals to endpoints and checks admissible traces would be able to enforce precisely these parts.

What this does *not* preserve. Two aspects of K-FOG do *not* survive this projection:
1. **Outcome-as-value.** In K-FOG the terminal yields a $\{0,1\}^K$ vector which we later feed to solution concepts (forceability, IR-NE, brittleness). In the refined MPST image this becomes a family of per-role *obligations* on the trace, not a vector we can later quantify over.
2. **Counterfactual / strategy-based queries.** K-FOG lets us ask "if player j had played differently, would i still have obtained 1?" This is a property of the *game* induced by the prefix, not of the set of admissible traces. Refined MPST, being trace-based, does not encode these counterfactuals.

In other words, the projection is faithful on the protocol side and on all constraints that are local-at-some-point, but it necessarily flattens the game-theoretic layer to trace admissibility.

On expressiveness. This construction does *not* prove that K-FOG is more expressive than refined MPST; it shows that every K-FOG formula admits a refinement-like reading for its protocol and local constraints. The extra material that we keep on the K-FOG side (multi-headed outcome, solution concepts) is extra *analysis* over the same script, not an extra control construct in the protocol itself.

12.8 Embedding a Refined MPST Fragment into K-FOG

To complete the comparison, we describe the converse direction: how a refined–MPST-style specification can be represented in our calculus. Since K-FOG works with a globally described interaction *but* executes it in a way that respects only the visibility/causal order (i.e. strategies make choices based on what they are allowed to see, and different compatible strategy profiles may yield different runs), we restrict attention to the choreography-style fragment of refined MPST in which the whole interaction is given as one global sequence of actions, each owned by one participant, optionally decorated with a logical refinement, and delivered to a set of receivers. This is precisely the part of MPST that mirrors our "global prefix" view.

Source fragment. Consider a refined MPST global of the following shape:

$$G \;=\; a_1;\; a_2;\; \ldots;\; a_n;\; \mathbf{end}$$

where each action

$$a_k \equiv p_k \to \{q_{k,1}, \ldots, q_{k,m_k}\} : x_k : U_k\{\psi_k\}$$

says: participant p_k originates a value x_k of sort U_k, to be delivered to the listed receivers, and the action is subject to a first-order refinement ψ_k over values introduced by earlier actions. We assume refinements mention only previously introduced data, as is standard.

Participants. Let $\mathcal{P} = \{0, \ldots, K-1\}$ be the participant set of G. We assume this is the same participant set as in K-FOG; otherwise we fix a bijection from names to indices.

Action-to-binder translation. We translate each global action to a K-FOG binder. For the k-th action $p_k \to \{q_{k,1}, \ldots, q_{k,m_k}\} : x_k : U_k\{\psi_k\}$ we introduce the binder

$$P_{p_k} x_k \setminus V_k . \cdot$$

where

$$V_k \;:=\; \mathcal{P} \setminus \big(\{p_k\} \cup \{q_{k,1}, \ldots, q_{k,m_k}\}\big)$$

is exactly the set of roles that are *not* supposed to observe this value. Intuitively, player p_k now chooses x_k, and all non-slashed roles may condition their later moves on it. This matches the MPST visibility pattern.

Refinements. There are two placements for the step refinement ψ_k:

(i) If we allow an optional guard on binders, we write

$$P_{p_k} x_k \text{ where } \psi_k \setminus V_k . \cdot$$

which mirrors the MPST step exactly.

(ii) In the core K-FOG syntax (no binder guards), we push ψ_k to the terminal and conjoin it to those players' components that must ensure (or rely on) that condition. Because K-FOG is prenex, $x_1, \ldots, x_k$ are in scope at the terminal.

In both variants, the MPST predicate becomes an FO condition over the same variables.

Terminal. MPST ends in `end`. In K-FOG we must end in a K-tuple, so the simplest embedding is

$$\langle \top, \ldots, \top \rangle.$$

If we want the MPST refinements to be seen as "all must hold," we can instead take

$$\langle \psi_1 \wedge \cdots \wedge \psi_n, \ldots, \psi_1 \wedge \cdots \wedge \psi_n \rangle,$$

i.e. every participant records that all step conditions were satisfied.

Execution correspondence. In MPST, the global script is usually read as prescribing the whole run sequence. In K-FOG, the same script is read as a *visibility-guided* interaction: each binder introduces data, makes it available to its non-slashed observers, and strategies for later binders may respond differently depending on what they saw. Thus, different strategy profiles over the K-FOG-encoded script may realize different admissible runs of the original MPST script, but always in a way that respects the same causal/visibility structure.

Expressiveness boundary. This embedding is faithful for the linear, non-branching, choreography-style fragment of refined MPST. Features that go beyond this fragment do not embed without extending K-FOG:

- *Branching/choice* would require a choice construct in the prefix or an encoding via values and guards.
- *Recursion/infinite sessions* would require iterative or recursive structure.
- *Asynchronous channel behaviour* is not represented, since our execution model is strategy-driven under a single global interaction description, not FIFO-channel–driven.

Within the overlapping fragment, however, the translation preserves participants, causal/visibility dependencies, and the first-order content of step refinements.

13 Complexity of Forceability

We analyze the complexity of the forceability problem: given a finite structure $\mathcal{M}$, a closed K-FOG formula Φ, and a player i, decide if $F_i(\Phi, \mathcal{M})$ is true. We assume $\mathcal{M}$ is fixed and the formula Φ is the input.

13.1 The Broadcast Core is PSPACE-complete

First, we observe that the core calculus (without $\backslash V$ binders) is PSPACE-complete, by a standard reduction from Quantified Boolean Formulas (QBF).

Proposition 13.1 (PSPACE-completeness of Core)**.** *For fixed $K \geq 2$, the forceability problem F_i for the broadcast fragment (no $\backslash V$ binders) over finite structures is PSPACE-complete.*

Proof sketch. Membership in PSPACE is by a standard alternating Turing machine. We can evaluate $F_i(\Phi, \mathcal{M})$ by a recursive algorithm that simulates the game. For a binder $P_i x.\, \Phi'$, it existentially guesses $a \in |\mathcal{M}|$ and recurses. For $P_j x.\, \Phi'$ (where $j \neq i$), it universally checks all $a \in |\mathcal{M}|$ and recurses. Since the domain $|\mathcal{M}|$ is finite, this takes polynomial space in the depth of the formula.

Hardness is by reduction from QBF. Let ψ be a QBF

$$Q_1 x_1 \ldots Q_n x_n.\, f(x_1, \ldots, x_n)$$

over Boolean variables. We build a $K = 2$ formula Φ_ψ over a structure $\mathcal{M}$ with domain $\{0, 1\}$. This is a **one-hot** (**zero-sum** or **conflicting**) game, as defined in Section 3.2.

$$\Phi_\psi := P_{q(1)} x_1.\, P_{q(2)} x_2.\, \ldots\, P_{q(n)} x_n.\, \langle f, \neg f \rangle$$

where $q(j) = 0$ if $Q_j = \exists$ and $q(j) = 1$ if $Q_j = \forall$. (Assuming $K = 2$; for $K > 2$ we pad with $\top$). The QBF ψ is true if and only if player P_0 can force their objective, i.e., $F_0(\Phi_\psi, \mathcal{M})$ holds. Since QBF is PSPACE-complete, so is our problem. □

13.2 IF-style Independence is NEXPTIME-hard

When we add the IF-style binders, the complexity jumps. The problem is now at least as hard as Dependency QBF (DQBF), which is NEXPTIME-complete. This construction needs variable-level independence (each existential binder excluding an arbitrary, binder-specific subset of prior variables); Section 5 makes this the primitive form of $\backslash V$ for exactly this reason.

Proposition 13.2 (NEXPTIME-hardness with IF-binders)**.** *For fixed $K \geq 2$, the forceability problem F_i with $\backslash V$ binders is NEXPTIME-hard.*

Proof sketch. We reduce from DQBF. A DQBF instance consists of a formula

$$\forall y_1 \ldots \forall y_n\, \exists x_1 \ldots \exists x_m.\, f(\vec{y}, \vec{x})$$

and for each x_j, a dependency set $D_j \subseteq \{y_1, \ldots, y_n\}$. The instance is true iff there exist Skolem functions $f_1, \ldots, f_m$, where each f_j *only* depends on the variables in D_j, such that $\forall \vec{y}, f(\vec{y}, f_1(D_1), \ldots, f_m(D_m))$ holds.

We build a $K = 2$ formula Φ over $\mathcal{M} = \{0, 1\}$. Let P_1 be the $\forall$ player and P_0 be the $\exists$ player. For each x_j, define its "hidden set" H_j as the variables x_j is *not* allowed to see: $H_j := \{y_1, \ldots, y_n\} \setminus D_j$.

The formula Φ is:

$$\begin{aligned}\Phi := P_1 y_1.\ \ldots\ P_1 y_n.\ & P_0 x_1 \setminus H_1. \\ & P_0 x_2 \setminus H_2. \\ & \ldots \\ & P_0 x_m \setminus H_m.\ \langle f, \neg f \rangle\end{aligned}$$

The DQBF instance is true if and only if $F_0(\Phi, \mathcal{M})$ holds.

($\Rightarrow$) If DQBF is true, the Skolem functions f_j exist. P_0's strategy σ_0 is defined as $\sigma_0(P_0 x_j \setminus H_j, s) := f_j(s|_{D_j})$. This strategy is valid because it is independent of H_j. By construction, it forces f to be true.

($\Leftarrow$) If $F_0(\Phi, \mathcal{M})$ holds, then a valid strategy σ_0 exists. The binder $P_0 x_j \setminus H_j$ *forces* σ_0 to be independent of H_j. This means σ_0's choice for x_j can only depend on variables *not* in H_j, which is exactly D_j. This strategy σ_0 *is* the set of Skolem functions $f_j(D_j)$ that witnesses the DQBF. □

Remark 13.3 (This construction is a $K = 2$, single-role phenomenon). The hardness in Proposition 13.2 already appears at $K = 2$, and every x_j on the existential side is bound by the *same* role P_0: the independence driving the DQBF reduction is between binders owned by *one* role ($P_0 x_j \setminus H_j$ for varying j), not between distinct roles or coalitions of roles. So this result shows that a single role can fail to act as one self-visible agent once its own moves are mutually restricted; it does not by itself show that $K \geq 3$ roles, or coalitions of roles, do anything a two-role game cannot already do. We return to this distinction below, where it turns out to matter for how to read the next subsection.

13.3 Coalitional Forceability

Having $K > 2$ roles raises a natural further question: can a *coalition* of roles jointly force all of its members' objectives at once, using strategies that may depend on each other's earlier moves however the visibility structure allows?

Definition 13.4 (Coalitional Forceability)**.** Let $S \subseteq \{0, \ldots, K-1\}$ be a coalition. S can jointly force its objectives if

$$F_S(\Phi, \mathcal{M}) := \exists \{\sigma_i\}_{i \in S}\ \forall \{\sigma_j\}_{j \notin S}\ \bigwedge_{i \in S} \left([\![\Phi]\!]_{\mathcal{M}, \sigma} \right)_i = 1.$$

Relation to effectivity functions and Quantified Coalition Logic. The map $S \mapsto F_S(\Phi, \mathcal{M})$ is an instance of an *effectivity function* [61]: a specification of which outcomes each coalition can guarantee. Pauly's representation theorem shows Coalition Logic is the modal logic of *playable* effectivity functions [65], later corrected to *truly* playable functions to handle infinite outcome spaces [37]. Closer still, quantifying over coalitions directly in the object language — "some coalition satisfying property P can force ϕ" — is already Quantified Coalition Logic's central operator [5]. We are not aware of a prior treatment of coalitional forceability specifically inside a first-order, player-labeled quantifier-prefix calculus like K-FOG's, but the coalition-quantification idea itself is not new, and any future extension of K-FOG with an object-language coalition-quantifying operator should be scoped against this literature rather than presented as novel on that count alone.

A previous version of this section claimed F_S was a hard problem "not present in 2-player logic," with a proof that in fact only ever instantiated $S = \{0\}$: a singleton coalition, i.e. exactly

the individual-forceability problem of Proposition 13.1 under a different name, padding every other role with a vacuous objective $\top$. That argument says nothing about coalitions with $|S| \geq 2$, the only case in which "coalition" means anything beyond "individual." We withdraw it. Using the coalition-forming apparatus already defined in Definition 6.10, the honest statement is close to the opposite of what was claimed:

Proposition 13.5 (Coalition merging is lossless under broadcast visibility)**.** *Let Φ be a closed* K-FOG *formula with no $\setminus V$ binders (broadcast fragment), let $S \subseteq \{0, \ldots, K-1\}$, and let π be the surjection onto $\{0,1\}$ with $\pi^{-1}(0) = S$ and $\pi^{-1}(1) = \{0, \ldots, K-1\} \setminus S$. Then*

$$F_S(\Phi, \mathcal{M}) \iff F_0\big(\mathsf{Co}^{\wedge}_{\pi}(\Phi), \mathcal{M}\big).$$

Proof. By Definition 2.1, a strategy σ_i for role i is already a single function ranging over *every* binder owned by i, keyed by the history visible at that binder. Under broadcast visibility, relabeling a binder from P_i to $P_{\pi(i)}$ (as $\mathsf{Co}^{\wedge}_{\pi}$ does) changes neither the binder's position in the prefix nor the history visible at it: it only renames which strategy fills it. Hence a family $\{\sigma_i\}_{i \in S}$ of per-role strategies and a single merged strategy for role 0 in $\mathsf{Co}^{\wedge}_{\pi}(\Phi)$ are the same underlying data, viewed either split by original owner or joined into one function on the disjoint union of domains; likewise for $\{\sigma_j\}_{j \notin S}$ against role 1. The terminal of $\mathsf{Co}^{\wedge}_{\pi}(\Phi)$ has first component $\bigwedge_{i \in S} \varphi_i$, matching the conjunction in F_S exactly, and F_0 inspects only that component. The two sides therefore quantify over the same strategy spaces and test the same condition. □

Corollary 13.6 (Complexity of coalitional forceability, broadcast fragment)**.** *For any fixed $K \geq 2$ and any $S \subseteq \{0, \ldots, K-1\}$, F_S without $\setminus V$ binders is PSPACE-complete: both membership and hardness follow from Proposition 13.1 applied to the two-role formula $\mathsf{Co}^{\wedge}_{\pi}(\Phi)$, via Theorem 13.5. No new reduction is needed.*

Remark 13.7 (Where, if anywhere, coalitions have separate content)*.* Theorem 13.5 is specific to the broadcast fragment, and the proof shows exactly why: merging relies on relabeling not changing any binder's visible history. This breaks as soon as $\setminus V$ binders are present, because a slash-set can name a *specific* role ("independent of role j"), and after merging S into one role that name either refers to a member of the merged role itself or vanishes along with the role it named. Concretely, if Φ has a binder $P_i x \setminus \{j\}.\, \Phi'$ with $i, j \in S$, $i \neq j$ — role i's move must not depend on role j's earlier move, even though i, j are coalition partners — there is no faithful way to fold this into a single merged strategy for S: dropping the restriction overstates what the coalition can jointly achieve, and keeping it means the "merged" role is not really merged.

This last claim needs a correction. Stated this bluntly it is too strong: a $\setminus V$-crossing within a coalition does not, by itself, force a loss. Theorem 6.13 exhibits exactly this crossing pattern inside a coalition (the cross-diagonal product $\circledast$, e.g. Battleship) and shows the merge is nonetheless lossless there, because the two members' objectives are separable — neither could have used the withheld information anyway. What should matter is not the syntactic crossing but whether it *couples* the coalition's objectives: whether some counterfactual strategy exploiting the withheld information would provably do better. We do not have a general criterion for when that is the case, so the conjecture below should be read as requiring coupled objectives, not merely a $\setminus V$-crossing.

With that correction, this still suggests, but does not prove, where genuine $K \geq 3$ content in this calculus would have to live: coalitions whose own members are mutually information-restricted by IF-binders *and* whose objectives are coupled across that restriction. Proposition 13.2 already shows that independence between binders owned by a *single* role breaks naive analysis (Theorem 13.3), but, as noted there, that construction uses $K = 2$ and one role's self-restriction, not restrictions between distinct roles of a coalition with coupled goals. We do not have a construction that separates coalitional forceability with IF-binders from its two-coalition projections for $K \geq 3$. We conjecture one exists, along the lines sketched above, and we pose it

as the main open technical question raised by this section (see also Section 15). This conjecture is stated information-theoretically throughout; whether it survives, strengthens, or collapses once independence is realized by (and relaxed to) computational cryptographic primitives rather than assumed exactly is a separate question we have not addressed, posed precisely in Section B.14.

One prerequisite of this conjecture is now checked rather than assumed: the grammar must actually be able to express cross-role, per-binder information restriction of the kind a construction along these lines would need. Section 5's correction to variable-level exclusion confirms it can — a binder owned by one role of a coalition can exclude an arbitrary subset of prior variables, including ones bound by another role of the *same* coalition, which is exactly the pattern Theorem 6.13 shows is *not* sufficient by itself (coupling is still required) but which role-level exclusion could not even have expressed. This does not make the conjecture true; it removes one way it could have been vacuous.

Remark 13.8 (What a general coalition-quantifying operator would need, if built). Suppose K-FOG were extended with an object-language operator quantifying over coalitions, in the style of Quantified Coalition Logic [5]. Such an operator would have (at least) four parameters: (i) a *coalition family* $\mathcal{S} \subseteq 2^{\{0,\dots,K-1\}}$ to quantify over — note that a monotone coalition family is itself a *simple game* in Shapley's sense [74], so this parameter connects directly to voting-theoretic structure; (ii) a *goal-aggregation* rule per coalition (conjunctive, as $\mathsf{Co}^{\wedge}$ already does, disjunctive, or threshold); (iii) an *environment attitude* — demonic/forceability, or one of the admissible/rational/secure alternatives of Section C; and (iv) an *information/merge policy* for how a coalition's own members are allowed to share visibility, i.e. which lens (Section 14.5) realizes the merge. For fixed K, parameters (i) and (ii) alone add no expressive power beyond a finite Boolean combination of the F_S primitives already in this section: an operator varying only those two is syntactic sugar. Parameters (iii) and (iv) are exactly the two axes this paper keeps finding to be load-bearing — the solution-concept spectrum of Section C and the lens-realizability question of Section 14.5 — so a general operator's real content, if it has any, is concentrated there. We have not built this operator; this remark records what its design space looks like so a future attempt is not mistaken for new content when only (i) and (ii) are varied.

Remark 13.9 (An informal illustration: balance of power). Classical balance-of-power theory offers a natural-language test of the previous remark's distinction. Read *statically* — no coalition is currently strong enough to profitably dominate the rest — balance of power is analyzed via the core or von Neumann–Morgenstern *stable sets* [82], and in "might makes right" settings with no external enforcement the core is typically empty or degenerate, which is the formal echo of such systems being dynamically deterred rather than statically stable [53]; the dedicated international-relations formalization is Niou–Ordeshook–Rose's [63]. Read this way, balance of power is a Boolean property of the coalition landscape — $\forall S.\ \neg F_S(\text{hegemony})$ — and by Theorem 13.8 it is sugar: parameters (i)–(ii) only, hence a two-coalition fact under this paper's flattening results. Read *dynamically* instead — stability sustained because an attempted power grab would trigger a *responding* counter-coalition, the mechanism of alliance politics — the relevant concepts are *bargaining sets* [7] and *farsighted stability* [25, 71], where the deterring response is chosen *after*, and in reaction to, information about the deviation — parameter (iv): who learns what, when, before reacting. We do not know whether the dynamic reading resists flattening in K-FOG's sense; we note only that the literature itself needed a strictly more complex concept to capture the same intuition once information timing entered the picture, which is suggestive rather than conclusive.

Remark 13.10 (A third complexity regime: hierarchical independence). The two complexity results so far are extremes: no $\backslash V$ restriction at all (Proposition 13.1, PSPACE-complete) and fully general, possibly-incomparable $\backslash V$ restriction (Proposition 13.2, NEXPTIME-hard via DQBF). Call a formula's $\backslash V$-annotations *hierarchical* if, for every two binders, the sets of earlier binders each is permitted to see are $\subseteq$-comparable — no two binders have genuinely incomparable

views of the prior history, only nested ones. This is a purely syntactic, checkable condition, strictly weaker than "no restriction" and strictly stronger than "arbitrary $\backslash V$."

Peterson and Reif's multiperson alternation model [68], and its refinement to *hierarchical* multiplayer games of incomplete information by Peterson, Reif, and Azhar [66, 67], is the direct precedent for this restriction: in their model, players whose information forms a nested (clique) hierarchy of depth d have a *decidable* outcome problem with a $(d{+}1)$-fold-exponential time bound, strictly between the alternation-only case and the fully general case, which they show can be *undecidable* — the same wall distributed-synthesis hits for non-hierarchical architectures [70].

Three separable claims, of different strength. We were imprecise about this in an earlier draft of this remark, so we spell it out.

1. **Is the general (non-hierarchical) IF fragment even decidable?** Proposition 13.2 proves only a lower bound (hardness via a reduction *from* DQBF); it does not show K-FOG's $\backslash V$ is *no more* expressive than DQBF, only that it is at least as hard. Because $\backslash V$ permits arbitrary interleaving and multiple rounds of hiding, unlike DQBF's fixed one-block-each shape, it is a live possibility — not excluded by anything proved here — that unrestricted K-FOG forceability is harder than NEXPTIME, or even undecidable, matching Peterson–Reif–Azhar's own finding that the fully general (non-hierarchical) case can be undecidable. This is currently an unaddressed gap in this section, not merely an open conjecture, and it is more basic than either item below.
2. **Is the hierarchical sub-fragment decidable at all?** Almost certainly yes, and probably by a routine argument: guess a Skolem-function table at each hierarchy tier and verify, the same proof shape that already shows plain DQBF is decidable (NEXPTIME-complete, not merely NEXPTIME-hard), applied tier-by-tier instead of in one block. We have not written this out, but we do not consider it a deep unknown.
3. **Does the complexity match the Peterson–Reif–Azhar $(d{+}1)$-fold-exponential bound exactly, with a matching lower bound specific to K-FOG's syntax?** This is the genuinely open part: we have attempted no reduction in either direction, their hardness constructions are for their own abstract model and may or may not embed into K-FOG's prenex, terminal-tuple syntax, and there is no reason the bound has to transfer unchanged rather than shift with K-FOG-specific structure.

This sharpens Section 13.3's conjecture rather than answering it: hierarchical $\backslash V$ is, by construction, exactly the regime where visibility sets are pairwise comparable, and Theorem 6.13 already shows that a merely *comparable* crossing pattern is not enough to break the flattening argument on its own (coupling was also required there). If the $K \geq 3$ separation conjecture is true, its witness must therefore use *non*-hierarchical $\backslash V$-annotations; testing whether coupled objectives plus hierarchical (but nontrivial) independence already separates, before attempting the harder non-hierarchical construction, is a natural intermediate step we have not carried out.

13.4 Discussion

Taken together, these results are more informative about *where* K-role structure matters than a bare list of hardness results would suggest:

- The broadcast K-role fragment is PSPACE-complete, uniformly in K (Proposition 13.1). This uniformity is itself a signal: if $K > 2$ roles carried strategic content beyond two-coalition alternation, we would expect some dependence of the complexity on K, and there is none.
- Coalitional forceability adds nothing further in the broadcast fragment: it reduces to ordinary two-role forceability of a merged game (Theorem 13.5), and is PSPACE-complete for exactly that reason (Theorem 13.6).
- Adding IF-style independence increases hardness to at least NEXPTIME (Proposition 13.2), but the construction we give for this is itself a $K = 2$, single-role phenomenon (Theorem 13.3).

So every hardness result in this section is, on inspection, a two-player or single-role phenomenon stated in K-role notation. We think this is worth saying plainly rather than leaving

implicit: by themselves, these results show that K-FOG forceability is at least as expressive as, and no harder than, the two-player machinery it is built from — they do not show that the K-role structure is doing separate strategic work. Whether there is a genuinely $K \geq 3$ phenomenon, in complexity or elsewhere in the calculus, is open; the IF-binder coalition question above (and Section 15) is where we think it is most likely to be found.

14 Related Work

Scope. There exist broad surveys of logic-and-games that cover structural, epistemic, and dynamic perspectives on interaction, ranging from semantic games for classical logic to logics of powers, preferences, and play updates; see especially the entries *Logic and Games*, *Logics for Analyzing Games*, and *Games, Full Abstraction and Full Completeness* in the *Stanford Encyclopedia of Philosophy* [80, 48, 19]. To keep the comparison focused, we review here only those strands that (i) offer an explicit syntactic account of game structure, or (ii) are close to our first-order, role-indexed presentation.

14.1 Global Specifications of Interaction

A first cluster of formalisms describes *the whole interaction* among several named participants.

Multiparty session types (MPST) and choreographies specify a single global protocol over roles and obtain role-local views by projection; well-formedness ensures the locals compose [50]. The logical underpinning for this line of work is the **"Propositions-as-Sessions"** correspondence, which gives a rigorous mapping from linear logic to session-typed processes [83, 14]. This work, and its extension to **Dependent Session Types (DST)**, primarily focuses on using the logical foundation to guarantee protocol-level properties like **type preservation** and **global progress** (i.e., deadlock-freeness) [75].

The goal of these systems is typically not game-theoretic, but rather to ensure **protocol safety** or **relational verification**. For instance, recent work uses dependent session types to prove that a concurrent program (like a queue) correctly implements the semantics of its sequential model (e.g., proving it is FIFO, not LIFO) [33]. This entire line of work is overwhelmingly focused on *binary* (two-party) sessions; indeed, recent work on dependent session types for verified programming explicitly lists integration with *multi-party* session types as future work [33].

There is also work on enriching global types with logical/refinement conditions and carrying them through projection to verified endpoints [51, 86]. Even in systems that *are* multi-party, such as the "Multiparty Dependent Session Types" of Wu and Xi, the goal remains protocol safety, with the main soundness result being **deadlock-freeness** [11, 85, 81].

Our calculus builds on a similar "session-like" global structure, but differs in two crucial respects: (i) our calculus is **natively K-player**, a direction explicitly noted as future work in the binary logical-foundations literature, and more importantly (ii) evaluation produces a per-role outcome tuple, on top of which we can define game-theoretic success (forceability, coalitions). In other words, we use a session-like global form as a *game logic*, not only as a source of *protocol-safety types*.

The logical account closest to what we would need to make this precise is the **coherence** reading of MPST: Carbone, Montesi, Schürmann, and Yoshida show that multiparty global types correspond to proofs in a generalization of classical linear logic where binary duality is replaced by an n-ary *coherence* condition [18]. That correspondence is fully cooperative: coherence certifies that K local behaviors fit together into one consistent global protocol, with a single (implicit) notion of success. The open question in Section 15 — whether K-FOG admits an abstraction into a K-player fragment of linear logic where cut corresponds to game composition — is best understood as asking for an *adversarial* analogue of coherence, where the K local types carry separate, possibly conflicting objectives rather than a shared one. We have not attempted this; it is the most directly relevant piece of literature we are aware of for that open question, and we flag it as the place to start rather than as something we have shown a relationship to.

Declarative game-description languages such as GDL [59] and later systems like Ludii [69] also give global specifications, but they axiomatize transition, observation, and goal relations of a grounded game, whereas we define games *compositionally* inside a first-order syntax. Unlike Castellan–Yoshida [20], who show that binary session protocols and game-semantic strategies can be given a common event-structure model for the (session) π-calculus, we work with a first-order, K-role, payoff-producing calculus and treat viewpoint/rotation as syntactic operators. Our system is not a model of the session π-calculus, but a specification language whose global shape is session-like.

14.2 Strategic and Coalition Logics

Modal and temporal logics for strategic ability—Parikh's game logic [64], Coalition Logic [65], ATL [6], Strategy Logic [22], and its imperfect-information extensions [9]—interpret formulas over concurrent game structures and reason about what agents or coalitions can enforce. These systems are natively multi-agent but remain modal and essentially Boolean: objectives are truth conditions, not structured outcomes, and the syntax does not expose the move/quantifier level that our calculus uses as its core. Classical game theory [82, 62, 73, 42, 43, 44] provides the semantic background for multi-player interaction, but not a first-order, operational specification language of the kind we develop.

We should be precise about the strength of this distinction. Semantically, individual forceability F_i (Definition 2.3) is exactly a one-shot instance of ATL's $\langle\langle i \rangle\rangle$ coalition modality: "role i has a strategy that guarantees the objective against every strategy of the rest." The syntactic difference we point to — that a K-FOG prefix *is* the arena, quantifier by quantifier, rather than a formula interpreted over an externally given concurrent game structure — is real, but we have not turned it into a translation or a separation result in either direction. We also do not know the precise relationship between K-FOG's PSPACE-complete/NEXPTIME-hard forceability problem (Section 13) and the model-checking complexity of ATL/ATL*/Strategy Logic: these are stated over different complexity parameters (formula size against a fixed finite structure here, versus formula and explicit game graph together there), so the numbers are not directly comparable without doing the reduction. We leave working out this relationship, in either direction, as open.

14.3 Information-Sensitive First-Order Logics

Independence-Friendly Logic and its descendants [47, 46, 60, 77] expose informational independence directly in the syntax and give semantics via two-player imperfect-information games. This line is the closest antecedent for our treatment of information flow: it shows that first-order binding can be read as game moves with (in)dependence conditions. However, it is inherently two-player and polar (Verifier vs. Falsifier) and does not provide role rotation or multiway outcome vectors. Our system can be seen as taking that mechanism and lifting it to K explicitly named roles.

A complexity-measure caveat, parallel to the ATL one above. Classical IF logic's headline complexity fact is Fagin's theorem: IF sentences are exactly as expressive as existential second-order logic, hence (over finite structures) capture NP [29, 77]. This is *data/structure* complexity — the sentence is fixed, the finite structure varies. Proposition 13.2's NEXPTIME-hardness is *formula* complexity — the structure is fixed (domain $\{0, 1\}$), the formula varies, exactly as in the DQBF reduction it uses. These are different axes, and the paper should not be read as claiming IF logic is "secretly NEXPTIME-hard": the correct comparison is between K-FOG's formula complexity and DQBF's own formula complexity, which does match. We had already flagged this same fixed-vs-varying distinction as unresolved for the ATL comparison (Section 14); it applies here for the same reason and we did not originally state it for IF logic specifically.

A negation parallel worth noting. IF logic's own strong (dual, game-interpreted) negation $\sim$ and contradictory negation $\neg$ provably do not coincide outside the FOL-equivalent (determined) fragment, and closure of IF logic under negation over finite models is equivalent to $\mathsf{P} = \mathsf{NP}$ [77]. This is structurally the same shape of result as Theorem 3.10: a natural role-swap/negation-like operator ($\sim$ there, $\tilde{}$ here) that agrees with genuine logical negation only on a restricted fragment (determined sentences there, the zero-sum fragment here), and fails outside it. We do not know of a direct technical connection beyond this shared shape, but it is independent evidence that this kind of failure is a structural feature of information/role-sensitive logics generally, not an artifact of K-FOG's specific construction.

The reformulation of IF logic as *team semantics*, and its development into Dependence Logic [78], is the modern presentation of this same two-player, imperfect-information territory, and is arguably closer to our IF-binder extension (Section 5) than the original Hintikka–Sandu game semantics is. We have not worked out the precise relationship. A natural way to state our contribution once that work is done would be: team semantics is the $K = 2$ slice of K-FOG's information-sensitive fragment, and the DQBF-hardness of Proposition 13.2 is already a $K = 2$ result in this sense (see the remark following that proof); what a genuinely $K \geq 3$ extension of team semantics buys is a separate question we pose but do not answer.

14.4 Game Semantics and Compositional Interaction

Game-semantical and dialogical accounts of logic and computation model proofs or programs as two-player games and connectives as game combinators [10, 35, 58, 79], and are surveyed in [19]. This is distinct from the "Propositions-as-Sessions" (P-a-S) line of work, which uses linear logic as a *type system* for *protocols* (e.g., π-calculus processes) to ensure safety properties [83, 14]. We borrow the compositional stance from classical game semantics—formulas *are* interaction structures—but we make it syntactic, multi-role, and tied to first-order binders, which standard game semantics does not. In other words, we internalize the game structure at the level of the object language, rather than only using games as semantic models for an independent language.

The closest direct antecedent for a multi-player game-semantics program as such is Abramsky's proposal for a "socially responsive" logic with games among more than two players and an explicit conflict/cooperation structure [4]. That proposal is, to our knowledge, still largely a research program rather than a developed calculus, which is itself informative: the gap between "games with $K > 2$ players have an obvious informal reading" and "games with $K > 2$ players have a calculus with the metatheory a 2-player game logic has" is exactly where the difficulty lies, and it is the gap this paper is trying to close for one specific (prenex, first-order) shape of calculus. We do not claim a formal relationship to Abramsky's proposal; we flag it because a reader asking "has this general idea been tried before" should be pointed to it before anything else in this section.

This also distinguishes K-FOG from Japaridze's Computability Logic (CoL) [51], a broad game-semantic framework where formulas represent computational problems (games) and validity means the existence of a machine (strategy) that always wins. CoL is a comprehensive logic of interactive computability, whereas K-FOG is a logic of multi-role strategic ability with explicit information constraints and distinct per-role objectives.

14.5 Compositional Game Theory, Optics, and Selection Functions

A separate line of work gives games a categorical, fully compositional semantics: *open games* are the morphisms of a symmetric monoidal category, composed sequentially and in parallel and drawn as string diagrams, with a distinguished equilibrium structure singling out optimal play [34, 45]. The wiring these diagrams describe is built from *lenses*, a get/put pair describing a lossless view of, and update to, a larger state [32], generalized to the more permissive notion of *optics* [72] to accommodate branching and stochastic structure [12].

This is close enough to Section 6's structural-combinator algebra that it is worth flagging, though we have not done the work of relating them. Two places we suspect a connection, neither

of which we have checked:

1. **The wiring layer.** Sequencing ($|\rangle$), tensor ($\otimes$), and splicing (Section 6.2) all describe how to interleave prefixes and merge terminals under a visibility/information discipline — exactly the kind of compositional wiring lenses and optics are built to formalize. If K-FOG's structural combinators are (a fragment of) open games in different notation, that would hand Section 6's normal-form lemma (Theorem 6.4) a categorical semantics for free. We have not checked whether the correspondence holds, or where it would break.
2. **The coalition-merging question, stated as a lens.** Theorem 13.5 says exactly that role-merging under broadcast visibility is a *lawful* lens: the get reading a coalition's per-member strategies off as one merged strategy, and the put writing a merged strategy back out to the members, are mutually inverse, so no information is invented or destroyed. The open conjecture in Section 13.3 (as sharpened by Theorem 6.13: it needs coupled objectives, not merely a $\backslash V$-crossing) then has a clean restatement: for some IF-restricted, objective-coupled coalition, no lawful lens realizes the merge, and the two naive candidates — give the merged role the *union* of its members' visibility, or the *intersection* — fail in different directions (one invents coordination the coalition did not have, the other destroys coordination it did), with distributed coalitional forceability sitting somewhere in between. We have not pinned down which of the standard lens laws (get-put vs. put-get) each candidate violates, or whether a weaker, quantified notion (lawful only up to the gap) is the right model; that formalization is what would turn the conjecture into a checkable statement.
3. **Is forceability even (lax) monoidal?** Before either of the above is attempted, the open-games framework would ask a more basic question first: is the map from a formula to its forceability vector compatible with $\otimes$ at all — does $F(\Phi \otimes \Psi)$ decompose into $F(\Phi)$ and $F(\Psi)$, or only receive a lossy map from them? Two small results already in this paper bear on it: forceability is insensitive to the extra visibility of the leakier sequencing product for disjoint arenas (Theorem 6.9), and the cross-diagonal product's forceability factors exactly into its two components' forceability when their objectives are separable (Theorem 6.13). Both are *strict*, not merely lax, but both are also restricted to the disjoint-arena, separable-objective combinators we define by hand. We have not checked whether strictness survives for structural combinators with coupled cross-component objectives (the terminal-combining Boolean function B_ℓ of Theorem 6.1's (SC2), which is explicitly allowed to mix several components' formulas) — which is exactly where we would also expect the coalition-merging question above to bite, and for the same underlying reason: information flow the syntax restricts, that a naive value-level composition could only assume away or invent.

There is a mismatch to flag before either of these is pursued. The standard open-games equilibrium structure selects *best-response* play — it is built to have the same Nash equilibria as the games it represents [34] — which is exactly the solution concept K-FOG's forceability F_i was built to avoid (Section 2). Grafting K-FOG's semantics onto the open-games categorical structure without addressing this would silently reintroduce the equilibrium-composability problem the rest of this paper works to sidestep. We looked for existing work generalizing the open-games equilibrium functor to a worst-case/adversarial (minimax-style) selection instead of a Nash one and did not find any; if none exists, that generalization would itself be new work, not a citation we are missing.

A closer semantic match for F_i specifically, and one that does not detour through Nash equilibria at all, is Escardó and Oliva's *selection functions* [28]: a selection function picks a preferred input for a given continuation, existential and universal quantification are both instances of the pattern, and they compose (via a monad structure) into exactly the optimal-strategy, backward-induction semantics of alternating-turn sequential games. That alternation — $\exists \sigma_i \, \forall \sigma_{-i} \ldots$ — is structurally what Definition 2.3's forceability already is. We flag selection functions as the more promising candidate for a *semantic* (rather than only structural/wiring) categorical account of forceability, and open games/optics as the more promising candidate

for the *compositional algebra* of Section 6; whether the two combine into one framework for K-FOG, or pull in different directions, is a question for a scoped investigation, not something this paragraph settles.

14.6 Quantitative and Multi-Outcome Extensions

Work on richer winning conditions and quantitative objectives in games and verification (e.g. parity and stochastic games [54, 23], multi-agent model checking [57]) shows how to move beyond purely Boolean outcomes, but typically stays modal and does not offer a first-order, role-indexed syntax with explicit symmetry. Our outcome tuples live at the object level of the calculus and are aligned with the roles of the global description.

The terminal K-tuple $\langle \varphi_0, \ldots, \varphi_{K-1} \rangle$ itself, in isolation, is not new: it is exactly the payoff structure of a *Boolean game* [41], where each player controls some variables and has a propositional goal formula over the shared variable set. We should be explicit that this is where our terminal comes from and what we add on top of it: (i) the goals are first-order, not propositional, and range over a shared first-order structure rather than a shared set of Boolean variables; and (ii) the variables are not simply "controlled" by a player in the sense of an unordered assignment — they are bound by an explicit, ordered, visibility-annotated quantifier prefix, so that who moves when and who sees what is part of the syntax rather than left implicit. Neither addition is deep on its own; the calculus's content, if any, is in fixing this specific combination and working out its metatheory (Sections 2, 3 and 13), not in the terminal payoff shape.

14.7 Position of This Work

The calculus here can be read as: (i) a global, role-indexed, dependent type with per-player refinement in the spirit of MPST/choreographies [50] and of declarative game-description languages [59, 69], but with a focus on game-theoretic payoffs rather than *only* protocol safety or *relational verification* [33]; (ii) internal to first-order, information-sensitive syntax as in IF/Dependence [47, 60], and (iii) producing per-role outcomes and supporting explicit symmetry operations (adversarial projection to a viewpoint, rotation of roles, and an explicit embedding of classical FOL). These operations should be understood as making the embedding of classical FOL into the multi-role setting precise, not as a separate algebraic contribution.

14.8 Applications

The structures in K-FOG are general enough to model a wide range of multi-agent scenarios. While detailed case studies are beyond the scope of this paper, we briefly sketch several application domains:

- **Blockchain Protocols:** K-FOG can serve as a core calculus for domain-specific languages targeting smart contracts. IF-binders (Section 7) naturally encode commit-reveal schemes for on-chain games with hidden information. This is a focus of a concurrent work.
- **Game-Theoretic Analysis:** The same compiler targets Gambit, translating specifications into extensive-form games for automated equilibrium analysis and strategy synthesis.
- **Partially-Cooperative Specifications:** K-FOG can model software systems where multiple modules cooperatively satisfy a global invariant (shared objective) while also achieving their own local postconditions (individual objectives). The forceability semantics lets developers implement modules defensively, accounting for incomplete or adversarial behavior by other components or test harnesses.
- **Interactive Proof Systems:** The information hiding of $\backslash V$ binders directly models non-communication constraints in multi-prover interactive proofs (MIP). The distinction between visible and hidden moves corresponds to public (AM) versus private (IP) randomness in complexity-theoretic protocols. Secret-sharing might be incorporated via fractional permission.

15 Conclusion

We have presented K-FOG, a symmetric K-player calculus with a minimal, prenex core and a $K = 2$ specialization that is algebraically rich and serves as a retract of classical FOL. By internalizing player roles, vectorized outcomes, and explicit S_K symmetry, the calculus provides a foundation for analyzing multi-agent games. We extended this core with IF-style binders, which raises the complexity of the forceability problem from PSPACE-complete to NEXPTIME-hard, and allows the logic to syntactically capture information constraints.

We want to be explicit about the limits of what this paper shows. In particular, the complexity results should *not* be read as evidence that $K > 2$ roles carry strategic content beyond the two-player machinery the calculus is built from: as discussed in Section 13, the PSPACE-completeness of the broadcast core is uniform in K, coalitional forceability under broadcast visibility reduces to ordinary two-role forceability of a merged game (Theorem 13.5), and even the NEXPTIME-hardness result is witnessed by a $K = 2$ construction (Theorem 13.3). What we have established is that a specific K-role, prenex, first-order game syntax admits clean semantics, a retract onto classical FOL at $K = 2$, a nontrivial (if only partially understood) symmetry algebra, and MPST-style protocol readings, with the expected metatheory worked out. We have *not* established that this syntax does anything a family of two-player games could not already do; that is the open question we consider most important, first below.

Open questions.

- **Genuine $K \geq 3$ separation.** Is there a K-FOG phenomenon — in forceability, complexity, or elsewhere — that is not reducible to a family of two-player or two-coalition facts? Section 13.3 conjectures that coalitional forceability under IF-binders, where a coalition's own members are mutually information-restricted *and* whose objectives are coupled across that restriction, is the place to look: role-merging is provably lossless once visibility is broadcast (Theorem 13.5), a mere $\setminus V$-crossing within a coalition is not by itself enough (Theorem 6.13), and the existing NEXPTIME-hardness construction does not touch this case (Theorem 13.3). No separating construction is currently known; finding one — or proving that none exists, i.e. that K-FOG forceability always reduces to two-coalition facts — would be the paper's missing central theorem. This same fork has now recurred three times under three different readings of the calculus — here, in the tensor/coalition-merging discussion of Section 14.5, and in the Byzantine-robustness remark of Section C — and collapsed to two-coalition facts in the perfect-information case each time. That convergence is itself evidence about where to look: every attempt so far suggests genuine $K \geq 3$ content, if it exists in this calculus, comes from information asymmetry between roles rather than from coalition counting. All of this is posed information-theoretically; Section B.14 poses, but does not resolve, the same question once independence is only assumed to hold computationally.
- **A third complexity regime between the two known ones — and a more basic gap underneath it.** Theorem 13.10 separates three claims of different strength. The most basic is not about the hierarchical restriction at all: Proposition 13.2 only proves NEXPTIME-*hardness* of the general IF fragment, with no matching upper bound, so unrestricted K-FOG forceability is not currently known to be decidable — Peterson–Reif–Azhar's own finding that fully general (non-hierarchical) multiplayer incomplete-information games can be undecidable [66, 67] leaves this as a live possibility, not merely a formality. Restricting to *hierarchical* (pairwise-comparable) $\setminus V$ patterns is very likely decidable by a routine Skolem-table argument (not a deep unknown); what is genuinely open is whether its complexity matches their $(d{+}1)$-fold-exponential bound exactly, with a matching lower bound specific to K-FOG's own syntax. This is a well-posed, independently interesting complexity question even if the $K \geq 3$ separation conjecture above turns out to be false, and a natural warm-up for it: it isolates nesting depth as a complexity parameter distinct from K, and — because Theorem 6.13 already shows comparability alone does not break flattening — confirms that a separating construction, if one

exists, must specifically use non-hierarchical independence, not merely nontrivial independence.

- **Relationship to ATL, Coalition Logic, and Strategy Logic — check this one first.** Individual forceability F_i is semantically a one-shot instance of ATL's $\langle\langle i \rangle\rangle$ modality (Section 14). We have not worked out a translation in either direction, nor compared complexity results on equal footing (formula complexity here vs. combined complexity there). Unlike the other items in this list, a bad answer here is not merely a missing result but a threat to the rest of them: if forceability turns out to be a routine encoding of $\langle\langle i \rangle\rangle$, several other open questions in this list inherit that reduction for free (in either a good or a bad way, depending on which direction it runs) and should be revisited in its light. It is also comparatively cheap to check. This should be done before any claim of novelty relative to that literature is treated as established rather than plausible, and before investing in the harder items below.
- **Team semantics / Dependence Logic.** The IF-binder extension overlaps with Väänänen's dependence logic [78], the modern reformulation of the same two-player, imperfect-information territory Hintikka–Sandu IF logic occupies. We conjectured team semantics corresponds to the $K = 2$ slice of K-FOG's information-sensitive fragment, consistent with Theorem 13.3; on reflection a clean biconditional is unlikely to be the right target, since IF logic and dependence logic are already known to diverge at the open-formula level (despite sentence-level equivalence) and team semantics itself has lax/strict variants. The more realistic — and still worthwhile — shape of this item is a *taxonomy*: which variant corresponds to K-FOG's $K = 2$ IF-fragment under which translation, and exactly where the others diverge from it. We have not attempted this in either form.
- **A full proof theory — scope this before attempting it.** We have no sequent calculus, cut rule, or cut-elimination result for K-FOG; Section 8's relational-algebra reading gives compositional intuition (profiles as joins) but is not a substitute for one. This should not be stated as flatly open, though: classical IF logic is provably *not* axiomatizable (validity is not recursively enumerable, via Trakhtenbrot's theorem [76, 77]), and our IF-binder extension is close enough to IF logic (Sections 5 and 14) that a complete proof calculus for the *full IF-extended* fragment may be impossible in principle, not merely unbuilt. What remains open — and what a proof theory should actually be scoped to — is the broadcast core (which retracts onto classical FOL at $K = 2$ and inherits its axiomatizability) and whichever restricted IF fragments (if any) stay decidable or recursively axiomatizable; the conjecture that the full IF-extended fragment resists complete axiomatization should itself be checked, not assumed, since we have not verified K-FOG's IF-binders are expressive enough to actually embed the constructions Trakhtenbrot's argument needs.
- K-FOG **as a semantic model of a K-player fragment of linear logic.** This would mean an abstraction from multi-party linear propositions into K-FOG games such that cut corresponds to game composition. Carbone, Montesi, Schürmann, and Yoshida's coherence-proofs correspondence for MPST [18] is the closest existing result along these lines we are aware of, though it is fully cooperative (coherence, not our adversarial forceability); an adversarial analogue of coherence is, as far as we know, still to be worked out (Section 14).
- **Classifying the complement operator.** Theorem 3.11 notes that our negative results about $\widetilde{\cdot}$ (failure to preserve zero-sumness, failure of De Morgan duality) are witnessed by single counterexamples, not a classification of when these properties hold. Relatedly (Theorem 3.12): what group of transformations of the K-player outcome space interpolates between the one-hot (zero-sum) and diagonal (fully cooperative) extremes for $K > 2$, generalizing the $K = 2$ rotation/complement symmetries of Section 3, and do its orbits correspond to meaningful degrees of cooperativeness or coalition structure? A categorical axiomatization of what the group action preserves (in the spirit of $*$-autonomous categories for classical duality) would likely answer both questions at once. A more speculative version of the same question, not derived here: the S_K action also acts on coalition families $\mathcal{S} \subseteq 2^{\{0,\ldots,K-1\}}$ (Theorem 13.8), and its orbits are a candidate notion of "type of power structure" — "no singleton dominates"

is S_K-invariant, "role 0 is hegemon" is not — which would make "unbiased" formal as S_K-equivariance of a coalition predicate. We have not computed a single orbit or posed the corresponding representation question (which S_K-invariant labelings of the coalition lattice are realized by an actual K-FOG formula's forceability landscape); this is a research direction, not a claim.

- **Compositional game theory as candidate machinery.** Section 14.5 surveys open games, lenses/optics, and Escardó–Oliva selection functions as a possible source for both the categorical axiomatization above and a precise version of the coalitional-merging conjecture (do coalitions correspond to lenses, with the over-/under-approximation gap identifying exactly where $K \geq 3$ content lives?). One specific obstacle: the standard open-games equilibrium functor is Nash-style, which is the composability failure mode K-FOG's forceability semantics was built to avoid, so this machinery cannot be adopted wholesale without first substituting an adversarial/minimax selection notion — itself apparently unstudied in that literature. This is scoped as a side investigation to try, not a claim about what it will find.
- **Probabilistic and quantitative extensions.** Combining this work with mixed strategies, graded payoffs, or probabilistic reasoning is future work; Section B.15 notes specifically that the $\{0,1\}$-valued setting collapses the utility-gap arguments used in rational-proof and scoring-rule literature, which graded payoffs would restore.

A Proofs

Proposition A.1 (Canonical 2-player embedding)**.** *Fix $i \in \{0,1\}$. Assume a 2-player calculus with binders $P_0x.$ and $P_1x.$ and an adversarial projection operation*

$$\underset{j}{\exists\forall} : \textit{2-FOL} \to \mathrm{FOL} \qquad (j = 0, 1)$$

such that

$$\underset{i}{\exists\forall}(\langle\psi_0, \psi_1\rangle) = \psi_i, \qquad \underset{i}{\exists\forall}(P_i x.\, \Phi) = \exists x.\, \underset{i}{\exists\forall}(\Phi), \qquad \underset{i}{\exists\forall}(P_{1-i} x.\, \Phi) = \forall x.\, \underset{i}{\exists\forall}(\Phi),$$

and similarly for $j = 1-i$ with the roles of $\exists$ and $\forall$ swapped. Then there is at most one operation

$$\mathsf{A}_i : \mathrm{FOL} \to \textit{2-FOL}$$

satisfying

1. $\exists\forall_i \circ \mathsf{A}_i = \mathrm{id}_{\mathrm{FOL}}$,
2. $\exists\forall_{1-i} \circ \mathsf{A}_i = \neg(\,\cdot\,)$,
3. A_i *is compositional over the FOL constructors (atoms, $\neg$, $\wedge$, $\vee$, $\exists$, $\forall$).*

Moreover, this unique operation is given inductively by

$$\mathsf{A}_i(\alpha) = \langle\alpha, \neg\alpha\rangle, \quad \mathsf{A}_i(\neg\varphi) = \langle\neg\, \underset{i}{\exists\forall}(\mathsf{A}_i(\varphi)),\ \underset{i}{\exists\forall}(\mathsf{A}_i(\varphi))\rangle,$$

$$\mathsf{A}_i(\varphi \wedge \psi) = \mathsf{A}_i(\varphi) \wedge \mathsf{A}_i(\psi), \quad \mathsf{A}_i(\varphi \vee \psi) = \mathsf{A}_i(\varphi) \vee \mathsf{A}_i(\psi),$$
$$\mathsf{A}_i(\exists x.\varphi) = P_i x.\, \mathsf{A}_i(\varphi), \qquad \mathsf{A}_i(\forall x.\varphi) = P_{1-i} x.\, \mathsf{A}_i(\varphi),$$

where $\wedge, \vee$ on the right-hand side are the componentwise game-combinators compatible with adversarial projection. Up to the player-rotation that swaps 0 and 1, this embedding is canonical.

Proof sketch. Proceed by structural induction on the FOL formula φ.

For atoms α: by (1) we must have $\exists\forall_i(\mathsf{A}_i(\alpha)) = \alpha$. By (2) we must also have $\exists\forall_{1-i}(\mathsf{A}_i(\alpha)) = \neg\alpha$. But adversarial projection on terminal pairs is just projection, so $\mathsf{A}_i(\alpha)$ must be $\langle\alpha, \neg\alpha\rangle$.

For Boolean connectives: by (3) we must define $\mathsf{A}_i(\varphi \circ \psi)$ from $\mathsf{A}_i(\varphi)$ and $\mathsf{A}_i(\psi)$ using the same constructor $\circ$. Because adversarial projection commutes with these constructors (by definition of the calculus), (1) and (2) are preserved.

For quantifiers: consider $\exists x.\varphi$. By (1), $\exists\forall_i(\mathsf{A}_i(\exists x.\varphi))$ must be $\exists x.\varphi$. By the specification of $\exists\forall_i$ on binders, this is possible only if $\mathsf{A}_i(\exists x.\varphi)$ is of the form $P_i x.\,\Phi$ with $\exists\forall_i(\Phi) = \varphi$, hence $P_i x.\,\mathsf{A}_i(\varphi)$. Similarly, for $\forall x.\varphi$, condition (2) forces the other player's binder: $\exists\forall_{1-i}(\mathsf{A}_i(\forall x.\varphi)) = \neg\forall x.\varphi = \exists x.\neg\varphi$, and under the given adversarial projection convention this means $\mathsf{A}_i(\forall x.\varphi)$ must be $P_{1-i}x.\,\mathsf{A}_i(\varphi)$.

Since each constructor is forced in this way, no alternative definition of A_i can satisfy (1)–(3). Finally, choosing $i = 0$ or $i = 1$ just swaps the two components, so the two embeddings are related by the role-rotation. □

B Cryptographic Primitives: Capabilities, Axioms, and Relevance

This appendix summarizes standard cryptographic primitives in a way that is useful for reasoning about information flow and coordination. For each primitive we list: (i) what it provides in standard terms; (ii) canonical security notions (assumptions/axioms); (iii) an intuitive/abstract reading; and (iv) potential relevance to K-FOG.

Notation. We write i, j for principals/roles, pk_i, sk_i for public/secret keys, c for commitments/ciphertexts, and Π for public assertions/certificates.

B.1 Commitment Schemes

What it provides (standard). Two-phase interface: $\mathsf{Com}(x; r) \to c$ (commit) and $\mathsf{Open}(c; x; r)$ (reveal). Pedersen, hash-based, etc.

Canonical security notions. *Hiding* (computationally, c leaks no information about x). *Binding* (no adversary can open c to two distinct values). *Correctness* (honest opens verify).

Intuitive / abstract meaning. Separation of *declaration* from *definition*: "fixed now, opaque until chosen to reveal." Lets you assert relations later (equality, membership) without retroactively changing the value.

Potential relevance to K-FOG**.** Model "causal without value" dependencies; schedule *when* a value becomes usable; link hidden choices across branches by later equalities.

B.2 Zero-Knowledge Proofs (Commit-and-Prove)

What it provides (standard). Prover convinces verifier that a statement $\phi(w; z)$ is true about a hidden witness w (often committed) and public input z, revealing nothing beyond validity (Sigma protocols, SNARKs/STARKs).

Canonical security notions. *Soundness* (false statements are unprovable); *Zero-Knowledge* (no leakage beyond truth); optionally *(Knowledge-)soundness* (a valid proof implies existence of a witness, sometimes extractable in the ideal model).

Intuitive / abstract meaning. "Declaration + properties": you can publish any property of a hidden value without revealing the value itself; selectively disclose.

Potential relevance to K-FOG**.** Express rich constraints on hidden choices (range, membership, equality, functional relationships) while preserving independence/visibility structure.

B.3 Digital Signatures

What it provides (standard). $\mathsf{Sign}_{sk_i}(m) \to \sigma$, $\mathsf{Verify}_{pk_i}(m, \sigma) \in \{\top, \bot\}$.

Canonical security notions. *EUF-CMA unforgeability*; *transferable verifiability*; (non-*deniable*) *non-repudiation*.

Intuitive / abstract meaning. Attribution that survives transfer: anyone can check who endorsed what.

Potential relevance to K-FOG**.** Attach statements to roles; make public assertions accountable; distinguish authorized from informal claims.

B.4 Public-Key Encryption (PKE)

What it provides (standard). $\mathsf{Enc}_{pk_j}(m) \to c$, $\mathsf{Dec}_{sk_j}(c) \to m$. Typically via KEM+AEAD.

Canonical security notions. *IND-CCA confidentiality* (best practice); authenticity only if combined with signatures.

Intuitive / abstract meaning. *Capability split*: public *write* to j's mailbox; private *read* by j only. Adds a private edge to j without informing others.

Potential relevance to K-FOG**.** Implements private visibility / IF-style independence; enables hidden coordination without disturbing public knowledge.

B.5 Symmetric Encryption (AEAD)

What it provides (standard). Authenticated encryption with associated data: $\mathsf{Enc}_k(m, ad) \to c$, $\mathsf{Dec}_k(c, ad) \to m/\perp$.

Canonical security notions. *IND-CPA/CCA confidentiality* and *INT-CTXT integrity*; nonce-discipline assumptions.

Intuitive / abstract meaning. A secure channel once a key exists; *ad* binds context (session, role, epoch).

Potential relevance to K-FOG**.** Implements reliable private subchannels after key agreement; preserves global structure while enabling local secrecy.

B.6 Hashes, Merkle Trees, and Vector Commitments

What it provides (standard). Binding, succinct handles for data; short membership/non-membership proofs.

Canonical security notions. *Collision resistance* (hashes); *position-binding* and proof soundness (Merkle/VC).

Intuitive / abstract meaning. Names for large structures; selective disclosure of parts.

Potential relevance to K-FOG**.** Attach short certificates to public state; avoid flooding the public script when only a property matters.

B.7 Public Randomness: Beacons and VRFs

What it provides (standard). Common unpredictable randomness (*beacon*); per-identity, privately sampled but publicly verifiable randomness (*VRF*).

Canonical security notions. *Unpredictability*, *uniqueness*, *bias-resistance*.

Intuitive / abstract meaning. Stochastic moves that are either common-knowledge (beacon) or privately verifiable (VRF).

Potential relevance to K-FOG**.** Model probabilistic binders without introducing side information; coordinate random choices.

B.8 Secret Sharing and Threshold Cryptography

What it provides (standard). t-of-n reconstruction for secrets (Shamir); threshold signatures/decryption.

Canonical security notions. *t-privacy* ($< t$ learn nothing), *t-correctness* ($\geq t$ succeed); *monotonicity*.

Intuitive / abstract meaning. Fractional permissions / joint custody; coalitions gate learning or action.

Potential relevance to K-FOG**.** Coalition-controlled reveals; "broadcaster/choke-point" nodes realized as threshold events.

B.9 Secure Multiparty Computation (MPC)

What it provides (standard). Compute $y = f(x_1, \ldots, x_n)$ while keeping inputs private; reveal only agreed outputs.

Canonical security notions. *Correctness*, *privacy* against coalitions up to a threshold; optionally *robustness*/*fairness*.

Intuitive / abstract meaning. Compute on secrets; disclose only what the specification allows.

Potential relevance to K-FOG**.** Publish complex functions of hidden choices without violating independence; keep value-visibility minimal.

B.10 Append-Only Ledgers / Bulletin Boards

What it provides (standard). Public, tamper-evident, globally ordered log of statements.

Canonical security notions. *Append-only*; *consistency* (common prefix); liveness under network assumptions.

Intuitive / abstract meaning. A shared timeline; prevents backdating and equivocation.

Potential relevance to K-FOG**.** Implements a globally agreed causal order for public assertions; supports time-based reasoning.

B.11 Verifiable Delay Functions (VDFs)

What it provides (standard). Enforces a minimum delay Δ to learn $y = \mathsf{VDF}^{\Delta}(x)$ with a short, quickly verifiable proof.

Canonical security notions. *Sequentiality*, *uniqueness*, *fast verification*.

Intuitive / abstract meaning. Time-locks: information becomes available only after Δ.

Potential relevance to K-FOG**.** Model latency constraints ("reveal only after time t") without adding new players.

B.12 Anonymous Credentials; Ring and Group Signatures

What it provides (standard). Authorized statements with hidden author (ring); membership-certified anonymity with optional opening/revocation (group/credentials).

Canonical security notions. *Unforgeability*, *anonymity*; (un)linkability depending on scheme; accountable opening for group sigs.

Intuitive / abstract meaning. Authorization decoupled from identity; speak as a class without revealing who.

Potential relevance to K-FOG**.** Model "someone with role-rights said this" without disturbing role anonymity.

B.13 Oblivious Transfer (OT), Mixnets, and ORAM

What it provides (standard). OT: receiver learns one of many items, sender learns not which. Mixnets: unlink sender/receiver; ORAM: hide access patterns to memory.

Canonical security notions. Standard indistinguishability definitions against traffic/choice observers.

Intuitive / abstract meaning. Hides meta-information (who talks to whom; which index was used).

Potential relevance to K-FOG**.** If independence should hold even against traffic analysis, these support private edges without meta-leakage.

Design takeaway. For most information-flow arguments one can assume the following *axioms* without committing to a concrete scheme:

- *Declaration vs. definition:* values can be fixed yet hidden until revealed (commitment).
- *Selective disclosure:* properties of hidden values can be asserted without revealing the values (zero-knowledge).
- *Capability split:* public write / private read channels exist (PKE); private authenticated channels exist given a key (AEAD).
- *Attribution:* transferable accountability for statements (signatures).
- *Coalition gates:* certain reveals/actions require threshold participation (secret sharing/threshold crypto).
- *Order/time:* public append-only order and enforceable delays can be assumed (ledger, VDF).

These are sufficient to justify common K-FOG constructions around visibility, independence, and coordination *without* embedding cryptographic details in the language.

B.14 A Soundness Obligation This Appendix Has Not Discharged

The axioms above justify K-FOG's $\backslash V$ constructions as *realizable*, but not as *sound*: $\backslash V$-independence (Section 5) is an exact, unconditional claim about a strategy function's argument list, of exactly the same shape as the Dolev–Yao "perfect encryption" idealization in symbolic protocol analysis — this is an identification, not merely an analogy, since both say a value is *totally* unavailable to an excluded party rather than merely hard to extract. Whether an idealized independence claim survives once the primitives above realize it computationally is exactly the question computational-soundness results (relating symbolic and computational models of cryptography) and universal composability (relating a real protocol to an ideal functionality up to indistinguishability by any efficient environment) exist to answer [2, 16]. We have not checked this for any K-FOG construction in this appendix.

The direction such a check would go is not obvious, and depends on a definitional fork we have not resolved: does $\backslash V$ prohibit *any* private communication among the excluded parties (a genuine isolation constraint, under which a coalition sharing secret correlated randomness — the exact mechanism secure multiparty computation and correlated-equilibrium-without-a-mediator constructions provide [27] — is simply outside the model by fiat), or only communication an efficient *outside observer* could detect (under which such a coalition might satisfy a relaxed, computational notion of independence while still correlating its choices with the excluded value, since the winning condition in Section 13.3-style constructions checks realized bits directly and is not itself a statistical test a coalition needs to fool). Under the first reading we see no obvious mechanism for computational assumptions to change the flattening/separation landscape of Section 13 at all, since Theorem 13.5's argument is a pure relabeling that preserves efficiency automatically. Under the second, standard results give real reason for caution rather than optimism: Halpern, Pass, and Reichman show Nash equilibrium can fail to exist outright for resource-bounded players [39], and the closest existing treatment of bounded computation inside extensive-form games [38, 40] — surveyed together with the rest of this intersection in [55] — is exactly the place to check whether forceability's maximin shape (rather than Nash's fixed-point shape) is resilient to that hazard before conjecturing anything about coalitions specifically. We pose the fork itself, not either resolution, as the open question.

B.15 Payoff as Evidence of (In)Capability or Knowledge

A recurring theme across economics, learning theory, and cryptography is using *payoffs* (or scores) as operational evidence about an agent's information state or computational capability. The informal reading is: if an agent is *rational* (utility-maximizing) and nevertheless fails to achieve a certain payoff threshold that would be attainable by someone with a specific piece of knowledge or computational ability, then the observed shortfall is evidence that the agent lacks that knowledge/ability. This idea appears under several names:

Rational (Interactive) Proofs and Utility Gaps. In *rational proofs* (a.k.a. *rational interactive proofs*), the verifier designs payments so that a prover maximizes expected payoff if and only if it supplies correct information (e.g., a correct computation). The key notion is a *utility gap*: there exists $\delta > 0$ such that any strategy producing the correct answer yields an expected payoff at least δ higher than any incorrect strategy. Persistently earning less than the optimal payoff is then formal evidence of *incapability* (or irrationality). This is the closest formalization of "payoff $\Rightarrow$ (lack of) capability."

Proper Scoring Rules and Peer Prediction. With *strictly proper scoring rules*, truthful reports of one's belief uniquely maximize expected score. Systematically suboptimal scores thus serve as evidence that the reporter *lacks* accurate information (or chose to misreport against interest). When ground truth is unavailable, *peer prediction* and *Bayesian Truth Serum* engineer

payoffs so that honest, well-informed reports form (the unique) best response in equilibrium.

Signaling and Separating Equilibria. In *costly signaling* (e.g., Spence), only high-ability or well-informed types find it profitable to take a particular action. Failure to take the costly signal is Bayesian evidence of *not* being the high-ability/informed type, again reading payoff-differences as evidence about capability.

Algorithmic Knowledge (Resource-Bounded Knowing). *Algorithmic knowledge* models explicit knowledge via a knowledge algorithm: an agent "knows" φ precisely when its algorithm can derive φ within given resources. Coupled with payoff design (as above), inability to produce a derivation/witness that would strictly improve payoff is evidence of non-knowledge under the stipulated resource bound.

Remark on payoff granularity. The literature typically relies on *numeric* payoffs or scores to create a strict improvement margin (the utility gap δ). In the current $\{0, 1\}$-valued setting, such margins collapse: "profitable deviation" reduces to turning a 0 into a 1, and you cannot use a *larger* loss in one game to certify the ability to secure a *smaller* but positive payoff elsewhere. If you later extend to graded payoffs, you can leverage cross-game arguments of the form: "accepting a higher loss here credibly certifies an ability that secures a smaller gain there," which is the standard currency in scoring-rule and rational-proof analyses.

C A Game-Theoretic Framework for Modular Specifications

A Spec-Agnostic Semantic Lens for Architectural Analysis

Abstract. We present a semantic, specification-agnostic framework for analyzing modular software architectures through *multi-player coordination games*. Component implementations act as strategies; the composed system induces demonic nondeterministic behaviors; and local specifications define binary obligation-satisfaction utilities. A chosen solution concept SC selects architecturally relevant profiles.

Architectural properties such as coupling, substitutability, minimality, and information hiding arise by comparing these SC-selected profiles with globally desirable behaviors. As a motivating illustration rather than a general theorem (see Section C.4 below), some solution concepts—notably Pure Nash Equilibrium (PNE)—can select equilibrium profiles that are **undesired** stability traps: points where no component can locally repair its obligation even though the global system behavior is undesirable.

The approach is semantic, lightweight, and applies to any specification formalism with a behavioral interpretation.

C.1 Motivation

In modular architectures, components interact through a shared semantic environment while each carries its own local obligations. Models such as game semantics, assume–guarantee/rely-guarantee reasoning [52, 1], interface theories [26], or contract-based design [8] typically use *two-player* perspectives (a component against its environment). Some approaches (MPST, certain assume-guarantee frameworks [24]) extend to multi-role perspectives, but here we adopt an even more general view:

> *A tuple of specifications induces a multi-player game. Implementations act as strategies; the composed system induces behaviors; a solution concept selects key profiles for architectural analysis.*

Crucially, the framework is *agnostic* about what those profiles *mean*: depending on the solution concept, they may represent desirable configurations or undesirable ones.

C.2 System Model

Let $C = \{c_0, \ldots, c_{n-1}\}$ be the set of components.

Specifications and Behaviors.

- A behavior domain Beh.
- For each component c:
  - a specification space Spec_c,
  - an implementation space Σ_c,
  - a semantics $[\![-]\!]_c : \mathsf{Spec}_c \to \mathcal{P}(\mathsf{Beh})$.

The satisfaction set may be empty. If $[\![S_c]\!]_c = \varnothing$, then component c is permanently unsatisfiable: $u_c(\sigma) = 0$ for all σ.

Demonic Composition Semantics. A joint profile is $\sigma = (\sigma_c)_{c \in C}$.

$$\mathsf{out} : \prod_{c \in C} \Sigma_c \to \mathcal{P}(\mathsf{Beh})$$

out is *demonic*: every behavior in $\mathsf{out}(\sigma)$ must satisfy obligations. Outputs are assumed non-empty. Utilities therefore depend on the *entire* behavior set.

This demonic reading is appropriate for safety-style reasoning ("every possible behavior must be safe") and pessimistic architectural analysis. For models of intentional internal nondeterminism or optimization under uncertainty, one would instead use angelic or probabilistic readings; these can be accommodated by changing the definition of u_c and the behavior model, but we do not pursue that here.

Observation Granularity. Each component observes via

$$\pi_c : \mathsf{Beh} \to \mathsf{Obs}_c, \qquad \pi_c(B) = \{\pi_c(b) \mid b \in B\}.$$

The projection π_c may collapse behaviors; it defines the abstraction level and visible nondeterminism.

Global Desirability. A global correctness predicate:

$$\mathsf{Valid} \subseteq \mathsf{Beh}, \qquad \mathsf{ValidBeh} = \{\sigma \mid \mathsf{out}(\sigma) \subseteq \mathsf{Valid}\}.$$

No relationship between Valid and local obligations is assumed.

C.3 The Induced Game

Given $S = (S_c)_{c \in C}$:

- players: components;
- strategies: implementations;
- outcomes: $\mathsf{out}(\sigma)$;
- utilities:

$$u_c(\sigma) = \begin{cases} 1 & \text{if } \mathsf{out}(\sigma) \subseteq [\![S_c]\!]_c \\ 0 & \text{otherwise.} \end{cases}$$

Utilities are binary, demonic, and obligation-based: any violating behavior forces utility 0, and satisfying behaviors are not ranked.

C.4 Solution Concepts

A solution concept gives a set

$$\mathsf{SC}(S) \subseteq \prod_{c \in C} \Sigma_c$$

of architecturally relevant profiles.

Different solution concepts correspond to different architectural questions. Pure Nash Equilibrium (PNE) captures unilateral repairability; dominant strategies capture robustness to

any environment behavior; max–min or worst-case concepts capture hardening against adversarial contexts; correlated or cooperative equilibria model designed coordination. In this paper we work explicitly with PNE as a running example, but the definitions are parametric in SC and can be instantiated with these standard notions.

Pure Nash Equilibrium (PNE). Let (σ_{-c}, σ'_c) replace the c-component of σ.

$$\sigma \in \mathsf{PNE}(S) \iff \neg\exists c, \sigma'_c.\ u_c(\sigma_{-c}, \sigma'_c) > u_c(\sigma).$$

Because utilities are binary and demonic:

- only $0 \to 1$ improvements matter;
- profiles where all utilities are 0 may still be PNE;
- PNE can identify *undesired stability traps*.

This undesirability is specific to PNE under binary demonic utilities; other solution concepts may select desirable configurations.

Caveat. The claim that PNE "identifies" undesired stability traps is an informal observation motivating the framework, illustrated by the all-zero-utility example above, not a general theorem: we have not characterized which specification tuples S admit an all-desirable PNE, which admit only trap-like PNE, or how common each case is. Formalizing that characterization (e.g. as a condition on Valid, out, and the u_c) is left open; until it exists, this subsection should be read as a worked motivating example for the framework's vocabulary rather than as evidence that PNE analysis of module specifications reliably surfaces bad states.

Before treating that characterization as a fresh open problem, it should be checked against rational verification's existing complexity theory (see the paragraph below), which already studies "which games admit desirable equilibria" in a closely related form [31, 3]; "self-contained" is precisely the property that makes a question likely to already exist in a neighboring literature.

Named intermediate solution concepts. Forceability-style coupling (Theorem C.1 below) is sound and compositional but paranoid: it certifies safety by assuming every other component behaves adversarially, discarding any information about what the other components actually are or want. PNE is close to the opposite failure: it can fail to rule out profiles that are not even individually compliant. The formal-methods and multi-agent-systems literature has already engineered several points strictly between these extremes, each trading paranoia for compositionality in a controlled way:

- **Secure equilibria** [21] use lexicographic objectives (primarily achieve your own goal, secondarily hurt the opponent) to obtain equilibria with a unique value, engineered specifically to support compositional verification — a targeted fix for the stability-trap problem of Section C.4.
- **Assume-admissible synthesis** [13] has each component assume the others play *admissible* (non-dominated) strategies, rather than assuming either full adversarial behavior or a fixed-point equilibrium. This composes, unlike PNE, while being strictly less paranoid than forceability's $\forall\sigma_{-D}$.
- **Rational synthesis / rational verification** [31, 3] certify a specification under the assumption that the environment plays *some* equilibrium of its own objectives, with a developed complexity theory for when this is decidable.

Ordering these by how much they assume about the environment gives a monotone family, in the sense of SC Monotonicity below, sitting between $\mathsf{SC}_{\mathrm{All}}$ (Theorem C.1: assumes nothing beyond raw compliance) and forceability (assumes the worst). Each of these is a solution concept in this appendix's sense and generates its own coupling, substitutability, and minimality notions by the schema above; we have not carried out that instantiation, nor checked which points of this spectrum survive K-FOG's symmetry-group action (Section 3) or remain two-coalition facts under coalition merging (Section 13.3). Both are left open.

SC Monotonicity. If $\mathsf{SC}_1(S) \subseteq \mathsf{SC}_2(S)$, then

$$\mathsf{SC}_2(S) \subseteq \mathsf{ValidBeh} \;\Rightarrow\; \mathsf{SC}_1(S) \subseteq \mathsf{ValidBeh}.$$

No monotonicity holds for specification weakening or behavior refinement.

C.4.1 Implementation Independence

The coupling notions above are defined *relative to a solution concept*: they only quantify over profiles in $\mathsf{SC}(S)$. Sometimes we want a stronger, *solution-concept–independent* notion that says: once a set of components meets its local obligations, the system is globally well-behaved, regardless of which particular implementations were chosen.

Let $D \subseteq C$ be a set of components. We say that the specification tuple S is *implementation independent on* D if any profile that satisfies the obligations of all components in D is globally desirable:

$$\mathsf{Indep}_D(S) \;\triangleq\; \forall \sigma \in \textstyle\prod_{c \in C} \Sigma_c.\; \left(\bigwedge_{c \in D} u_c(\sigma) = 1 \right) \implies \mathsf{out}(\sigma) \subseteq \mathsf{Valid}.$$

Intuitively, $\mathsf{Indep}_D(S)$ states that implementations for the components in D can be chosen *independently*: any combination of implementations that meets their local specifications yields globally desirable behavior, irrespective of the particular choices made for the remaining components in $C \setminus D$ (provided they are included in σ and participate in out as usual).

Reading via assume-guarantee contracts. This quantifier shape — $\forall$ over D's own compliant implementations, $\forall$ over the environment — is exactly the satisfaction relation of assume-guarantee/rely-guarantee reasoning [52, 1], interface theories [26], and their modern synthesis in contract-based design [8]: a component's contract is satisfied demonically on *both* sides, not only the environment's. Forceability (the main text's Definition 2.3) is angelic on the player's own side ($\exists \sigma_i$: some strategy works) and demonic on the environment's side ($\forall \sigma_{-i}$); Indep_D is demonic on *both* sides ($\forall \sigma_D$ compliant, $\forall \sigma_{-D}$), which is exactly what makes it a statement about the *contract itself* rather than about the existence of a witness — you should not need to find the clever compliant implementation, any of them must do. That body of theory already develops the composition operators (quotient/residuation), refinement orders, and compatibility conditions this appendix does not; what follows should be read as an instance of that theory inside this paper's vocabulary, not as new theory.

Two special cases are worth highlighting:

- **Pairwise independence.** For two components $a, b \in C$, implementation independence on $D = \{a, b\}$ says that whenever both a and b satisfy their local obligations in some profile, the resulting global behavior is valid. This matches the informal idea that the modules for a and b can be implemented and combined without introducing architectural failures.
- **Full independence.** Implementation independence on $D = C$ says that as soon as *all* components satisfy their local specifications in a profile, the system is globally desirable:

$$\mathsf{Indep}_C(S) \;\equiv\; \forall \sigma.\; \left(\bigwedge_{c \in C} u_c(\sigma) = 1 \right) \implies \mathsf{out}(\sigma) \subseteq \mathsf{Valid}.$$

  This is a very strong architectural property: any locally compliant joint implementation is guaranteed to be globally acceptable.

Unlike the SC-relative coupling notion $\mathsf{SC}(S) \subseteq \mathsf{ValidBeh}$, implementation independence quantifies over *all* profiles that satisfy the obligations of the chosen components D, regardless of whether they are selected by the solution concept. It therefore captures a stricter, solution-concept–agnostic form of "independent implementability".

A compatibility (nonemptiness) side condition. As stated, $\mathsf{Indep}_D(S)$ is vacuously true whenever D's locally-compliant profile set is empty — i.e. whenever D's own local specification is unsatisfiable in isolation. This is the same vacuity contract theory calls out under the name *compatibility*. We do not build a nonemptiness requirement into the definition above, so that $\mathsf{Indep}_D(S)$ remains a pure implication matching the rest of this appendix's style; but any use of $\mathsf{Indep}_D(S)$ as evidence of *safe* modularity, rather than merely *vacuously consistent* modularity, should check nonemptiness of D's compliance set separately.

Proposition C.1 (Full independence is coupling at the compliance solution concept)**.** *Define* $\mathsf{SC}_{\mathrm{All}}(S) := \{\sigma \in \prod_c \Sigma_c : \forall c \in C,\ u_c(\sigma) = 1\}$*, the degenerate solution concept that imposes no strategic or rationality assumption beyond raw local compliance. Then*

$$\mathsf{Indep}_C(S) \iff \mathsf{SC}_{\mathrm{All}}(S) \subseteq \mathsf{ValidBeh}.$$

Moreover, if σ is a profile in which every $c \in C$ plays a dominant strategy for its own local specification (i.e. $u_c(\sigma_{-c}, \sigma_c) = 1$ for every σ_{-c}), then $\sigma \in \mathsf{SC}_{\mathrm{All}}(S)$; so the dominant-strategy solution concept is a subset *of $\mathsf{SC}_{\mathrm{All}}$, not in general equal to it (a dominant strategy is compliant in every environment, but a profile can be compliant in the one environment actually realized without any component's strategy being dominant). By SC Monotonicity, $\mathsf{Indep}_C(S)$ therefore implies loose coupling under the dominant-strategy solution concept, of which forceability is the binary-utility instance — but the implication runs only in that direction.*

Proof. Immediate from unfolding both definitions: $\mathsf{Indep}_C(S)$ says $\forall\sigma.\ (\bigwedge_c u_c(\sigma) = 1) \Rightarrow \mathsf{out}(\sigma) \subseteq \mathsf{Valid}$, which is literally $\mathsf{SC}_{\mathrm{All}}(S) \subseteq \mathsf{ValidBeh}$ once $\mathsf{SC}_{\mathrm{All}}(S)$ is unfolded. The dominant-strategy inclusion and the monotonicity consequence are as stated above. □

Remark C.2 (Why PNE escapes this hierarchy). $\mathsf{PNE}(S)$ is *not* in general a subset of $\mathsf{SC}_{\mathrm{All}}(S)$: as already noted, an all-zero-utility profile can be a PNE. So Theorem C.1's monotonicity argument says nothing about PNE-coupling in either direction — $\mathsf{Indep}_C(S)$ can hold while some PNE is still an undesired stability trap, because that PNE need not be locally compliant at all and so falls entirely outside what Indep_C constrains. This is the precise mechanism behind Section C.4's caveat, not merely an informal observation about it.

Remark C.3 (A tempting but likely-flat K-ary candidate). Byzantine-style robustness — "valid under arbitrary deviation by any $\leq f$ of the K components" — is the notion in this framework that most looks like genuine K-ary content, since it quantifies over a whole family of coalitions rather than one fixed coalition. But it is exactly a finite conjunction of two-coalition facts, one forceability-style statement per coalition of size $\leq f$, and the flattening result of Theorem 13.5 applies to each conjunct separately. This is the third time in this paper that a plausible-looking source of K-ary content collapses into a family of two-coalition facts, after the tensor/coalition-merging discussion of Section 14.5 and the IF-binder conjecture of Section 13.3. If this calculus has genuine $K \geq 3$ content anywhere, the pattern across all three attempts points the same way: it will come from information asymmetry between components, not from counting coalitions.

C.5 Architectural Notions

Architectural properties compare $\mathsf{SC}(S)$ with desirable profiles.

Coupling. A system is *loosely coupled* under SC when

$$\mathsf{SC}(S) \subseteq \mathsf{ValidBeh}.$$

Otherwise some SC-profile is globally undesirable. If $\mathsf{SC}(S) = \varnothing$, the condition holds vacuously.

Substitutability Relative to the Global Property. Fix component B. For all environments σ_{-B} and all σ_B, σ'_B with $u_B(\sigma_{-B}, \sigma_B) = u_B(\sigma_{-B}, \sigma'_B) = 1$:

$$\mathsf{out}(\sigma_{-B}, \sigma_B) \subseteq \mathsf{Valid} \iff \mathsf{out}(\sigma_{-B}, \sigma'_B) \subseteq \mathsf{Valid}.$$

Quantification is over *all* environments, not only SC-reachable ones. This expresses Valid-preserving substitutability, not refinement or observational equivalence.

Minimality of Specifications. Semantic weakening:

$$S'_c \leq S_c \Rightarrow [\![S_c]\!]_c \subseteq [\![S'_c]\!]_c.$$

S_B is SC-*minimal* in context S_A when:

1. $\mathsf{SC}(S) \neq \varnothing$ and $\mathsf{SC}(S) \subseteq \mathsf{ValidBeh}$;
2. no strictly weaker $S'_B < S_B$ preserves *all* desirable SC-stable profiles.

Information Hiding (Stability-Oriented).

$$\sigma_{-c} = \sigma'_{-c} \ \wedge \ \pi_c(\mathsf{out}(\sigma)) = \pi_c(\mathsf{out}(\sigma')) \ \Rightarrow \ (\sigma \in \mathsf{SC}(S) \Rightarrow \sigma' \in \mathsf{SC}(S)).$$

A strict extensional criterion: equality of projected behavior sets is required; adding new projected behaviors or splitting one into multiple distinguishable observations breaks hiding. This is stricter than trace- or simulation-based hiding unless accounted for in π_c.

C.6 Adversarial Test Components

In a modular setting, *tests* are naturally modeled as *adversarial* components: they are written with the explicit intention of catching bad behavior of other components. This is slightly at odds with everyday testing terminology: here a "good" test is one that is *bad for* at least one incorrect implementation.

Let $\mathcal{I}$ be the set of implementations of a given interface or component role, and let $\mathcal{G} \subseteq \mathcal{I}$ be the set of *intended* implementations. For example, one may take $\mathcal{G}$ to be those implementations that both satisfy a local specification S_c and lead to globally desirable behavior (with respect to Valid) under some chosen class of environments.

Definition C.4 (Test suite)**.** Let $\mathcal{I}$ be the implementation space for a given interface, and let $\mathcal{G} \subseteq \mathcal{I}$ be the intended implementations. A *test suite* is a function

$$T : \mathcal{I} \to \{\mathsf{pass}, \mathsf{fail}\}.$$

It is

- *non-vacuous* if $\exists I \in \mathcal{I} \setminus \mathcal{G} : \ T(I) = \mathsf{fail}$;
- *satisfiable* if $\exists I \in \mathcal{G} : \ T(I) = \mathsf{pass}$;
- *complete* if $\forall I \in \mathcal{I} : \ T(I) = \mathsf{pass} \iff I \in \mathcal{G}$.

Thus:

Non-vacuous At least one incorrect implementation is actually detected:

$$\exists I \in \mathcal{I} \setminus \mathcal{G} : \ T(I) = \mathsf{fail}.$$

Satisfiable At least one intended implementation survives:

$$\exists I \in \mathcal{G} : \ T(I) = \mathsf{pass}.$$

Complete The suite perfectly separates intended from unintended implementations:

$$\forall I \in \mathcal{I} : \ T(I) = \mathsf{pass} \iff I \in \mathcal{G}.$$

This is a *counterfactual* view: a test suite classifies *all* implementations in $\mathcal{I}$, including those never actually deployed or executed in any particular run. It asks which incorrect implementations *would* be rejected, not only what happens on the single implementation actually in use.

In an architectural setting, a *test component* together with its surrounding harness induces such a test suite T on $\mathcal{I}$: for each implementation I, $T(I)$ records whether the composed system, with that implementation plugged in, causes the test component to pass or fail.

Consequently, what we specify here is not "a system that is already correct" in isolation, but rather

> a system whose interaction structure contains meaningful, adversarial tests that separate intended implementations $\mathcal{G}$ from unintended ones $\mathcal{I} \setminus \mathcal{G}$; correctness itself is then obtained only under the additional assumption that the tests do in fact succeed on the implementations actually deployed.

C.7 Why This Lens Is Useful

The framework separates:

- specification semantics,
- implementations,
- joint behavior,
- observation level,
- architectural stance (solution concept),
- global acceptability.

These parameters are orthogonal; small semantic changes can alter the architecture's stability landscape. This fragility reveals mismatches between obligations, composition semantics, and global aims.

C.8 Summary

A tuple of component specifications induces a multi-player game with binary demonic obligation semantics. A solution concept SC selects profiles of architectural interest. Comparing these with globally desirable behaviors yields precise notions of coupling, substitutability, specification minimality, and information hiding.

Some solution concepts (notably PNE) reveal undesired stability traps, while others may select desirable or optimal configurations. The framework accommodates both without committing to a single interpretation.

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Game-calculus notion	**DB/RA reading (intuition)**
Binder $P_i x_k$ with type D_{x_k}	A *typed column* $x_k{:}D_{x_k}$ in the eventual row
Syntactic (prenex) order	The order we fill columns left-to-right
Visibility edge $x_j \to x_k$	x_j is in the *key* for x_k (allowed dependency)
Slash / independence for x_k	*Key exclusion*: certain earlier columns are *forbidden* from x_k's key
Player i controls x_k	Ownership: who supplies the value for column x_k given its key
Game specification (no strategies)	Schema + key policy: column types and allowed dependencies
Pure strategy for x_k	Deterministic *fill-rule* (graph of a function) from key(x_k) to x_k
Strategy profile	*Natural join* of all per-binder strategy tables
Play (deterministic case)	The *single completed row* produced by the join
Terminal objective for player i	A Boolean predicate on the completed row (selection/filter)
Sequential composition $\Phi;\Psi$	Join on an *interface* of shared columns, then (optionally) project to hide internals
Clock (unit choke-point)	Add a dummy unit column at the boundary; require it in the next phase's keys
Broadcast	Put all prior columns into initial keys of the next phase (wide join)
α-renaming / player permutation	Column rename / ownership relabel; schema isomorphism
Equivalence up to DAG isomorphism	Same dependency structure between columns (schema isomorphism)
Legal histories at x_k	Reachable partial rows projected to key(x_k)
Feasibility (well-posedness)	There exist per-column fill-rules total on legal histories
Forceability of player i	Anti-join pattern: no opponent completion produces a row failing i's predicate
Pure Nash (binary outcomes)	"No $0\to1$ unilateral improvement" expressed as self-join + anti-join

Figure 1: **Relational cheat-sheet.** A compact correspondence between the game calculus and a database/relational-algebra reading. Use this as a waypoint if you get lost in the details.

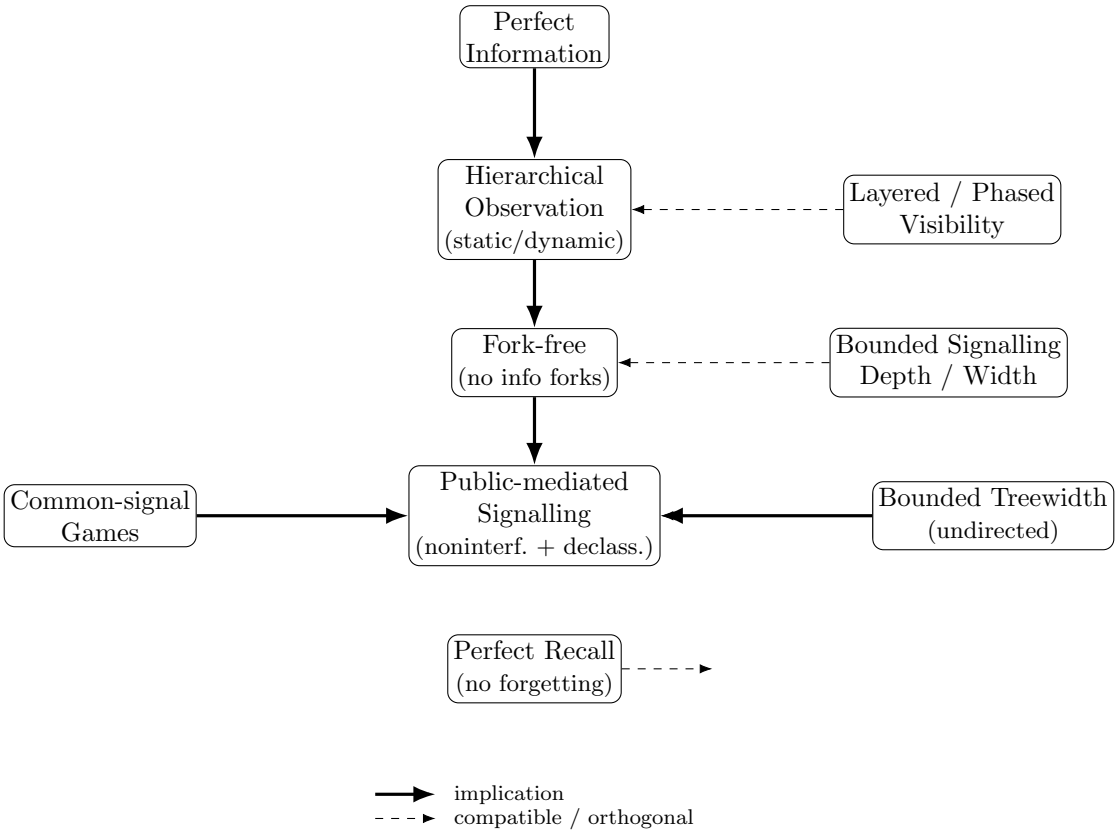


Figure 2: Information-structure relationships.